arXiv:submit/8279132 [math.QA] 25 Oct 2026

CROSSED MODULARITY OF TWISTED VOA MODULES

SHUN XU

Abstract. Let V be a strongly rational vertex operator algebra and $G \leq \mathrm{Aut}(V)$ a finite group. Using the G-crossed braided fusion structure of twisted modules constructed in [7], we prove that the conformal twist $e^{2\pi i L(0)}$ gives a crossed ribbon structure and that the crossed category is modular in Kirillov's sense. Consequently, its equivariantization is a modular tensor category. We also obtain dimension and simple-object counting formulas for the twisted sectors.

1. Introduction

Let V be a strongly rational vertex operator algebra and $G \leq \mathrm{Aut}(V)$ a finite group. For each $g \in G$, let $\mathcal{C}_g$ denote the category of grading-restricted g-twisted V-modules, and put $\mathcal{C}_G = \bigoplus_{g\in G} \mathcal{C}_g$, with the elements of G retained as sector labels. Orbifold theory suggests that tensor products of twisted modules should respect multiplication of these labels, while exchanging a g-twisted field with another field should involve the action of g. The appropriate categorical framework is therefore a G-crossed, rather than an ordinary braided, tensor category.

The representation theory underlying this expectation was developed by Dong, Li and Mason, who established fundamental results on finite-order twisted modules and twisted Zhu algebras [2]; their modular-invariance theorem concerns trace functions in orbifold theory under its stated finiteness and rationality hypotheses [3]. For untwisted modules, the tensor-product theory of Huang and Lepowsky realizes products through intertwining operators and their analytic continuation [12]. Huang's differential-equation results provide convergence and extension tools for this theory [13], and his rigidity and modularity theorem gives a modular tensor category of ordinary modules under the standard semisimplicity, self-duality and cofiniteness assumptions [14]. The crossed tensor structure on all twisted sectors is the additional categorical structure needed for the problem considered here.

Kirillov developed the theory of G-equivariant fusion categories, with twisted modules as a motivating example, and introduced the extended Verlinde algebra and its S-operator [15, Example 5.5 and Sections 7–10]. In this setting modularity means invertibility of the extended S-operator; it is stronger than requiring only the neutral component to be modular. The corresponding VOA problem was formulated by Huang in [4, Conjecture 6.2]: for a finite automorphism group and a vertex operator algebra satisfying the specified self-contragredience, complete-reducibility and C_2-cofiniteness conditions, the twisted sectors should form a G-crossed tensor category. Huang's preceding Conjecture 6.1 separately addresses associativity, commutativity and modular invariance of twisted intertwining operators. The modularity question considered here is the invertibility of Kirillov's extended S-operator.

An approach through the fixed-point algebra V^G gives important instances of the predicted crossed structure, subject to additional hypotheses. Carnahan and Miyamoto proved regularity of fixed points of a finite-order automorphism of a simple, nonnegatively graded, regular VOA with a nondegenerate invariant bilinear form, and thereby obtained the corresponding result for finite solvable groups [5]. McRae constructed a braided G-crossed category of twisted modules and identified its equivariantization with an ambient tensor category of V^G-modules, assuming the requisite vertex tensor category and algebra-object setting [6]. These constructions rely on the additional vertex-tensor-category and algebra-object hypotheses described above.

2020 *Mathematics Subject Classification.* 17B69, 18M15, 18M20.

Key words and phrases. Vertex operator algebra; twisted module; crossed ribbon category; modular tensor category; equivariantization.

Du and Huang subsequently developed an analytic theory of general twisted intertwining operators, including skew-symmetry and contragredient isomorphisms, $P(z)$-tensor products under suitable conditions, and crossed commutativity and braiding maps [8]. They restated Huang's crossed tensor-category conjecture and proposed a stronger crossed *modular* tensor-category conjecture in a suitable sense. They expressly noted that the modularity notions considered by Kirillov and Turaev differ, so the stronger assertion requires a choice of categorical definition. In the present paper, modularity is understood in Kirillov's sense.

Xu [7, Theorem 1.1] constructs, for strongly rational V and finite G, a faithfully G-graded G-crossed braided fusion category of twisted V-modules. Its tensor products represent the complete Du–Huang intertwining operators, with specified parallel transports, positive-half-turn crossed braidings, action tensorators and contragredient pairings. We use this established structure as the categorical input. The remaining question is whether these concrete operators carry the additional ribbon and nondegeneracy structures required for Kirillov's modularity.

There are genuine compatibility issues beyond the existence of a crossed braiding. To apply Huang's modularity theorem to the neutral component of the category in [7], one must compare the ordinary intertwining operators defined by the Jacobi identity with the *complete* intertwining operators of Du and Huang; the latter impose absolute convergence and multivariable continuation conditions that require verification. Moreover, the conformal monodromy $e^{2\pi i L(0)}$ must be interpreted as a crossed twist and checked against the chosen positive-half-turn braiding, the action tensorators and categorical duality. Only after constructing this crossed ribbon structure can the equivariantization be treated as a ribbon category. Finally, the nondegeneracy criterion of [16, Proposition 4.56(ii)] uses *both* nondegeneracy of the neutral component and faithfulness of the grading; the comparison with Kirillov's extended S-operator then uses [15, Theorem 10.5].

Starting from Xu's established crossed-category construction, we prove this categorical modular refinement. The neutral component is identified with Huang's ordinary modular category, the conformal twist is shown to satisfy crossed balancing and duality, and the equivariantization becomes a nondegenerate ribbon fusion category. Kirillov's theorem then gives invertibility of the extended S-operator. Our modularity statement is categorical: it concerns the crossed category and Kirillov's extended S-operator. The genus-one transformation laws of twisted trace functions form a separate analytic question.

Theorem 1.1. *Let V be strongly rational and $G \leq \mathrm{Aut}(V)$ finite, and let $\mathcal{C}_G$ be Xu's faithfully G-graded G-crossed braided fusion category of grading-restricted finite-order twisted V-modules* [7, Theorem 1.1]. *We use its binary $P(z)$-representing property and canonical operators* [7, Theorem 4.3, equations (4.3)–(4.4)], *its pullback tensorators and monoidal action coherence* [7, equation (4.7), Proposition 8.5, equations (8.15), (8.16), (8.18), (8.19)], *its product–iterate normalization* [7, Theorem 8.3, equations (8.10)–(8.11)], *its positive-half-turn crossed commutors and parallel-transport identities* [7, Proposition 9.6, equations (9.27)–(9.30)], *and its contragredient pairings and rigidity* [7, Lemma 10.1, equations (10.1)–(10.2), Proposition 10.3]. *Then:*

(i) *The maps $\theta_X = e^{2\pi i L_X(0)} : X \longrightarrow T_g X$ for $X \in \mathcal{C}_g$ define a canonical crossed ribbon structure in Kirillov's conventions.*

(ii) *The equivariantization $\mathcal{C}_G^G$, with its induced ribbon twist, is a modular tensor category, and $\mathcal{C}_G$ is modular as a G-equivariant category in the sense of Kirillov. In particular the extended $\widetilde{S}$-operator is invertible.*

(iii) *The Frobenius–Perron dimensions satisfy*

$$\mathrm{FPdim}(\mathcal{C}_g) = \mathrm{FPdim}(\mathrm{Rep}(V)), \qquad \mathrm{FPdim}(\mathcal{C}_G^G) = |G|^2\,\mathrm{FPdim}(\mathrm{Rep}(V)).$$

The comparison of ordinary and complete intertwining operators is proved in Theorem 3.3, and the resulting identification of the neutral tensor category appears in Proposition 3.4. Sections 4 and 5 establish the crossed balancing and duality conditions, respectively. The equivariantization and nondegeneracy arguments, together with the Frobenius–Perron dimension formulas, occupy Section 6. That section also derives a count of the simple objects in each twisted sector in terms of the g-fixed simple untwisted modules (Corollary 6.5).

2. Conventions and the canonical operators

2.1. **Categorical, VOA, and twisted-module conventions.** A *fusion category* is a $\mathbb{C}$-linear semisimple rigid monoidal category with simple unit, finite-dimensional morphism spaces and finitely many isomorphism classes of simple objects. For a *braided* fusion category $\mathcal{D}$ with braiding c, its *Müger center* $\mathcal{Z}_2(\mathcal{D})$ is the full subcategory of objects A for which $c_{B,A}c_{A,B} = \mathrm{id}_{A\boxtimes B}$ for all objects B. Such a category is *nondegenerate* if $\mathcal{Z}_2(\mathcal{D})$ is braided equivalent to Vec. Here a *modular tensor category* means a nondegenerate ribbon fusion category.

We use the standard notion of a vertex operator algebra $(V, Y, \mathbf{1}, \omega)$ over $\mathbb{C}$ [9, 2]. In particular, $V = \bigoplus_{d\in\mathbb{Z}} V_d$ has finite-dimensional homogeneous subspaces and $V_d = 0$ for $d \ll 0$; $\mathbf{1} \in V_0$, $\omega \in V_2$, and $Y(u,x) = \sum_{n\in\mathbb{Z}} u_n x^{-n-1}$ is lower truncated on each $v \in V$. The vacuum and creation properties are $Y(\mathbf{1},x) = \mathrm{id}_V$ and $Y(u,x)\mathbf{1} \in u + xV[[x]]$. The VOA Jacobi identity and the Virasoro relations for $Y(\omega,x) = \sum_{n\in\mathbb{Z}} L(n)x^{-n-2}$ are understood as in [9]; in particular, $L(0)|_{V_d} = d\,\mathrm{id}$ and $Y(L(-1)u,x) = \partial_x Y(u,x)$. Write $\operatorname{wt} u = d$ for $u \in V_d$. An automorphism preserves Y, $\mathbf{1}$ and ω. The VOA is *of CFT type* if $V_d = 0$ for $d < 0$ and $V_0 = \mathbb{C}\mathbf{1}$; it is *simple* if it has no proper nonzero ideal. Define $C_2(V) = \operatorname{span}_{\mathbb{C}}\{u_{-2}v : u, v \in V\}$; C_2-cofiniteness means $\dim_{\mathbb{C}} V/C_2(V) < \infty$. The *restricted contragredient* V' is its graded dual with the contragredient vertex operators [9]. The condition $V \simeq V'$ as V-modules is called *self-contragredience*; equivalently, V admits a nondegenerate invariant bilinear form. Here invariance means $(Y(u,x)v,w) = (v, Y(e^{xL(1)}(-x^{-2})^{L(0)}u, x^{-1})w)$.

Fix a finite-order $g \in \mathrm{Aut}(V)$, let $N = \mathrm{ord}(g)$, and write

$$V^{(r)} = \{u \in V : gu = e^{2\pi i r/N}u\}, \qquad V = \bigoplus_{r=0}^{N-1} V^{(r)}.$$

We recall the finite-order *formal* twisted-module convention of [2, Definition 3.1]. A weak g-twisted module is a vector space M with a linear state–field map Y_M^g, such that for $u \in V^{(r)}$ one has

$$Y_M^g(u,x) = \sum_{n\in r/N+\mathbb{Z}} u_n^M x^{-n-1}, \qquad u_n^M w = 0 \quad (n \gg 0,\ w \in M),$$

$Y_M^g(\mathbf{1},x) = \mathrm{id}_M$, and the *twisted Jacobi identity*

$$\begin{aligned} &x_0^{-1}\delta\!\left(\frac{x_1-x_2}{x_0}\right) Y_M^g(u,x_1)Y_M^g(v,x_2) - x_0^{-1}\delta\!\left(\frac{x_2-x_1}{-x_0}\right) Y_M^g(v,x_2)Y_M^g(u,x_1) \\ &\quad = x_2^{-1}\left(\frac{x_1-x_0}{x_2}\right)^{-r/N} \delta\!\left(\frac{x_1-x_0}{x_2}\right) Y_M^g(Y(u,x_0)v,x_2), \qquad u \in V^{(r)},\ v \in V. \end{aligned} \tag{2.1}$$

Here $\delta(x) = \sum_{j\in\mathbb{Z}} x^j$ and the formal binomial expressions are expanded in their prescribed nonnegative powers of the second variable. The fractional factor in (2.1) agrees with the *positive* eigenvalue convention above: $gu = e^{2\pi i r/N}u$ gives $n \in r/N + \mathbb{Z}$, not $-r/N + \mathbb{Z}$. Since $\omega \in V^{(0)}$, the field $Y_M^g(\omega,x) = \sum_{m\in\mathbb{Z}} L_M(m)x^{-m-2}$ has integral modes; (2.1) implies the usual Virasoro relations and, for $u \in V_d$, $[L_M(0), u_n^M] = (d-n-1)u_n^M$ [2, Section 3]. A twisted-module morphism is a linear map commuting with all the fields $Y_M^g(u,x)$ (equivalently, all twisted modes).

For $\lambda \in \mathbb{C}$, set $M_{[\lambda]} = \bigcup_{k\geq 1} \ker(L_M(0) - \lambda)^k$. A *generalized* g-twisted module is a weak twisted module with $M = \bigoplus_{\lambda\in\mathbb{C}} M_{[\lambda]}$. It is *lower bounded* in the grading-restricted class used here if $M_{[\lambda]} = 0$ whenever $\Re\lambda < B$ for some $B \in \mathbb{R}$, and *grading restricted* if in addition $\dim M_{[\lambda]} < \infty$ for every λ. It is *ordinary* if $L_M(0)$ acts semisimply; in that case $M_\lambda = M_{[\lambda]}$ is the genuine eigenspace. For general M we reserve brackets for generalized eigenspaces, and write $\pi_\lambda^M : M \to M_{[\lambda]}$ for the canonical projection. Thus $u_n^M M_{[\lambda]} \subseteq M_{[\lambda+d-n-1]}$ when $u \in V_d$. For a grading-restricted generalized module put $\overline{M} = \prod_\lambda M_{[\lambda]}$; when M is ordinary, this is $\prod_\lambda M_\lambda$, and π_λ^M projects to M_λ.

For comparison with the preceding generalized-weight terminology, an *admissible* g-twisted module in the sense of [2, Definition 3.3] is a weak twisted module with a grading $M = \bigoplus_{j\in N^{-1}\mathbb{Z}_{\geq 0}} M(j)$ satisfying $u_n^M M(j) \subseteq M(j+d-n-1)$ for $u \in V_d$; no semisimplicity of $L_M(0)$

or finite dimensionality of $M(j)$ is required in this definition. *g-rationality* means complete reducibility of admissible g-twisted modules, whereas *g-regularity* means that *every weak* g-twisted module is an algebraic direct sum of ordinary irreducible g-twisted modules. For $g = 1$ these are called *rationality* and *regularity*, respectively. We call V *strongly rational* if it is simple, of CFT type, self-contragredient, rational and C_2-cofinite. In particular, rationality here concerns admissible untwisted modules, not arbitrary weak modules.

Du–Huang's generalized g-twisted modules without a g-action are defined by analytic product–iterate duality, branch equivariance, $L(0)$-grading and the $L(-1)$-derivative property [8, Section 2]. The comparison with the formal convention above is valid in the grading-restricted finite-order setting, as follows.

Proposition 2.1. *Let g have finite order and let M be equipped with a direct sum of finite-dimensional generalized $L_M(0)$-weight spaces whose real parts have a common lower bound. For the same state–field map, the following are equivalent:*

(i) *M is a grading-restricted generalized analytic g-twisted module without an auxiliary g-action in the sense of* [8, Section 2]*;*

(ii) *M is a grading-restricted generalized module satisfying the finite-order formal weak g-twisted Jacobi identity of* [2, Definition 3.1]*, with its indicated generalized weight grading.*

The correspondence leaves the vertex operators and module maps unchanged. In particular, it applies to ordinary irreducible finite-order formal twisted modules.

Proof. First consider (i). For $u \in V^{(r)}$, compare the two sides of Du–Huang's branch identity $(Y_M^g)^{p+1}(gu, z) = (Y_M^g)^p(u, z)$. For a fixed power x^{-n-1}, its logarithmic coefficient is a polynomial $P(t)$ in $t = \log z$ satisfying $e^{2\pi i(r/N-n-1)}P(t + 2\pi i) = P(t)$. The highest nonzero coefficient first forces $e^{2\pi i(r/N-n-1)} = 1$; comparison of the next coefficient then forces P to be constant. Consequently $Y_M^g(u, x) = \sum_{n\in r/N+\mathbb{Z}} u_n^M x^{-n-1}$ has no logarithmic terms. The $L(0)$-bracket identity follows from analytic duality and the Virasoro field, while the stated lower bound and the weight-shift formula give lower truncation; see [8, Section 2, following the definition of generalized twisted modules].

We apply the analytic–Jacobi comparison in [10, Theorems 2.6, 2.7 and 2.10]. Its auxiliary $\overline{\mathbb{C}}_+$- and g-gradings may be supplied here from the given $L_M(0)$-decomposition: translate the weight indices by a sufficiently small real number so that their real parts are nonnegative, and take the generalized eigenspace grading of $g_M := e^{-2\pi i L_M(0)}$. This auxiliary operator supplies the grading compatibility required in the analytic–Jacobi comparison; objects of $\mathcal{C}_g$ retain their given twisted-module data. The $L(0)$-bracket and the integral weights of V give $g_M Y_M^g(u, x)g_M^{-1} = Y_M^g(gu, x)$ and the required grading compatibility. Thus the grading and equivariance conditions of [10, Definition 2.1] are satisfied. In their notation the nilpotent part $\mathcal{N}$ of g on V is zero, so their logarithm-free field Y_0^g is precisely Y_M^g. Theorem 2.10, (1)⇒(2)⇒(3), then supplies the full twisted Jacobi identity. Their expression (2.29) is the standard twisted delta identity, equivalent to (2.1) with $a = r/N$. The comparison is made by extracting formal coefficients from the absolutely convergent product, commuted-product and iterate branches.

Conversely, start with (ii). The formal twisted Jacobi identity supplies the $L(0)$-bracket, Virasoro relations and $L(-1)$-derivative, as in [2, Section 3]. The same construction of the auxiliary grading and g_M applies; the formal g-eigenspace condition gives branch equivariance. Theorem 2.10 of [10], now in the direction (3)⇒(4)⇒(2)⇒(1), gives analytic duality, including *absolute* convergence, rather than only formal rationality. More explicitly, its Theorem 2.6 establishes convergence in $|z_1| > |z_2| > 0$, $|z_2| > |z_1| > 0$ and $|z_2| > |z_1 - z_2| > 0$, with the prescribed compatible branches. For $u \in V^{(r)}$, $v \in V^{(s)}$ and vectors paired with the restricted dual, the common multivalued continuation has the form

$$z_1^{-r/N} z_2^{-s/N} \frac{P(z_1, z_2)}{z_1^a z_2^b (z_1 - z_2)^c}, \qquad P \in \mathbb{C}[z_1, z_2], \quad a, b, c \in \mathbb{Z}_{\geq 0},$$

with finite sums for inhomogeneous inputs. Expansions of its rational factors in the respective chambers are absolutely convergent on compact subsets. The restricted-dual contractions

therefore yield the required projected weight sums, not merely an identity of formal Laurent series. Finite-dimensional generalized weight spaces and the lower bound give the other grading requirements of [8, Section 2]. The proofs of [10, Theorems 2.6 and 2.10] supply the branch and product–iterate comparison used here. Since the two realizations use exactly the same operators, their intertwining morphisms also coincide. □

Theorem 2.4 of [11] supplies a complementary formal criterion: for finite-order g, the vacuum and lower-truncation properties together with weak associativity *with the exponent in* $r/N+\mathbb{N}$ imply the full twisted Jacobi identity. The absolute-convergence and branch statements needed to pass between the analytic and formal formulations above are provided by [10].

2.2. **The twisted-module category, duals, and dimensions.** Under the present strong-rationality hypotheses the required semisimplicity has a precise external source. By [1, Theorem 4.5], V is regular, since it is of CFT type, rational and C_2-cofinite. The simple VOA V has nonnegative $L(0)$-spectrum and a nondegenerate invariant form; hence [5, Corollary 5.26] applies to every finite-order g and gives both g-rationality and g-regularity for the formal weak twisted modules. Then [2, Theorem 8.1(b),(c)] gives finitely many irreducible admissible g-twisted types, all ordinary; conversely ordinary irreducibles admit admissible gradings by [2, Lemma 3.4]. In particular, a grading-restricted generalized analytic object is, by Proposition 2.1, a *finite* direct sum of ordinary irreducibles: weak-module regularity decomposes it into simples, and finite dimensionality of a chosen nonzero weight space for each of the finitely many simple types bounds the multiplicity of that type. The nonemptiness of each sector follows separately from [3, Theorem 9.1], which assumes simplicity, C_2-cofiniteness and finite order of g. The module-level formal–analytic comparison is supplied here by Proposition 2.1; the crossed tensor products and their coherence are provided by Xu's construction [7, Theorem 1.1].

Let $\mathcal{C}_g$ be the category of grading-restricted generalized g-twisted V-modules in the analytic sense used by [8, 7], with the finite-order formal realization established by Proposition 2.1. Its morphisms are the weight-preserving twisted-module maps; for objects in one sector we also denote these spaces by $\mathrm{Hom}_V(X,Y)$. Define the *external* direct sum $\mathcal{C}_G=\bigoplus_{g\in G}\mathcal{C}_g$ to have objects finite direct sums of pairs (g,X), $X\in\mathcal{C}_g$, and to have $\mathrm{Hom}_{\mathcal{C}_G}((g,X),(h,Y))=0$ when $g\neq h$ and $\mathrm{Hom}_{\mathcal{C}_g}(X,Y)$ when $g=h$, extended additively to sums. The label g is part of the object and is not determined by an identification of the underlying vector spaces. The G-crossed tensor product, action and braiding on this additive category are *additional structures* imported from [7], not part of the definition of a twisted module. Finally, $\mathrm{Rep}(V)$ denotes the category of ordinary untwisted V-modules, with the Huang–Lepowsky vertex tensor product [12]. Under the hypotheses above this is a fusion category, and in fact a modular tensor category, by Huang's rigidity and modularity theorem [14]. For a grading-restricted generalized g-twisted module $X=\bigoplus_\lambda X_{[\lambda]}$, its *restricted contragredient* is the algebraic direct sum $DX=X'=\bigoplus_\lambda X^*_{[\lambda]}$, where $X^*_{[\lambda]}=\mathrm{Hom}_{\mathbb{C}}(X_{[\lambda]},\mathbb{C})$. The weight-$\lambda$ generalized eigenspace of DX is $X^*_{[\lambda]}$; in particular, if X is ordinary, then $DX=\bigoplus_\lambda X^*_\lambda$. With the evaluation pairing $\langle\ ,\ \rangle : X'\times X\to\mathbb{C}$, its twisted vertex operator is specified by

$$\langle Y^{g^{-1}}_{DX}(u,x)\varphi,w\rangle=\langle\varphi,Y^g_X(e^{xL(1)}(-x^{-2})^{L(0)}u,x^{-1})w\rangle,\quad u\in V,\ \varphi\in DX,\ w\in X. \tag{2.2}$$

Here $(-x^{-2})^{L(0)}u=(-1)^dx^{-2d}u$ for $u\in V_d$; the formula is expanded formally with the opposite coordinate x^{-1}. The factor $e^{xL(1)}$ is a finite sum on each homogeneous u. The contragredient construction of [9, Theorem 5.2.1], its admissible twisted analogue [2, Lemma 3.7], and the analytic generalized twisted construction of [8, Section 2] give a grading-restricted g^{-1}-twisted module in the present setting. Indeed, for $u\in V^{(r)}$ the powers in (2.2) have mode indices in $-r/N+\mathbb{Z}$, as required for g^{-1}. This construction uses the weightwise algebraic dual, not the full linear dual X^* or the completion $\overline{X}$.

For a twisted-module morphism $f:X\to Y$ in a fixed sector, $Df:Y'\to X'$ is its weightwise transpose, $\langle Df(\psi),w\rangle=\langle\psi,f(w)\rangle$. Thus D is contravariant and reverses the sector degree. The canonical evaluation into the restricted double dual,

$$\delta_X : X\longrightarrow D^2X,\qquad \delta_X(w)(\varphi)=\varphi(w),$$

is a natural twisted-module isomorphism: each generalized weight space is finite-dimensional and the second application of (2.2) recovers the original vertex operators, as in the double-contragredient identity [9, Proposition 5.3.1]. The comparison with the *categorical* right dual $X^\vee$ is established from pairing and rigidity in Proposition 5.1.

The pullback T_k preserves $L(0)$ and the underlying graded vector space. Consequently $D(T_kX)$ and $T_k(DX)$ both belong to $\mathcal{C}_{kg^{-1}k^{-1}}$, and the identity on their common restricted-dual vector space defines a natural isomorphism $\tau_{k,X} : D(T_kX) \xrightarrow{\sim} T_k(DX)$. For $u \in V$ and $w \in X$, this is an intertwiner since

$$\langle Y_{D(T_kX)}(u,x)\varphi, w\rangle = \langle \varphi, Y^g_X(k^{-1}e^{xL(1)}(-x^{-2})^{L(0)}u, x^{-1})w\rangle = \langle Y_{T_k(DX)}(u,x)\varphi, w\rangle,$$

where k^{-1} commutes with $L(1)$ and $L(0)$ because k fixes ω. With the underlying functor identity $T_kT_l = T_{kl}$, its coherence is $\tau_{kl,X} = T_k(\tau_{l,X})\,\tau_{k,T_lX}$. These are natural comparisons of twisted-module functors. The monoidal constraints enter in the later comparison with categorical duals.

For a grading-restricted generalized module M, the *weight completion* is the direct product $\overline{M} = \prod_\lambda M_{[\lambda]}$, as defined above, with coordinate projections $\pi^M_\lambda : \overline{M} \to M_{[\lambda]}$ extending the projections on M. The restricted pairing extends to $M' \times \overline{M}$ by $\langle (\varphi_\lambda)_\lambda, (m_\lambda)_\lambda\rangle = \sum_\lambda \varphi_\lambda(m_\lambda)$; this sum is finite because M' is a *direct sum*. It separates points of $\overline{M}$. Since a twisted-module map $f : M \to N$ commutes with $L(0)$, it preserves each generalized weight space and has the canonical componentwise extension

$$\overline{f} : \overline{M} \longrightarrow \overline{N}, \qquad (m_\lambda)_\lambda \longmapsto (f|_{M_{[\lambda]}} m_\lambda)_\lambda.$$

For ordinary modules, the same statements use eigenspaces M_λ in place of $M_{[\lambda]}$. The completed codomains of the canonical intertwining operators below are understood in this sense, not as restricted double duals.

For a fusion category $\mathcal{D}$, the *Frobenius–Perron dimension* $\mathrm{FPdim}_{\mathcal{D}}(X)$ of an object is obtained from the positive dimension character on its Grothendieck ring: on a simple object X it is the Perron–Frobenius eigenvalue of left multiplication by $[X]$, and it is extended additively (and multiplicatively) to objects [17, Definition 3.3.3 and Proposition 3.3.6]. In particular, $\mathrm{FPdim}_{\mathcal{D}}(X) > 0$ for $X \neq 0$, and $\mathrm{FPdim}(\mathcal{D}) = \sum_{[X]\in \mathrm{Irr}(\mathcal{D})} \mathrm{FPdim}_{\mathcal{D}}(X)^2$. The homogeneous component $\mathcal{C}_g$ need not itself be monoidal. Throughout, its dimension means $\mathrm{FPdim}(\mathcal{C}_g) := \sum_{[X]\in \mathrm{Irr}(\mathcal{C}_g)} \mathrm{FPdim}_{\mathcal{C}_G}(X)^2$, with X regarded as a labeled object of the *whole* fusion category $\mathcal{C}_G$ furnished by [7]. This is a finite positive real number because the grading is faithful and $\mathcal{C}_G$ is a fusion category. On the neutral fusion subcategory the dimension character restricts to its intrinsic Frobenius–Perron dimension. The later equality of homogeneous-component dimensions uses the *faithful* finite G-grading in [18, Proposition 8.20], while the corresponding equivariantization formula uses a finite group acting by tensor autoequivalences on a fusion category [16, Proposition 4.26].

2.3. **The G-crossed categorical structure.** The G-grading of the fusion category $\mathcal{C}_G$ means that its homogeneous summands $\mathcal{C}_g$ are full abelian subcategories, $\mathrm{Hom}_{\mathcal{C}_G}(X_g, Y_h) = 0$ for $g \neq h$, every object is a finite direct sum of homogeneous objects, and the tensor unit V belongs to $\mathcal{C}_1$. The tensor product and the pullback functors have the following conventions:

$$\mathcal{C}_g \boxtimes \mathcal{C}_h \subseteq \mathcal{C}_{gh}, \quad T_k : \mathcal{C}_g \to \mathcal{C}_{kgk^{-1}}, \quad Y_{T_kX}(v,z) = Y_X(k^{-1}v, z), \quad T_kT_l = T_{kl}. \tag{2.3}$$

The grading is *faithful* if every $\mathcal{C}_g$ is nonzero. The categorical right dual of a homogeneous object of degree g can be chosen in $\mathcal{C}_{g^{-1}}$, by rigidity and the grading. These grading and tensor structures are those established in [7, Theorem 1.1, Proposition 8.5, Proposition 9.6]; the external sector label is retained even when the underlying vector spaces of objects agree.

The functor T_k acts as the identity on the underlying vector spaces and linear module morphisms, but pulls back the vertex operators as in (2.3). Indeed, if $X \in \mathcal{C}_g$ and v is an eigenvector of kgk^{-1} with eigenvalue $e^{2\pi i r/N}$, then $k^{-1}v$ is a g-eigenvector with the same eigenvalue. Thus the fractional modes and the twisted-module identities are preserved. The

inverse functor is $T_{k^{-1}}$, so T_k is a $\mathbb{C}$-linear exact autoequivalence. Applying two pullbacks gives $Y_{T_kT_lX}(v,z) = Y_X(l^{-1}k^{-1}v,z)$, and the equality $T_kT_l = T_{kl}$, as well as $T_1 = \mathrm{id}$, holds literally on objects and morphisms of the externally labeled category. This strictness concerns the *underlying functors* only.

Each T_k is equipped with a strong monoidal structure. We use the tensorators and unit maps

$$J^k_{X,Y} : T_kX \boxtimes T_kY \xrightarrow{\sim} T_k(X \boxtimes Y), \qquad j_k : \mathbf{1} \xrightarrow{\sim} T_k\mathbf{1},$$

where $\mathbf{1}$ in categorical formulas denotes the tensor unit V. On V the canonical unit comparison of [7] is $j_k(v) = k^{-1}v$. In general neither J^k nor j_k is the identity. Write $a_{X,Y,Z} : (X \boxtimes Y) \boxtimes Z \to X \boxtimes (Y \boxtimes Z)$ for the associator, and $l_X : \mathbf{1} \boxtimes X \to X$ and $r_X : X \boxtimes \mathbf{1} \to X$ for the unitors. The tensorators and unit maps are natural isomorphisms and obey

$$J^{kl}_{X,Y} = T_k(J^l_{X,Y})J^k_{T_lX,T_lY}, \qquad j_{kl} = T_k(j_l)j_k, \qquad J^1 = \mathrm{id}, \quad j_1 = \mathrm{id}\,. \tag{2.4}$$

Compositions are read from right to left. Both sides of the first identity belong to $\mathrm{Hom}(T_{kl}X \boxtimes T_{kl}Y, T_{kl}(X \boxtimes Y))$; both sides of the second belong to $\mathrm{Hom}(\mathbf{1}, T_{kl}\mathbf{1})$. The associativity and left- and right-unit conditions for each strong monoidal functor are, with the same tensorator direction,

$$\begin{aligned}
&T_k(a_{X,Y,Z})J^k_{X\boxtimes Y,Z}(J^k_{X,Y} \boxtimes \mathrm{id}_{T_kZ}) \\
&\qquad = J^k_{X,Y\boxtimes Z}(\mathrm{id}_{T_kX} \boxtimes J^k_{Y,Z})a_{T_kX,T_kY,T_kZ}, \\
&T_k(l_X)J^k_{\mathbf{1},X}(j_k \boxtimes \mathrm{id}_{T_kX}) = l_{T_kX}, \\
&T_k(r_X)J^k_{X,\mathbf{1}}(\mathrm{id}_{T_kX} \boxtimes j_k) = r_{T_kX}.
\end{aligned}$$

The first identity compares morphisms $(T_kX \boxtimes T_kY) \boxtimes T_kZ \to T_k(X \boxtimes (Y \boxtimes Z))$; the unit identities have domains $\mathbf{1} \boxtimes T_kX$ and $T_kX \boxtimes \mathbf{1}$, respectively. In the general formalism the action has monoidal compositors $T_kT_l \Rightarrow T_{kl}$; here they are identity natural transformations, without imposing strict monoidality on T_k.

For a homogeneous $X \in \mathcal{C}_g$ and any $Y \in \mathcal{C}_G$, the crossed braiding is a natural isomorphism $c_{X,Y} : X \boxtimes Y \xrightarrow{\sim} T_gY \boxtimes X$. It is defined componentwise when Y is a direct sum and is extended additively in the first argument: for $X = \bigoplus_g X_g$, its target is the direct sum $\bigoplus_g (T_gY \boxtimes X_g)$, with each summand acted on by the corresponding functor T_g. If $Y \in \mathcal{C}_h$, both endpoints have degree gh, because $(ghg^{-1})g = gh$. The action-equivariance and two crossed hexagons are the following equations:

$$T_k(c_{X,Y})J^k_{X,Y} = J^k_{T_gY,X}c_{T_kX,T_kY}, \tag{2.5}$$

$$\begin{aligned}
&(\mathrm{id}_{T_gY} \boxtimes c_{X,Z})a_{T_gY,X,Z}(c_{X,Y} \boxtimes \mathrm{id}_Z) \\
&\qquad = a_{T_gY,T_gZ,X}((J^g_{Y,Z})^{-1} \boxtimes \mathrm{id}_X)c_{X,Y\boxtimes Z}a_{X,Y,Z},
\end{aligned} \tag{2.6}$$

$$\begin{aligned}
&(c_{X,T_hZ} \boxtimes \mathrm{id}_Y)a^{-1}_{X,T_hZ,Y}(\mathrm{id}_X \boxtimes c_{Y,Z}) \\
&\qquad = a^{-1}_{T_{gh}Z,X,Y}c_{X\boxtimes Y,Z}a^{-1}_{X,Y,Z}.
\end{aligned} \tag{2.7}$$

In the first identity the common source is $T_kX \boxtimes T_kY$ and the common target is $T_k(T_gY \boxtimes X)$, using $T_{kgk^{-1}}T_k = T_kT_g$. For $X \in \mathcal{C}_g$, $Y \in \mathcal{C}_h$ and arbitrary Z, the two hexagons have, respectively, source and target

$$(X \boxtimes Y) \boxtimes Z \;\longrightarrow\; T_gY \boxtimes (T_gZ \boxtimes X), \qquad X \boxtimes (Y \boxtimes Z) \;\longrightarrow\; (T_{gh}Z \boxtimes X) \boxtimes Y.$$

The inverse $(J^g_{Y,Z})^{-1}$ in the first hexagon is necessary to expand $T_g(Y \boxtimes Z)$, and the second uses $T_gT_h = T_{gh}$ without interchanging g and h.

The corresponding crossed unit identities are

$$r_Yc_{\mathbf{1},Y} = l_Y, \qquad l_X(j_g^{-1} \boxtimes \mathrm{id}_X)c_{X,\mathbf{1}} = r_X \quad (X \in \mathcal{C}_g).$$

where the first identity uses $T_1 = \mathrm{id}$. They are the unit comparisons of the canonical crossed commutors in [7]; in particular, j_g must not be discarded. Equations (2.5)–(2.7) are the non-strict forms of [16, Definition 4.41] and [17, Definition 8.24.1, (8.105)–(8.107)], with J^k in the

strong monoidal direction and identity underlying action compositors. The sources use action compositors and may suppress associators; the displayed formulas retain both tensorators and associators. Kirillov's G-equivariant fusion convention [15, Section 2] uses action functors R_k and suppresses canonical coherence isomorphisms; its additional pivotal and twist data are not part of the bare braided G-crossed definition.

For $X \in \mathcal{C}_g$ put $\deg(X) = g$. Fix right duals $X^\vee$ with $\mathrm{ev}_X : X \boxtimes X^\vee \to \mathbf{1}$ and $\mathrm{coev}_X : \mathbf{1} \to X^\vee \boxtimes X$, satisfying the two triangle identities. The right-dual transposition of a map $f : A \to B$ is $f^\vee : B^\vee \to A^\vee$. The action–dual comparison $b_{k,X} : T_k(X^\vee) \to (T_k X)^\vee$ will be uniquely defined by evaluations in Lemma 5.3.

Definition 2.2. A *crossed ribbon twist* in our (positive-half-turn, right-dual) convention is a natural collection of isomorphisms $\theta_X : X \to T_g X$ for $X \in \mathcal{C}_g$ such that

(i) $\theta_{\mathbf{1}} = j_1$ (under the canonical unit identifications), and $\theta_{T_k X} = T_k(\theta_X)$ with $T_{kgk^{-1}} T_k = T_k T_g$;

(ii) if $X \in \mathcal{C}_g$, $Y \in \mathcal{C}_h$ and $q = ghg^{-1}$, then

$$\theta_{X \boxtimes Y} = J^{gh}_{X,Y}(T_q \theta_X \boxtimes T_g \theta_Y) c_{T_g Y, X} c_{X,Y}; \tag{2.8}$$

(iii) for $X \in \mathcal{C}_g$, with the action–right-dual comparison $b_{g,X} : T_g(X^\vee) \to (T_g X)^\vee$ of Lemma 5.3, the dual-twist identity is

$$\theta_{X^\vee} = T_{g^{-1}}(\theta_X^\vee)\, T_{g^{-1}}(b_{g,X}) : X^\vee \longrightarrow T_{g^{-1}} X^\vee. \tag{2.9}$$

The equivalent identity on the restricted contragredient DX is proved in Proposition 5.1.

These are the typed forms of Kirillov's four conditions in [15, Lemma 2.3]; the unit, covariance, balancing and dual compatibility in the source correspond to (i), (i), (ii) and (iii) here. Kirillov writes the group action as R_k and suppresses monoidal constraints; replacing R_k by our T_k requires the displayed J^k, j_k and duality comparisons. When $G = \{1\}$, the crossed braiding and twist reduce to the usual ribbon-category data.

Definition 2.3. A *braided G-crossed fusion category* is a fusion category $\mathcal{C} = \bigoplus_{g \in G} \mathcal{C}_g$ with a monoidal action T_k of G, $T_k \mathcal{C}_g = \mathcal{C}_{kgk^{-1}}$, and natural crossed braidings $c_{X,Y} : X \boxtimes Y \to T_g Y \boxtimes X$ for $X \in \mathcal{C}_g$, satisfying the unit, action-equivariance and two crossed hexagon axioms. The grading is *faithful* if every $\mathcal{C}_g$ is nonzero. A *G-crossed ribbon category* has, in addition, a rigid structure and natural crossed twists $\theta_X : X \to T_g X$ ($X \in \mathcal{C}_g$), with the unit, G-equivariance, crossed balancing and duality conditions stated explicitly in Definition 2.2 above.

For this paper, such a G-crossed ribbon fusion category is *G-crossed modular* if its braided ribbon equivariantization $\mathcal{C}^G$ is nondegenerate. By [15, Theorem 10.5], nondegeneracy of the equivariantization is equivalent to invertibility of the extended $\widetilde{S}$-operator of [15, Definition 10.1 and (9.3)]. Kirillov calls this a "G-equivariant modular category"; here we use the synonymous phrase "G-crossed modular category" to emphasize the crossed braiding. Unlike the weaker condition that only $\mathcal{C}_1$ be modular, this condition requires nondegeneracy of $\mathcal{C}^G$.

More concretely, the underlying space of Kirillov's extended Verlinde algebra has bigraded pieces

$$\widetilde{\mathcal{V}}_{k,g} = \bigoplus_{[X] \in \mathrm{Irr}(\mathcal{C}_g)} \mathrm{Hom}_{\mathcal{C}}(X, T_k X),$$

where one representative of each simple is chosen and a summand vanishes unless $T_k X \simeq X$. The extended $\widetilde{S}$-operator constructed from crossed braiding and categorical traces in [15, (9.3)] exchanges $\widetilde{\mathcal{V}}_{k,g}$ and $\widetilde{\mathcal{V}}_{g^{-1},k}$. By [15, Theorem 10.5], its invertibility is equivalent to the completely intrinsic Müger-center criterion of Definition 2.3; that criterion is what we verify below. We use this categorical criterion throughout.

2.4. **Ordinary and complete intertwining operators.**

Definition 2.4. Let X, Y, Z be ordinary V-modules. Write $Z\{x\}$ for formal Z-valued series with complex powers of x; as usual, x^α is a formal symbol and distinct powers are compared coefficientwise. An *ordinary intertwining operator of type* $\binom{Z}{X\,Y}$ is a linear map $\mathcal{Y} : X \otimes Y \to Z\{x\}$, written $\mathcal{Y}(a,x)b = \sum_{n\in\mathbb{C}} a_n^{\mathcal{Y}} b\, x^{-n-1}$, which satisfies the following properties for $a \in X$, $b \in Y$ and $u \in V$:

(i) *Lower truncation*: for every $n \in \mathbb{C}$, $a_{n+l}^{\mathcal{Y}} b = 0$ for all sufficiently large integers $l \geq 0$ (with the bound allowed to depend on a, b, n).

(ii) The $L(-1)$-derivative identity $\partial_x \mathcal{Y}(a,x)b = \mathcal{Y}(L_X(-1)a, x)b$.

(iii) The Jacobi identity

$$\begin{aligned} &x_0^{-1}\delta\!\left(\frac{x_1 - x_2}{x_0}\right) Y_Z(u,x_1)\mathcal{Y}(a,x_2)b - x_0^{-1}\delta\!\left(\frac{x_2 - x_1}{-x_0}\right)\mathcal{Y}(a,x_2)Y_Y(u,x_1)b \\ &\qquad = x_2^{-1}\delta\!\left(\frac{x_1 - x_0}{x_2}\right)\mathcal{Y}(Y_X(u,x_0)a, x_2)b. \end{aligned} \tag{2.10}$$

Here $\delta(x) = \sum_{m\in\mathbb{Z}} x^m$ and each formal binomial is expanded in nonnegative powers of its second variable as prescribed by the usual formal-variable convention. This is the Jacobi-identity definition of [9, Definition 5.4.1]; we permit complex rather than only rational powers in the ambient notation $Z\{x\}$, as in the generalized formulation. The $L(0)$-bracket follows from the Jacobi identity and derivative property [9, (5.4.12)–(5.4.14)]. Thus, for homogeneous $a \in X_\alpha$ and $b \in Y_\beta$, $a_n^{\mathcal{Y}} b \in Z_{\alpha+\beta-n-1}$; the nonzero powers are constrained by the weights of Z, and for the ordinary modules considered here they lie in finitely many cosets of $\mathbb{Z}$. Pre- and postcomposition with module homomorphisms acts on intertwining operators in the evident variance. The resulting vector space is denoted $\mathcal{I}_{\mathrm{HL}}\binom{Z}{X\,Y}$.

Definition 2.5. Let X, Y, Z be ordinary untwisted V-modules. We call these operators *complete Du–Huang intertwining operators*; they are the twisted intertwining operators of [8, Definition 2.8] specialized to three untwisted modules. It is a linear map $\mathcal{Y} : X \otimes Y \to Z\{x\}[\log x]$, $\mathcal{Y}(a,x)b = \sum_{j=0}^{K} \sum_{n\in\mathbb{C}} a_{n,j}^{\mathcal{Y}} b\, x^{-n-1} (\log x)^j$, with a finite logarithmic bound K, subject to these four conditions:

(i) For all a, b, n, j as above, $a_{n+l,j}^{\mathcal{Y}} b = 0$ when $l \in \mathbb{Z}_{\geq 0}$ is sufficiently large.

(ii) For $u \in V$, $a \in X$, $b \in Y$ and $z' \in Z'$, there is a maximally extended multivalued analytic function $f(z_1, z_2)$ on $M^2 = \{(z_1, z_2) \in (\mathbb{C}^\times)^2 : z_1 \neq z_2\}$, with a preferred branch f^e on M_0^2. Here M_0^2 is obtained by cutting in the z_1-, z_2- and $(z_1 - z_2)$-planes along the positive real rays and taking the upper-side boundary values, as in [8, Section 2]. With π_λ^M the weight projection on M, the three projected matrix-coefficient series are respectively

$$\begin{aligned} &\sum_\lambda \langle z', Y_Z(u,z_1)\pi_\lambda^Z \mathcal{Y}(a,z_2)b\rangle, \\ &\sum_\lambda \langle z', \mathcal{Y}(a,z_2)\pi_\lambda^Y Y_Y(u,z_1)b\rangle, \\ &\sum_\lambda \langle z', \mathcal{Y}(\pi_\lambda^X Y_X(u,z_1 - z_2)a, z_2)b\rangle. \end{aligned}$$

They are absolutely convergent for, respectively, $|z_1| > |z_2| > 0$, $|z_2| > |z_1| > 0$ and $|z_2| > |z_1 - z_2| > 0$, and their sums equal the same preferred branch f^e on the corresponding subregions

$$\begin{aligned} &|\arg(z_1 - z_2) - \arg z_1| < \pi/2, \\ &-3\pi/2 < \arg(z_1 - z_2) - \arg z_2 < -\pi/2, \\ &|\arg z_1 - \arg z_2| < \pi/2, \end{aligned}$$

within these three convergence chambers, respectively. All powers and logarithms are evaluated on these prescribed branches; continuation around the punctures gives the other branches, as in [8, Definition 2.8 and the following remark].

(iii) For every integer $k \geq 3$, $u_1, \dots, u_{k-1} \in V$, $a \in X$, $b \in Y$ and $z' \in Z'$, the projected multiple product

$$\sum_{\lambda_1,\dots,\lambda_{k-1}} \left\langle z', Y_Z(u_1,z_1)\pi^Z_{\lambda_1} Y_Z(u_2,z_2)\pi^Z_{\lambda_2} \cdots Y_Z(u_{k-1},z_{k-1})\pi^Z_{\lambda_{k-1}} \mathcal{Y}(a,z_k)b \right\rangle$$

is absolutely convergent in $|z_1| > \cdots > |z_k| > 0$ and admits a maximal multivalued analytic continuation $f(z_1,\dots,z_k)$ on $M^k = \{z_i \neq 0,\ z_i \neq z_j\ (i \neq j)\}$. For every fixed $z_k \in \mathbb{C}^\times$ and every choice of $M_{ij} \in \mathbb{Z}_{\geq 0}$ $(1 \leq i < j \leq k-1)$ with $x^{M_{ij}} Y_V(u_i,x)u_j \in V[[x]]$, every component-isolated singularity of the function

$$\prod_{1\leq i<j\leq k-1} (z_i - z_j)^{M_{ij}} f(z_1,\dots,z_{k-1},\zeta)\Big|_{\zeta=z_k}$$

in the variables $z_1,\dots,z_{k-1}$, of the form $(z_1,\dots,z_{k-1}) - \beta = \delta$ with $\beta \in \mathbb{C}^{k-1}$ and $\delta \in \{0,\infty\}^{k-1}$, is regular in the sense of Definition 2.6. The factors cancel possible OPE poles among the additional V-insertions; no factor involving z_k is imposed by this clause.

(iv) The $L(-1)$-derivative identity $\partial_x \mathcal{Y}(a,x)b = \mathcal{Y}(L_X(-1)a,x)b$.

Write $\mathcal{I}_{\mathrm{DH}}\binom{Z}{X\ Y}$ for this space in the stated ordinary untwisted setting. The convergence and preferred-branch requirements distinguish this definition from Definition 2.4. The $L(0)$-commutator formula follows from the Du–Huang duality property ([8, Remark following Definition 2.8]); for ordinary modules it forces the logarithmic coefficients of positive degree to vanish, as reflected in (2.13).

Definition 2.6. We use the component-isolated and regular-singularity terminology of [8, Definition 2.6 and its preceding paragraph]. Let F be a multivalued analytic function in m independent variables. A choice $\delta = (\delta_1,\dots,\delta_m)$ with $\delta_i \in \{0,\infty\}$ is component-isolated if the domain of F contains a region $\Delta^\times(\delta,r)$ specified, for some $r_i > 0$, by $0 < |\eta_i| < r_i$ when $\delta_i = 0$ and $|\eta_i| > r_i$ when $\delta_i = \infty$. The same terminology applies after an invertible affine change of independent variables $\eta = (z_1,\dots,z_m)A - \beta$, $A \in \mathrm{GL}(m,\mathbb{C})$; component isolation is tested in the new independent η-coordinates, not in the old ones.

At a component-isolated singularity, take all local branches F^b with cuts along the positive real rays. The singularity is *regular* if there exist a common integer $B \geq 0$ and, for each i, an exponent set D_i independent of the branch, of the form $D_i = \bigcup_{q=1}^{N_i}(r_{iq} + \mathbb{Z}_{\geq 0})$ when $\delta_i = 0$ or $D_i = \bigcup_{q=1}^{N_i}(r_{iq} - \mathbb{Z}_{\geq 0})$ when $\delta_i = \infty$, with $r_{iq} \in \mathbb{C}$, such that for each branch b one has an absolutely convergent expansion

$$F^b(\eta) = \sum_{a_1\in D_1} \cdots \sum_{a_m \in D_m} \sum_{j_1,\dots,j_m=0}^{B} \alpha^{(b)}_{\boldsymbol{a},\boldsymbol{j}} \prod_{i=1}^{m} \eta_i^{a_i} (\log \eta_i)^{j_i}$$

on some $\Delta^\times(\delta, r_b)$ (which may depend on b). In the present ordinary untwisted setting, rational functions have zero logarithmic degree, but the definition is not restricted to rational functions. Definition 2.8 of [8] tests this regularity only after fixing its final insertion z_k and in the indicated translation coordinates; Definition 2.6 itself permits general invertible affine changes. A singularity along a diagonal divisor need not be component-isolated in the original coordinates.

We shall prove $\mathcal{I}_{\mathrm{HL}}\binom{Z}{X\ Y} = \mathcal{I}_{\mathrm{DH}}\binom{Z}{X\ Y}$. The analytic requirements in Definition 2.5 are those to be checked, rather than conditions inferred from a formal Jacobi identity alone.

2.5. **Canonical $P(z)$ products and normalizations.** Let $X \in \mathcal{C}_g$, $Y \in \mathcal{C}_h$ and $Z \in \mathcal{C}_{gh}$, and fix $z \in \mathbb{C}^\times$ and $L \in \mathbb{C}$ with $e^L = z$. The order of the inputs corresponds to the marked points $(\infty, z, 0)$. In this subsection an intertwining operator between twisted modules means an operator satisfying the full Du–Huang twisted intertwining-operator conditions [8, Definition 2.8]; in particular these include the multiple-insertion analytic conditions. The term *complete* will retain this meaning for such operators. This is not the Jacobi-identity notion of ordinary untwisted intertwining operator in Definition 2.4.

For a twisted intertwining operator $\mathcal{Y}$ of type $\binom{Z}{X\ Y}$, its evaluation at (z, L) is the linear map

$$\mathrm{ev}_L(\mathcal{Y}) = I_{\mathcal{Y},L} : X \otimes Y \longrightarrow \overline{Z}, \qquad \pi_\gamma^Z I_{\mathcal{Y},L}(x \otimes y) = \pi_\gamma^Z \mathcal{Y}(x,t)y\Big|_{t^\rho = e^{\rho L},\, \log t = L}.$$

Thus, when a projected coefficient contains $t^\rho(\log t)^j$, its value is $e^{\rho L}L^j$. The evaluation is taken separately in each finite-dimensional weight space Z_γ, with the resulting value in $\overline{Z}$. Here and below the ordinary sector hypothesis ensures that each fixed output-weight projection has a single exponent and no logarithmic terms, as shown below. We write

$$\mathcal{I}_{P(z),L}\binom{Z}{X\ Y} := \mathrm{im}\left(\mathrm{ev}_L : \mathcal{I}_{\mathrm{tw}}\binom{Z}{X\ Y} \longrightarrow \mathrm{Hom}_{\mathbb{C}}(X \otimes Y, \overline{Z})\right),$$

where $\mathcal{I}_{\mathrm{tw}}$ denotes the full Du–Huang twisted operator space for the indicated sectors. A *twisted $P(z)$-intertwining map* is precisely an element of this image, not an arbitrary linear map into $\overline{Z}$. For $g = h = 1$, the usual Huang–Lepowsky $P(z)$-maps arise from ordinary intertwining operators; their comparison with the above complete operators is established in Section 3. These definitions follow [8, Section 4] and, in the untwisted case, [12].

A *$P(z)$-product* of (X, Y) in $\mathcal{C}_{gh}$ is a pair (U, I) with $U \in \mathcal{C}_{gh}$ and $I \in \mathcal{I}_{P(z),L}\binom{U}{X\ Y}$; its morphisms satisfy $\overline{f} \circ I = I'$ for $f : U \to U'$. A *$P(z)$-tensor product* is an initial such product $(X \boxtimes_{P(z)} Y, I^z_{X,Y})$. Equivalently, for every $W \in \mathcal{C}_{gh}$ the map

$$\mathrm{Hom}_{\mathcal{C}_{gh}}(X \boxtimes_{P(z)} Y, W) \xrightarrow{\sim} \mathcal{I}_{P(z),L}\binom{W}{X\ Y}, \qquad f \longmapsto \overline{f} \circ I^z_{X,Y}$$

is a bijection natural in W. Naturality in X, Y gives the bifunctorial morphism $f \boxtimes_{P(z)} q$, characterized by $\overline{f \boxtimes_{P(z)} q}\, I^z_{X,Y} = I^z_{X',Y'}(f \otimes q)$. The universal pair is defined by the property in [8, Section 4] and is unique up to the unique isomorphism preserving its universal map. Xu's construction supplies its existence in every twisted sector and ensures that the representing operators satisfy the *complete* analytic conditions [7, Theorem 4.3, equation (4.3)].

We choose the representative $X \boxtimes Y = X \boxtimes_{P(1)} Y$ and its universal map $I^1_{X,Y}$ with $\log 1 = 0$. The inverse evaluation correspondence in the ordinary twisted sectors determines a unique complete formal operator

$$\mathcal{Y}_{X,Y} : X \otimes Y \longrightarrow (X \boxtimes Y)\{t\}[\log t], \qquad \mathrm{ev}_0(\mathcal{Y}_{X,Y}) = I^1_{X,Y};$$

its evaluated values lie in $\overline{X \boxtimes Y}$, not in the algebraic tensor-product module in general. Explicitly, for $x \in X_\alpha$, $y \in Y_\beta$ and any target weight γ, the $L(0)$-commutator identity

$$L_{X\boxtimes Y}(0)\mathcal{Y}_{X,Y}(x,t)y - \mathcal{Y}_{X,Y}(x,t)L_Y(0)y = \mathcal{Y}_{X,Y}(L_X(0)x,t)y + t\partial_t \mathcal{Y}_{X,Y}(x,t)y$$

shows that, writing the projected formal series as $\sum_\rho t^\rho P_\rho(\log t)$, each coefficient satisfies $P'_\rho(s) + (\rho - \gamma + \alpha + \beta)P_\rho(s) = 0$. A polynomial solution is zero unless $\rho = \gamma - \alpha - \beta$, in which case it is constant. Hence evaluation at one selected logarithm is injective on these ordinary-module operator spaces; this is the weightwise recovery used in the universal-property construction of [7, Sections 2 and 4].The resulting weightwise formula is stated in (2.13) below. The corresponding logarithmic degree is zero; for generalized (nonsemisimple) $L(0)$-modules, polynomial logarithmic factors from the nilpotent parts can occur.

For another (z, L), the chosen universal $P(z)$-product is identified with $X \boxtimes Y$ only by specified representing isomorphisms or parallel transport. If γ is a path from z_1 to z_2 in $\mathbb{C}^\times$, continue the logarithm used for the universal operator at z_2 backwards along γ to a value L_1^γ over z_1. The universal property determines

$$\mathsf{T}_\gamma : X \boxtimes_{P(z_1)} Y \xrightarrow{\sim} X \boxtimes_{P(z_2)} Y, \qquad \overline{\mathsf{T}_\gamma}\, I^{z_1}_{X,Y} = \mathrm{ev}_{L_1^\gamma}(\mathcal{Y}^{z_2}_{X,Y}),$$

where $\mathcal{Y}^{z_2}_{X,Y}$ is the formal operator associated with the chosen universal map at z_2. Transport depends on the endpoint-fixed homotopy class of γ; opposite and concatenated paths give inverse and composite transports. Winding may produce nontrivial monodromy. In the common

representatives of [7, Proposition 9.6, equation (9.28)], transport along the positive real interval with real logarithms is the identity. These transport conventions agree with [8, Section 4, parallel transport].

For homogeneous $X \in \mathcal{C}_g$ and $Y \in \mathcal{C}_h$, the crossed commutor $c_{X,Y} : X \boxtimes Y \to T_g Y \boxtimes X$ is the Du–Huang $P(1)$ crossed commutativity isomorphism followed by transport from -1 to 1 through the upper half-plane, as fixed in [7, Sections 2 and 9]. The target $T_g Y \boxtimes X$ has degree $(ghg^{-1})g = gh$. With y viewed as the same underlying vector of $T_g Y$, its normalization is the identity of complete formal operators

$$\overline{c_{X,Y}}\mathcal{Y}_{X,Y}(x,z)y = e^{zL(-1)}\mathcal{Y}_{T_gY,X}(y,e^{\pi i}z)x, \qquad \log(e^{\pi i}z) = \log z + \pi i. \tag{2.11}$$

Both sides lie in $\overline{T_g Y \boxtimes X}$, with $L(-1)$ on the right acting on this output module; the exponential is defined weightwise, since each fixed target-weight projection receives only finitely many contributions from lower weights. The notation $e^{\pi i}z$ specifies analytic continuation through a positive half-turn, with the continued logarithm fixed by $\log(e^{\pi i}z) = \log z + \pi i$. We use the branch-sensitive identity [7, Proposition 9.6, equation (9.30)] as an input from Xu.

For $k \in G$, the pullback functors preserve the underlying $L(0)$-graded vector spaces and use $Y_{T_kW}(v,t) = Y_W(k^{-1}v,t)$. The same canonical formal operator, with $X, Y, X \boxtimes Y$ replaced by their pullbacks, is a complete operator of type $\binom{T_k(X\boxtimes Y)}{T_kX\ T_kY}$: its defining analytic identities are those for the original operator with each V-insertion u replaced by $k^{-1}u$. By the twisted universal property this fixes the action tensorator $J^k_{X,Y} : T_kX \boxtimes T_kY \to T_k(X \boxtimes Y)$ via

$$\overline{J^k_{X,Y}}\mathcal{Y}_{T_kX,T_kY}(x,z)y = \mathcal{Y}_{X,Y}(x,z)y, \tag{2.12}$$

where the right side is regarded as a completed vector in $\overline{T_k(X \boxtimes Y)}$ through the unchanged underlying weight-space identification. It holds first at $P(1)$ by the representing property and then coefficientwise as a formal-operator identity by the injectivity of evaluation proved above; the same identity is valid with every consistently continued logarithm. This is the tensorator normalization [7, equation (4.7) and Proposition 8.5]. Uniqueness of the representing maps, together with the strict underlying identities $T_kT_l = T_{kl}$, gives $J^{kl}_{X,Y} = T_k(J^l_{X,Y})J^k_{T_lX,T_lY}$ as in (2.4); the tensorators remain part of the monoidal action data. The paper [7] also supplies the associator, crossed hexagons and contragredient pairings used below.

For the chosen $P(1)$ representative, the preceding $L(0)$ calculation gives, for $x \in X_\alpha$ and $y \in Y_\beta$,

$$\pi_\gamma \mathcal{Y}_{X,Y}(x,z)y = z^{\gamma-\alpha-\beta}\, v_\gamma(x,y), \qquad v_\gamma(x,y) \in (X \boxtimes Y)_\gamma, \tag{2.13}$$

where $v_\gamma(x,y) = \pi_\gamma I^1_{X,Y}(x \otimes y)$ is independent of z and $z^\rho = e^{\rho L}$ on the chosen branch. Finite-dimensionality of $(X \boxtimes Y)_\gamma$ comes from ordinary grading restriction, not from convergence in the full completion.

3. The neutral sector and mixed rationality

3.1. **A locality-to-rationality lemma.** Throughout this subsection a matrix coefficient containing a completed module vector is defined by homogeneous projections. For homogeneous inputs the conformal-weight shift fixes the intermediate weight at any specified tuple of mode indices, so the coefficient involves no infinite sum of weight components. We distinguish a coefficientwise formal distribution from its expansion in a prescribed ordering of formal variables.

Lemma 3.1. *Let X, Y, Z be ordinary grading-restricted V-modules and $\mathcal{Y}$ an ordinary Jacobi-identity intertwining operator of type $\binom{Z}{X\ Y}$. Fix homogeneous $a \in X_\alpha$, $b \in Y_\beta$, $z' \in (Z')_\gamma$ and $u_i \in V_{d_i}$, $1 \le i \le m$, and put $\rho = \gamma - \alpha - \beta - \sum_{i=1}^m d_i$. The nested matrix coefficient*

$$F(\boldsymbol{x};t) = \langle z', Y_Z(u_1,x_1)\cdots Y_Z(u_m,x_m)\mathcal{Y}(a,t)b\rangle$$

is the ordered formal expansion, for $|x_1| > \cdots > |x_m| > |t| > 0$, of

$$t^\rho R(x_1/t,\dots,x_m/t), \qquad R \in \mathbb{C}[s_1^{\pm1},\dots,s_m^{\pm1},(s_i-1)^{-1},(s_i-s_j)^{-1}\ (i<j)]. \tag{3.1}$$

After fixing $\log t$*, the ordered expansion is absolutely and normally convergent on compact subsets of this chamber. Its individual scalar terms agree, without infinite regrouping, with the homogeneous intermediate-weight terms in the multiple-product series of* [8, Definition 2.8]*. For fixed* $t \neq 0$ *the only possible finite polar divisors in the additional insertion variables are* $x_i = 0$*,* $x_i = t$ *and* $x_i = x_j$*.*

Proof. We first identify the coefficient spaces used below. The $L(0)$-bracket, obtained from the intertwining-operator Jacobi identity, gives

$$\pi^Z_\delta \mathcal{Y}(a,t)b = t^{\delta-\alpha-\beta} v_\delta, \qquad v_\delta \in Z_\delta,$$

with no logarithmic terms. The grading restriction bounds the possible exponents of t from below. For each fixed exponent, v_δ is an actual vector in a finite-dimensional weight space, and the matrix coefficient of the m ordinary vertex operators on it is a well-defined nested formal Laurent series. Thus, up to its fixed fractional t-power, F belongs to the ordered iterated Laurent field $\mathbb{C}((x_1))((x_2))\cdots((x_m))((t))$, with the expansion order $|x_1| > \cdots > |x_m| > |t|$. In this space the expansions of $(x_i - x_j)^{-1}$ and $(x_i - t)^{-1}$ are fixed, and multiplication by these differences is injective. The same need not hold in an unrestricted space of bilateral formal distributions.

Choose $A_i \geq 0$ with $u_i(n)a = 0$ for all integers $n \geq A_i$. The commutator form of the Jacobi identity for $\mathcal{Y}$ contains only the finitely many terms $\mathcal{Y}(u_i(n)a,t)$, $0 \leq n < A_i$, multiplied by derivatives of the formal delta distribution at $x_i = t$. Multiplication by $(x_i - t)^{A_i}$ kills each of these terms, giving, as a coefficientwise identity in the two expansion orders,

$$(x_i - t)^{A_i} Y_Z(u_i, x_i)\mathcal{Y}(a,t) = (x_i - t)^{A_i} \mathcal{Y}(a,t) Y_Y(u_i, x_i). \tag{3.2}$$

The exponent depends only on u_i, a, not on b or any other mode indices. Likewise, choose $A_{ij} \geq 0$ so that $u_i(n)u_j = 0$ for $n \geq A_{ij}$ (increasing A_{ij} if necessary). The vertex-algebra Jacobi identity gives, on both Y and Z,

$$(x_i - x_j)^{A_{ij}} Y_M(u_i, x_i) Y_M(u_j, x_j) = (x_i - x_j)^{A_{ij}} Y_M(u_j, x_j) Y_M(u_i, x_i), \qquad M = Y, Z. \tag{3.3}$$

These are formal-distribution identities after multiplication by the indicated polynomial, not assertions that the unmultiplied series in opposite order are equal. There are only finitely many pairs. Set

$$Q(\boldsymbol{x}, t) = \prod_i (x_i - t)^{A_i} \prod_{i<j} (x_i - x_j)^{A_{ij}}, \qquad D = \deg Q.$$

In particular Q is homogeneous of total degree D in $\boldsymbol{x}, t$.

For clarity about changing order, we perform the following coefficientwise construction before dividing by Q. A coefficient of $x_1^{p_1} \cdots x_m^{p_m}$ in F is obtained by using exactly the modes $u_i(-p_i - 1)$; its $\mathcal{Y}$-output weight is forced to be $\gamma - \sum_i (d_i + p_i)$, and hence its t-power is $\rho - \sum_i p_i$. Thus the coefficient is a single scalar, with no implicit infinite sum. Multiplication by a polynomial is a finite convolution of these scalar coefficients. Applying (3.2)–(3.3) to each adjacent interchange, with the intervening modes fixed, therefore proves that the *cleared* coefficient array QF agrees with the array obtained from any ordering of the u_i's, with a chosen u_i also moved past $\mathcal{Y}$ onto Y.

We now prove bounds independent of every other mode index. Fix i and, in the cleared array, place u_i to the far right, after $\mathcal{Y}$. The final field acts on the fixed vector b as $u_i(n)b$, which vanishes for every $n \geq N_i$ for some N_i depending only on u_i, b. Before multiplication by Q the resulting powers $p_i = -n - 1$ satisfy $p_i \geq -N_i$. As Q has no negative powers of x_i, the same lower bound holds for QF, regardless of all other indices.

For the upper bound, place u_i at the far left, keeping $\mathcal{Y}$ to its right. If $w \in Z_\lambda$ is any weight component supplied by the remaining operators, then $\langle z', u_i(n)w\rangle \neq 0$ requires $\lambda = \gamma - d_i + n + 1$. Since Z is lower bounded, there is $B_Z \in \mathbb{R}$ with $\Re\lambda \geq B_Z$ for all its weights. Consequently $n \geq B_Z - \Re\gamma + d_i - 1$, where $n \in \mathbb{Z}$. Hence the powers $p_i = -n - 1$ are uniformly bounded above; multiplication by Q increases an x_i-exponent by at most $\deg_{x_i} Q$. This bound too is independent of the remaining mode indices. Repeating this argument for every i confines the support of QF to a finite Cartesian product of intervals in $\mathbb{Z}^m$. Both bounds refer to the *same*

cleared coefficient array, so they may be imposed simultaneously. It follows that QF has only finitely many monomials in $\boldsymbol{x}$.

For the remaining t-powers, $u_i(n)$ shifts weight by $d_i - n - 1 = d_i + p_i$, while $\mathcal{Y}(a,t)b$ has a weight-δ component proportional to $t^{\delta-\alpha-\beta}$. Thus a monomial $\boldsymbol{x}^{\boldsymbol{p}}$ of F has t-degree $\rho - \sum_i p_i$. Multiplication by any monomial of Q raises total degree by D. The finite support just proved therefore yields $QF = t^{\rho+D}P(x_1/t, \dots, x_m/t), P \in \mathbb{C}[s_1^{\pm 1}, \dots, s_m^{\pm 1}]$. This is initially an equality of coefficient arrays. Its left side is also the product of Q with the original ordered iterated Laurent series. In that ordered series space Q is invertible, so division uniquely gives the ordered expansion of $t^{\rho+D}P/Q$. Since $Q = t^D \prod_i (s_i - 1)^{A_i} \prod_{i<j}(s_i - s_j)^{A_{ij}}$, this is precisely (3.1). This use of the ordered Laurent space is essential: in a bilateral distribution space, $QH = 0$ need not imply $H = 0$.

We verify analytic convergence without appealing to formal rationality as an analytic theorem. In the stated chamber $|t/x_i| < 1$ and $|x_j/x_i| < 1$ for $i < j$. Expand each $(x_i - t)^{-A_i}$ as $x_i^{-A_i}(1 - t/x_i)^{-A_i}$ and each $(x_i - x_j)^{-A_{ij}}$ as $x_i^{-A_{ij}} \sum_{n\geq 0} \binom{A_{ij}+n-1}{n}(x_j/x_i)^n (A_{ij} > 0)$, with the factor understood as 1 when the exponent is 0. All of these binomial series converge absolutely and locally uniformly in their respective ratios. There are finitely many factors, and P is a finite Laurent polynomial, so their product converges absolutely and normally on compact subsets of the chamber. The factor $t^\rho = e^{\rho \log t}$ uses the fixed branch. The uniqueness of the ordered formal expansion proved above identifies *each* scalar monomial coefficient with that of the original matrix coefficient, rather than only identifying a formal rational expression.

Finally, insert the projections in the multiple series of [8, Definition 2.8]. Write $\lambda_0 = \gamma$, and let λ_i be the intermediate weight immediately to the right of $Y_Z(u_i, x_i)$. Because the u_i have integral weights, a nonzero term with mode $n_i \in \mathbb{Z}$ obeys $\lambda_{i-1} = \lambda_i + d_i - n_i - 1, n_i = \lambda_i + d_i - 1 - \lambda_{i-1}$. Hence the m intermediate weights determine the m modes uniquely, and the modes recursively determine the weights, with λ_m specifying the single t-power of the intertwining operator. Each projection is onto the correct finite-dimensional weight component, so each tuple gives exactly the same scalar as its corresponding monomial in the ordered expansion. This is a bijection of nonzero scalar terms, not a regrouping of several terms into one. Their absolute convergence therefore follows from that of the ordered rational expansion. Arbitrary (not necessarily homogeneous) inputs are finite linear combinations of homogeneous vectors. Finally, the rational function has no polar divisors outside the factors displayed in (3.1); its dependence on the fixed t may carry the selected power branch but introduces no monodromy in the additional variables. □

Lemma 3.2. *Fix $t \in \mathbb{C}^\times$. Let R be a rational function of the remaining m variables. At every component-isolated corner in the sense of* [8, Definition 2.6]*, including one specified by an invertible affine change of variables and any mixture of zero and infinity components, R has an absolutely convergent regular-singularity expansion with integral exponents and no logarithmic terms. The exponent sets can be chosen independently of the branch. In particular, this applies to the component-isolated translation corners required by* [8, Definition 2.8].

Proof. Write $\eta = zA - \beta$, where $A \in \mathrm{GL}_m(\mathbb{C})$, and put $\xi_i = \eta_i$ if $\eta_i \to 0$ and $\xi_i = \eta_i^{-1}$ if $\eta_i \to \infty$. All changes are rational and invertible on the relevant coordinate tori. Thus the transformed function, denoted $f(\xi)$, belongs to $\mathbb{C}(\xi_1, \dots, \xi_m)$. Clearing negative powers in a numerator and denominator gives $f(\xi) = P(\xi)/Q(\xi)$ with $P, Q \in \mathbb{C}[\xi_1, \dots, \xi_m]$ and $Q \neq 0$. In particular, f defines a meromorphic germ at $\xi = 0$ in the sense of a quotient of holomorphic germs; this alone does not yet yield a coordinatewise Laurent expansion.

By component isolation, there are $\epsilon_i > 0$ such that f is holomorphic on the entire product of punctured discs $\Delta_\epsilon^\times = \{\xi : 0 < |\xi_i| < \epsilon_i \text{ for every } i\}$. Its multivariable Laurent coefficients are therefore the iterated Cauchy integrals

$$a_{\boldsymbol{n}} = \frac{1}{(2\pi i)^m} \int_{|w_1|=s_1} \cdots \int_{|w_m|=s_m} \frac{f(\boldsymbol{w})}{w_1^{n_1+1} \cdots w_m^{n_m+1}}\, dw_m \cdots dw_1, \qquad 0 < s_i < \epsilon_i,$$

independent of the radii. The corresponding Laurent series converges absolutely and locally uniformly on $\Delta_\epsilon^\times$. We show that its negative powers are bounded uniformly in each variable, rather than assuming this from meromorphy.

Set $d_i = \deg_{\xi_i} Q$. For fixed values of the other coordinates outside the proper algebraic set on which $Q(\xi_i, \widehat{\xi_i})$ vanishes identically as a polynomial in ξ_i, the one-variable rational function $\xi_i \mapsto f(\xi_i, \widehat{\xi_i})$ has at $\xi_i = 0$ a pole of order at most d_i. Its one-variable Laurent coefficients with exponent $n_i < -d_i$ consequently vanish. For any circle $|\xi_i| = s_i$ inside the punctured disc, these coefficients are Cauchy integrals and vary holomorphically with the remaining coordinates on their punctured polydisc. Their vanishing on the indicated dense set therefore implies vanishing everywhere. Integrating the resulting zero function in the remaining variables shows that $a_{\boldsymbol{n}} = 0$ whenever some $n_i < -d_i$. The multivariable Laurent series hence has the form

$$f(\xi) = \xi_1^{-N_1} \cdots \xi_m^{-N_m} H(\xi), \qquad N_i = d_i \geq 0, \quad H \in \mathbb{C}\{\xi_1, \ldots, \xi_m\}. \tag{3.4}$$

Indeed the Laurent series of $H = \prod_i \xi_i^{N_i} f$ has only nonnegative powers, so its power series extends holomorphically across every coordinate hyperplane. It is absolutely convergent on a sufficiently small polydisc and normally convergent on compact subsets. Multiplication by $\xi^{-\boldsymbol{N}}$ preserves absolute and normal convergence on the corresponding punctured polydisc.

This also verifies the polar-divisor step directly. The factorization in (3.4) shows that the polar divisor germ is supported on coordinate hyperplanes. Equivalently, if a reduced denominator in the local ring $\mathbb{C}\{\xi_1, \ldots, \xi_m\}$ had an irreducible non-coordinate polar factor through the origin, its hypersurface would have points with every $\xi_i \neq 0$ arbitrarily near the origin. At a generic such point the coprime numerator does not vanish on that factor, contradicting holomorphy on $\Delta_\epsilon^\times$. The irreducibility assertion follows from the local analytic Nullstellensatz and the fact that an irreducible hypersurface germ contained in a finite union of coordinate hyperplanes is one of those hyperplanes; the coordinate-factor conclusion also follows without these results from the preceding Laurent-coefficient argument. In particular, no additional codimension-two obstruction is left unaddressed.

Expand $H(\xi) = \sum_{\boldsymbol{q} \in \mathbb{Z}^m_{\geq 0}} h_{\boldsymbol{q}} \xi^{\boldsymbol{q}}$. For a zero component, the exponents in the original variable η_i belong to $-N_i + \mathbb{Z}_{\geq 0}$; for an infinity component they belong to $N_i - \mathbb{Z}_{\geq 0}$, since $\xi_i = \eta_i^{-1}$. These sets, chosen simultaneously for all components, have exactly the forms required in [8, Definition 2.6]. The series is absolutely convergent in a sufficiently small mixed zero/infinity region, with logarithmic degree $K = 0$. A rational function remains single-valued under the coordinate changes, so the same exponent sets and coefficients apply to all its branches. Finally, when R arises from Lemma 3.1, fixing the nonzero intertwining coordinate t makes t^ρ a scalar on each selected $\log t$ branch. Multiplication by this scalar, or by any of the integral V–V pole-clearing polynomials in [8, Definition 2.8], preserves rationality in the additional coordinates. The preceding argument then supplies branch-independent integral exponent sets and zero logarithmic degree for the resulting function. Thus every component-isolated translation corner tested there is covered. □

3.2. **Equality of the two intertwining-operator spaces.**

Theorem 3.3. *For a strongly rational V and all ordinary V-modules X, Y, Z, every ordinary intertwining operator satisfies the complete Du–Huang analytic requirements of* [8, Definition 2.8]. *In particular, for every $m \geq 2$ and every choice of additional states of V, the prescribed nested correlation functions converge absolutely, extend to the full configuration space and have regular component-isolated corners after multiplication by the prescribed V–V difference factors. Moreover, $\mathcal{I}_{\mathrm{DH}}\binom{Z}{XY} = \mathcal{I}_{\mathrm{HL}}\binom{Z}{XY}$.*

Proof. We first prove the inclusion from right to left. Let $\mathcal{Y} \in \mathcal{I}_{\mathrm{HL}}\binom{Z}{XY}$. Its coefficientwise lower truncation and $L(-1)$-derivative property are precisely the first and fourth conditions of [8, Definition 2.8]. For homogeneous $a \in X_\alpha$, $b \in Y_\beta$, the $L(0)$-bracket [9, (5.4.12)–(5.4.14)] implies $\pi^Z_\gamma \mathcal{Y}(a, x)b = x^{\gamma-\alpha-\beta} v_\gamma(a, b)$. Thus its exponents have the requisite weight-difference support, and it has logarithmic degree zero.

We verify the three parts of the one-V duality condition, including their prescribed branches. Take homogeneous $u \in V_d$ and $z' \in (Z')_\gamma$. The case $m = 1$ of Lemma 3.1, with $x_1 = z_1$ and $t = z_2$, gives a rational function R such that the product series with its intermediate Z-weight

projection is the convergent expansion of $f(z_1,z_2)=e^{\rho\ell(z_2)}R(z_1/z_2),\rho=\gamma-\alpha-\beta-d$, where $\ell(z_2)$ is the logarithm on the preferred branch of [8, Definition 2.8]. Here R has possible poles only at $z_1/z_2=0,1,\infty$. For fixed homogeneous data, f consequently has a multivalued analytic continuation on M^2, arising only from continuation of z_2^ρ. Its germs along paths in M^2 provide the maximal analytic continuation of the initial preferred germ.

The other two matrix coefficients are not obtained simply by rearranging a convergent infinite series. The ordinary intertwining-operator Jacobi identity [9, Definition 5.4.1, (5.4.4)] gives weak commutativity between $Y_Z(u,z_1)$ and $\mathcal{Y}(a,z_2)$ and the product–iterate identity [9, (5.4.17), (5.4.24)]. Clear a sufficiently high power of z_1-z_2 in these formal identities. Uniqueness of the corresponding ordered expansions identifies the reversed product and iterate, with the Y- and X-weight projections in Definition 2.5, as the expansions of this *same* f in the respective variables (z_1,z_2) and (z_1-z_2,z_2). In the second chamber $|z_2|>|z_1|>0$ expand $(z_1-z_2)^{-1}$ in z_1/z_2; in the third chamber $|z_2|>|z_1-z_2|>0$ expand z_1^{-1} in $(z_1-z_2)/z_2$. The remaining possible denominators are monomials. Each expansion is a finite Laurent sum multiplied by finitely many geometric or binomial series, and hence is absolutely and locally normally convergent. The $L(0)$ weight relations identify its individual terms with the required intermediate-weight projections, without an infinite regrouping. On the three subregions specified in Definition 2.5(ii), the formal substitutions use the preferred determinations of $\ell(z_2)$ and the associated logarithms of z_1 and z_1-z_2. Thus all three sums represent the same preferred analytic germ; its continuation along paths gives the other branches. This argument uses neither an adjoint intertwining operator nor the additional assumptions of [9, Theorem 5.6.2].

For the multiple-V condition, first take homogeneous $u_i\in V$, a,b and z'. Apply Lemma 3.1 with $m=k-1$ for each $k\geq 3$. Its proof identifies the $k-1$ intermediate output-weight projections of Definition 2.5(iii) term by term with the modes in the ordered expansion. In particular the *projected* multiple series converges absolutely in $|z_1|>\cdots>|z_k|>0$. The analytic germ has the form $z_k^\rho R(z_1/z_k,\dots,z_{k-1}/z_k)$, with the z_k power taken on the selected logarithmic branch. Every possible pole of R lies among $z_i=0$, $z_i=z_k$, and $z_i=z_j$ $(i<j<k)$; after substituting the nonzero variable z_k, these are precisely divisors excluded from M^k. Thus the expression, with analytic continuation of z_k^ρ around 0, gives a multivalued analytic continuation throughout M^k. It is the maximal continuation of the germ defined by the convergent multiple series, with the branch of z_k^ρ retained throughout.

Now fix $z_k\neq 0$, including one logarithmic branch, as required in [8, Definition 2.8(3)]. For *every* choice of nonnegative integers M_{ij} for $1\leq i<j\leq k-1$ with $x^{M_{ij}}Y_V(u_i,x)u_j\in V[[x]]$, multiplying the continued function by $\prod_{1\leq i<j\leq k-1}(z_i-z_j)^{M_{ij}}$ leaves a rational function of the additional $k-1$ positions, up to the fixed nonzero scalar z_k^ρ. By Lemma 3.2 every component-isolated translation corner required in Definition 2.8 has an absolutely convergent regular-singularity expansion, with integer exponent sets common to all branches of the additional variables and logarithmic degree zero. The fixed scalar can change with the chosen branch of z_k but does not change these exponent sets. No common choice of leading exponents as z_k or the admissible M_{ij} vary is asserted or required. Finite linearity in the input vectors proves the claims for arbitrary, not necessarily homogeneous, inputs. This establishes $\mathcal{I}_{\mathrm{HL}}\binom{Z}{X\,Y}\subseteq\mathcal{I}_{\mathrm{DH}}\binom{Z}{X\,Y}$.

For the reverse inclusion, take $\mathcal{Y}\in\mathcal{I}_{\mathrm{DH}}\binom{Z}{X\,Y}$. The lower-truncation and derivative axioms already have the ordinary form. The remark immediately following [8, Definition 2.8] derives the $L(0)$-commutator formula from analytic duality (the conformal vector is fixed by all twists). To make the absence of logarithms explicit, take $a\in X_\alpha$, $b\in Y_\beta$ and project to Z_γ. For the coefficient of x^{-n-1} write $P_{n,\gamma}(y)=\sum_{j=0}^K\pi_\gamma^Z(a_{n,j}^{\mathcal{Y}}b)y^j$ with $y=\log x$. The $L(0)$-bracket gives the polynomial differential equation $P'_{n,\gamma}(y)=(\gamma-\alpha-\beta+n+1)P_{n,\gamma}(y)$. A polynomial solution is zero unless $\gamma-\alpha-\beta+n+1=0$; in the latter case it is constant. Therefore all coefficients of $(\log x)^j$ for $j>0$ vanish, and each remaining term has the ordinary weight-difference exponent. In particular $\mathcal{Y}:X\otimes Y\to Z\{x\}$.

We now obtain the full ordinary Jacobi identity at the level of matrix coefficients, independently of a commutator-only criterion. Fix homogeneous a,b,u,z' as above and the three absolutely convergent sums from Definition 2.5(ii). The $L(0)$-bracket implies simultaneous homogeneity:

$(z_1\partial_{z_1}+z_2\partial_{z_2})f=\rho f$, where $\rho=\gamma-\alpha-\beta-d$. For fixed $z_2\neq 0$ and its logarithmic determination, the three convergent expansions show that f is meromorphic in z_1 near 0, z_2 and ∞, with finite pole orders at each point. Indeed near 0 the reversed product has only finitely many negative powers of z_1 by $u_nb=0$ for large n; near z_2 the iterate has only finitely many negative powers of z_1-z_2 by $u_na=0$ for large n; and near infinity the product has only finitely many positive powers of z_1, because z' has fixed weight and the output Z is lower bounded. The expansions have integral powers in the respective local variable, so they have no local monodromy as z_1 encircles these punctures. Since these loops generate the fundamental group of $\mathbb{P}^1\setminus\{0,z_2,\infty\}$, the preferred germ is single-valued in z_1 for fixed z_2 and extends meromorphically to the sphere. Hence it is rational in z_1. Homogeneity then identifies it with $z_2^{\rho}R(z_1/z_2)$ for a rational R with possible poles only at $0,1,\infty$.

The product, reversed product and iterate are its three uniquely determined formal expansions in $|z_1|>|z_2|$, $|z_2|>|z_1|$ and $|z_2|>|z_1-z_2|$, respectively. Apply the elementary formal delta-function identity for these three expansions (after clearing powers of z_1, z_2 and z_1-z_2); this is the rationality–commutativity–associativity to Jacobi calculation of [9, Proposition 3.4.1], applied to the present mixed matrix coefficients. The resulting identity is precisely (2.10) after contraction with z'. The restricted-dual pairings separate the homogeneous output coefficients, so the full formal intertwining-operator Jacobi identity holds. In particular, this supplies the formal justification of the assertion in [8, Remark 2.11]: that remark also states that a Definition 2.8 operator is an ordinary intertwining operator for $V^{\langle g_1,g_2\rangle}$, including Jacobi rather than merely the commutator formula. Here $g_1=g_2=g_3=1$, so that fixed-point algebra is precisely V. We have therefore established the Jacobi identity, lower truncation, derivative property and absence of logarithms, giving $\mathcal{Y}\in\mathcal{I}_{\mathrm{HL}}\binom{Z}{X\ Y}$.

Both inclusions preserve the actual mode coefficients: neither constructs a new intertwining operator. Composition with module maps in the two inputs and the output commutes with every projection and formal identity used above, so the equality is natural in X,Y,Z (contravariantly in the inputs, covariantly in the output). It also preserves weightwise evaluation at any $P(z)$ with the same prescribed logarithm; a change of path or branch acts on the same formal coefficients on both sides. □

In the next proposition, write $\boxtimes_D$, c^D and a^D for the tensor product, crossed commutor restricted to degree 1, and associator constructed in [7]; write $\boxtimes_H$, c^H and a^H for their Huang–Lepowsky counterparts. The notation distinguishes structures on the same underlying ordinary module category, not different classes of modules.

Proposition 3.4. *Let V be strongly rational and $G\leq\mathrm{Aut}(V)$ finite, and let $\mathcal{C}_G$ be the category constructed by Xu in* [7, Theorem 1.1]. *Its binary $P(z)$-representing property and canonical operators are given in* [7, Theorem 4.3, equations (4.3)–(4.4)]*; its pullback tensorators and action coherence are given in* [7, equation (4.7), Proposition 8.5]. *The product–iterate and unit normalizations are established in* [7, Theorem 8.3, equations (8.10)–(8.11), Proposition 8.5], *and the positive-half-turn commutor and parallel-transport identities in* [7, Proposition 9.6, equations (9.27)–(9.30)]. *Then there is a canonical braided monoidal equivalence*

$$F:(\mathcal{C}_1,\boxtimes_D,c^D,a^D)\xrightarrow{\sim}(\mathrm{Rep}(V),\boxtimes_H,c^H,a^H)$$

whose underlying functor is the identity on modules and morphisms. For every $z\in\mathbb{C}^*$ and every specified logarithm L with $e^L=z$, the comparison also preserves the corresponding $P(z)$-representing objects and parallel transports.

Proof. By the regularity and semisimplicity established in Section 2, the neutral sector $\mathcal{C}_1$ and $\mathrm{Rep}(V)$ have the same ordinary V-modules and the same module homomorphisms. The tensor products, however, are initially different structures: $\boxtimes_D$ is the binary product imported from [7], whereas $\boxtimes_H$ is the Huang–Lepowsky product. The $P(z)$ universal property on the Huang–Lepowsky side is the one in [12, Definition 3.1 and Proposition 3.4]; the standard associator and braiding belong to the vertex-tensor-category construction of [12, Theorem 4.4]. We compare the structures through their representing maps, not merely through their common underlying objects.

Fix $z \neq 0$ and a logarithm L of z. For $E = D, H$, let $I^{E,z}_{X,Y} : X \otimes Y \to \overline{X \boxtimes_{E,P(z)} Y}$ denote the corresponding universal $P(z)$-intertwining map. Here $\boxtimes_{D,P(z)}$ is the $P(z)$-product from [7] and $\boxtimes_{H,P(z)}$ is the Huang–Lepowsky product, with the same chosen evaluation logarithm. For each ordinary target W, Theorem 3.3 identifies the complete Du–Huang intertwining operators with the ordinary Huang–Lepowsky intertwining operators coefficient by coefficient. Evaluating both spaces at the same L therefore identifies their $P(z)$-map spaces as the same subspace of $\mathrm{Hom}_{\mathbb{C}}(X \otimes Y, \overline{W})$. The binary representing property in [7] and the Huang–Lepowsky universal property thus represent the same functor of W.

Consequently there is a unique V-module isomorphism

$$\Phi^z_{X,Y} : X \boxtimes_{D,P(z)} Y \xrightarrow{\sim} X \boxtimes_{H,P(z)} Y, \qquad \overline{\Phi^z_{X,Y}} I^{D,z}_{X,Y} = I^{H,z}_{X,Y}. \tag{3.5}$$

For completeness, the inverse is the unique map $\Psi^z_{X,Y} : X \boxtimes_{H,P(z)} Y \to X \boxtimes_{D,P(z)} Y$ with $\overline{\Psi^z_{X,Y}} I^{H,z}_{X,Y} = I^{D,z}_{X,Y}$. The two universal properties give

$$\overline{\Psi^z_{X,Y}\Phi^z_{X,Y}} I^{D,z}_{X,Y} = I^{D,z}_{X,Y}, \qquad \overline{\Phi^z_{X,Y}\Psi^z_{X,Y}} I^{H,z}_{X,Y} = I^{H,z}_{X,Y};$$

injectivity of the representing bijections forces both composites to be identities. This also proves uniqueness of $\Phi^z_{X,Y}$ among maps preserving the universal maps.

The comparison is natural in the inputs. If $f : X \to X'$ and $q : Y \to Y'$ are module maps, then $\Phi^z_{X',Y'} \circ (f \boxtimes_{D,P(z)} q) = (f \boxtimes_{H,P(z)} q) \circ \Phi^z_{X,Y}$, because both sides, after completion and precomposition with $I^{D,z}_{X,Y}$, give $I^{H,z}_{X',Y'} \circ (f \otimes q)$. Naturality in the target object is the naturality in W of the two representing bijections. In the ordinary sector, evaluation at a fixed logarithm is injective weight by weight, by the $L(0)$-bracket and the formula in (2.13). Hence the evaluated identity in (3.5) lifts to the identity of formal complete operators $\overline{\Phi^z_{X,Y}}\, \mathcal{Y}^D_{X,Y}(x,t)y = \mathcal{Y}^H_{X,Y}(x,t)y$, with the same formal coefficients and the same continued logarithm. Equivalently, $\overline{(\Phi^z_{X,Y})^{-1}} \mathcal{Y}^H_{X,Y} = \mathcal{Y}^D_{X,Y}$.

This comparison also commutes with parallel transport, not just with the existence of an object isomorphism. Let γ be a path from z_1 to z_2 in $\mathbb{C}^*$, and let L_1^γ be the continuation back to z_1 of the selected logarithm L_2 at z_2. Denote the transports in the two constructions by T^D_γ and T^H_γ. Their defining identities and (3.5) give

$$\overline{\Phi^{z_2}_{X,Y}\mathsf{T}^D_\gamma} I^{D,z_1}_{X,Y} = \mathrm{ev}_{L_1^\gamma}(\mathcal{Y}^H_{X,Y}) = \overline{\mathsf{T}^H_\gamma \Phi^{z_1}_{X,Y}} I^{D,z_1}_{X,Y}.$$

The representing property in [7] then yields the typed equality $\Phi^{z_2}_{X,Y} \circ \mathsf{T}^D_\gamma = \mathsf{T}^H_\gamma \circ \Phi^{z_1}_{X,Y}$. The identity is pathwise: if γ winds around the origin, both sides retain the same branch change and monodromy.

Take $z = 1$ and $L = 0$. The functor F is the identity on the common module category, its unit map is $F_0 = \mathrm{id}_V$, and its tensorator, in the convention $F_2 : F(X) \boxtimes_H F(Y) \to F(X \boxtimes_D Y)$, is $F_{2,X,Y} = (\Phi^1_{X,Y})^{-1} : X \boxtimes_H Y \longrightarrow X \boxtimes_D Y$. The strong-monoidal-functor convention fixes the tensorator as $F_{2,X,Y} = (\Phi^1_{X,Y})^{-1}$.

We next verify associativity. Put $U_H = (X \boxtimes_H Y) \boxtimes_H Z, U_D = (X \boxtimes_D Y) \boxtimes_D Z, V_D = X \boxtimes_D (Y \boxtimes_D Z)$. The two sides of the required coherence equation are the following maps with the same source and target:

$$U_H \xrightarrow{F_{2,X,Y} \boxtimes_H \mathrm{id}_Z} (X \boxtimes_D Y) \boxtimes_H Z \xrightarrow{F_{2,X \boxtimes_D Y,Z}} U_D \xrightarrow{F(a^D_{X,Y,Z})} V_D,$$

$$U_H \xrightarrow{a^H_{X,Y,Z}} X \boxtimes_H (Y \boxtimes_H Z) \xrightarrow{\mathrm{id}_X \boxtimes_H F_{2,Y,Z}} X \boxtimes_H (Y \boxtimes_D Z) \xrightarrow{F_{2,X,Y \boxtimes_D Z}} V_D.$$

In particular, each occurrence of F_2 is applied to a pair of objects in the common ordinary module category, so the intermediate objects above are well typed.

For $E = D, H$, the defining normalization of the corresponding associator is the product–iterate identity

$$\overline{a^E_{X,Y,Z}}\, \mathcal{Y}^E_{X \boxtimes_E Y,Z}(\mathcal{Y}^E_{X,Y}(x, z_1 - z_2)y, z_2)z = \mathcal{Y}^E_{X,Y \boxtimes_E Z}(x, z_1)\mathcal{Y}^E_{Y,Z}(y, z_2)z,$$

on the common positive overlap $z_1 > z_2 > z_1 - z_2 > 0$, with all logarithms real. The left side denotes the convergent sum of intermediate-weight projections defining the iterate, with values in the completion of $(X \boxtimes_E Y) \boxtimes_E Z$ before applying a^E. The right side is the product, taking values in the completion of $X \boxtimes_E (Y \boxtimes_E Z)$. These are precisely the associator normalization imported from [7, Theorem 8.3, equation (8.11)] and the Huang–Lepowsky associator normalization. All three modules and the intermediate tensor products are ordinary, so Theorem 3.3 applies to each binary operator appearing here.

Apply the two composites displayed above to the canonical iterated coefficients of the H-product U_H. The identity $\overline{F_{2,A,B}}\mathcal{Y}^H_{A,B} = \mathcal{Y}^D_{A,B}$, naturality of the binary products, and the defining product–iterate identity show that the first composite sends each such coefficient to the corresponding D-product coefficient. The second composite does the same, by first using the H product–iterate identity and then applying the same binary comparison to its two factors. Thus, after pairing with any vector in the restricted dual of the common target module, the two composites give the same scalar analytic function on the positive overlap. The restricted dual separates the weight completion, so their values on every canonical iterated coefficient agree. The coefficients of a universal binary map generate its representing module: otherwise the quotient by the submodule they generate would give a nonzero morphism annihilating the universal map, contrary to injectivity of the representing bijection. Applying this observation successively to the two binary products shows that the canonical iterated coefficients generate U_H. The two module maps are therefore equal. This proves

$$\begin{aligned} &F(a^D_{X,Y,Z})F_{2,X\boxtimes_D Y,Z}(F_{2,X,Y} \boxtimes_H \mathrm{id}_{FZ}) \\ &\qquad = F_{2,X,Y\boxtimes_D Z}(\mathrm{id}_{FX} \boxtimes_H F_{2,Y,Z})a^H_{FX,FY,FZ}. \end{aligned}$$

No associator coherence is used to prove this identity; only the two independently specified product–iterate normalizations and binary universality enter.

For braiding, both sides of

$$F(c^D_{X,Y})F_{2,X,Y} = F_{2,Y,X}c^H_{FX,FY} \tag{3.6}$$

are maps $X \boxtimes_H Y \to Y \boxtimes_D X$. The crossed commutor from [7] on the neutral sector is the positive-half-turn map of (2.11); the Huang–Lepowsky braiding is defined by the same Ω_+ skew transformation and the same upper-half-plane transport. In both formulas the continued logarithm is $\log(e^{\pi i}z) = \log z + \pi i$, and $e^{zL(-1)}$ acts on the output module $Y \boxtimes X$. Since $T_1 = \mathrm{id}$ and the two canonical formal operators are related by F_2, no nontrivial pullback comparison or action tensorator occurs in this neutral-sector calculation. Precomposing the two composites with $I^H_{X,Y}$ gives the same evaluated Ω_+ operator by (2.11) and the Huang–Lepowsky braiding formula. The $P(1)$ representing property, which is injective on morphisms, gives (3.6).

Finally, both unit constraints are normalized by the same vacuum intertwiners: the left unit uses Y_X and the right unit uses $\Gamma_X = \Omega_- Y_X$, with the branch fixed in Section 2. The unit object is V on both sides, so set $F_0 = \mathrm{id}_V : V_H \to F(V_D)$. Evaluating on the universal maps gives the full unit equations

$$F(l^D_X) \circ F_{2,V,X} \circ (F_0 \boxtimes_H \mathrm{id}_{FX}) = l^H_{FX}, \qquad F(r^D_X) \circ F_{2,X,V} \circ (\mathrm{id}_{FX} \boxtimes_H F_0) = r^H_{FX},$$

which are the left and right unit diagrams in the chosen strong-monoidal convention. Thus F is a strong monoidal functor, and (3.6) makes it braided. Since its underlying functor is the identity equivalence and every $F_{2,X,Y}$ and F_0 is an isomorphism, it is a braided monoidal equivalence. Its tensorator is unique among those carrying the specified universal maps to one another, and the displayed transport identity proves the asserted $P(z)$-branch and parallel-transport compatibility. □

Remark 3.5. Theorem 3.3 concerns ordinary intertwining operators. Its proof uses weak locality with the distinguished ordinary intertwining operator and integer-powered vacuum fields. The rationality argument applies to the variables of the additional V insertions with the intertwining-operator coordinate fixed.

4. The conformal crossed twist

Lemma 4.1. *For $g \in G$ and $X \in \mathcal{C}_g$, the conformal-weight operator induces an isomorphism of g-twisted modules $\theta_X = e^{2\pi i L_X(0)} : X \xrightarrow{\sim} T_g X$. It is natural in X, satisfies $\theta_{T_k X} = T_k(\theta_X)$ for every $k \in G$ under the strict underlying pullback convention, and obeys $\theta_V = \mathrm{id}_V = j_1$.*

Proof. By the regularity result recalled in Section 2, X is an ordinary grading-restricted g-twisted module. Thus $X = \bigoplus_{\lambda\in\mathbb{C}} X_\lambda$ and $\theta_X|_{X_\lambda} = e^{2\pi i\lambda}\,\mathrm{id}$ defines an invertible linear operator on the algebraic direct sum, with inverse $e^{-2\pi i L_X(0)}$. Both operators preserve each weight space. Since automorphisms of V fix ω, the pullback $T_g X$ has the same $L(n)$, in particular $L(0)$ and $L(-1)$, on the same graded vector space.

Let $N = \mathrm{ord}(g)$ and choose $v \in V_d$ in the g-eigenspace of eigenvalue $e^{2\pi i r/N}$, where $d \in \mathbb{Z}$ and $0 \le r < N$. The finite-order convention of Section 2, [2, Definition 3.1], and the equivariance convention in [8, Section 2, definition of generalized twisted modules] give $Y_X(v,z) = \sum_{n\in r/N+\mathbb{Z}} v_n z^{-n-1}$. Indeed, for an ordinary finite-order twisted field the latter analytic equivariance reads $Y_X(gv, e^{2\pi i}z) = Y_X(v,z)$ on continued branches, so the monodromy of z^{-n-1} is $e^{-2\pi i(n+1)} = e^{-2\pi i r/N}$. Since $Y_X(\omega,z)$ has integral modes, the twisted $L(0)$-bracket gives $[L_X(0), Y_X(v,z)] = (z\partial_z + d)Y_X(v,z), [L_X(0), v_n] = (d-n-1)v_n$. For $x \in X_\lambda$, the vector $v_n x$ lies in $X_{\lambda+d-n-1}$. Consequently, comparing the actions on this weight space yields the operator identity

$$e^{2\pi i L_X(0)} v_n e^{-2\pi i L_X(0)} = e^{2\pi i(d-n-1)} v_n = e^{-2\pi i r/N} v_n = (g^{-1}v)_n.$$

The same equation holds for arbitrary $v \in V$ by its finite decomposition into homogeneous conformal weights and g-eigenspaces. It follows coefficientwise in $\mathrm{Hom}(X,X)((z^{1/N}))$ (with the usual lower truncation on each vector) that $\theta_X Y_X(v,z) = Y_X(g^{-1}v,z)\theta_X = Y_{T_g X}(v,z)\theta_X$. Here $T_g X \in \mathcal{C}_{ggg^{-1}} = \mathcal{C}_g$ and the last expression uses the defining pullback field. Thus $\theta_X : X \to T_g X$ is a degree-preserving twisted-module morphism, rather than an endomorphism of the original labeled module. Multiplying this identity by θ_X^{-1} on the appropriate sides gives $\theta_X^{-1} Y_{T_g X}(v,z) = Y_X(v,z)\theta_X^{-1}$; therefore $\theta_X^{-1} : T_g X \to X$ is also a twisted-module morphism. Both maps commute with $L(0)$ and $L(-1)$ and extend weightwise to the corresponding completions.

For any g-twisted module morphism $f : X \to Y$ one has $f L_X(0) = L_Y(0) f$ by applying f to the conformal-vector field. Hence, as maps $X \to T_g Y$, $T_g(f) \circ \theta_X = \theta_Y \circ f$. On morphisms $T_g(f)$ is the same underlying linear map, but its source and target are the pulled-back labeled modules. For $k \in G$, put $h = kgk^{-1}$. The two maps

$$\theta_{T_k X} : T_k X \longrightarrow T_h T_k X, \qquad T_k(\theta_X) : T_k X \longrightarrow T_k T_g X$$

have the same target since $T_h T_k = T_{hk} = T_{kg} = T_k T_g$ as strict underlying pullback functors. Because $Y_{T_k X}(\omega,z) = Y_X(\omega,z)$, they act on each weight space by the same scalar $e^{2\pi i\lambda}$, proving $\theta_{T_k X} = T_k(\theta_X)$. No monoidal tensorator or action compositor is involved.

The assertion is sectorwise: for an object $\bigoplus_g X_g$ of the externally graded sum, the collection assembles into the morphism $\bigoplus_g \theta_{X_g} : \bigoplus_g X_g \to \bigoplus_g T_g X_g$, not a morphism to T_g of the entire object for a single g. Finally $V \in \mathcal{C}_1$ is the tensor unit, $T_1 = \mathrm{id}$, and $V = \bigoplus_{d\in\mathbb{Z}} V_d$. Hence $e^{2\pi i L_V(0)} = \mathrm{id}_V = j_1$, the unit normalization in (2.4). □

Lemma 4.2. *Let $X \in \mathcal{C}_g$, $Y \in \mathcal{C}_h$, and put $q = ghg^{-1}$. The composite*

$$c_{T_g Y, X} \circ c_{X,Y} : X \boxtimes Y \xrightarrow{\sim} T_q X \boxtimes T_g Y$$

is an isomorphism. For $x \in X$ and $y \in Y$ it satisfies the identity of completed canonical intertwining operators

$$\overline{c_{T_g Y, X} \circ c_{X,Y}}\, \mathcal{Y}_{X,Y}(x,z)y = \mathcal{Y}_{T_q X, T_g Y}(x, e^{2\pi i}z)y, \tag{4.1}$$

where the right-hand side is evaluated on the branch $\log(e^{2\pi i}z) = \log z + 2\pi i$ obtained by two consecutive positive half-turns.

Proof. Put $U = X \boxtimes Y$, $B = T_gY \boxtimes X$, and $C = T_qX \boxtimes T_gY$. Since $T_gY \in \mathcal{C}_q$ and $T_qX \in \mathcal{C}_{qgq^{-1}}$, both B and C are in $\mathcal{C}_{qg} = \mathcal{C}_{gh}$. In particular the two arrows

$$c_1 := c_{X,Y} : U \longrightarrow B, \qquad c_2 := c_{T_gY,X} : B \longrightarrow C$$

are composable isomorphisms in the displayed order of g, h. Write $\mathcal{Z} = \mathcal{Y}_{T_gY,X}$ and $\mathcal{W} = \mathcal{Y}_{T_qX,T_gY}$. The formal normalization (2.11), applied first to (X, Y) and then to (T_gY, X), gives the following two equalities, respectively in the completed outputs $\overline{B}$ and $\overline{C}$:

$$\overline{c_1}\mathcal{Y}_{X,Y}(x,z)y = e^{zL_B(-1)}\mathcal{Z}(y,w)x, \qquad w = e^{\pi i}z,$$

$\overline{c_2}\mathcal{Z}(y,w)x = e^{wL_C(-1)}\mathcal{W}(x,e^{\pi i}w)y$. Here y in the first equality is the underlying vector of T_gY, x in the second is the underlying vector of T_qX, and the logarithms are continued successively as $\log w = \log z + \pi i$ and $\log(e^{\pi i}w) = \log z + 2\pi i$. The second equality uses $\deg(T_gY) = q$, not h. It applies at the inherited logarithm because (2.11) is an identity of complete *formal* operators, with the indicated logarithmic continuation, rather than only an identity at a single $P(1)$ evaluation.

As a morphism of twisted V-modules, c_2 intertwines the action of the conformal vector, so $c_2L_B(-1) = L_C(-1)c_2$. This relation also holds after weightwise completion. Indeed, $L(-1)$ raises conformal weight by one, and for an ordinary lower-bounded output every fixed-weight projection of $e^{zL(-1)}H(z)$ has only finitely many contributions from lower weights. Thus, coefficientwise in the completed output, $\overline{c_2}\, e^{zL_B(-1)}H(z) = e^{zL_C(-1)}\overline{c_2}H(z)$. Applying this to the first equality and then substituting the second, we obtain an equality in $\overline{C}$:

$$\begin{aligned}\overline{c_2 \circ c_1}\,\mathcal{Y}_{X,Y}(x,z)y &= e^{zL_C(-1)}\overline{c_2}\,\mathcal{Z}(y,w)x \\ &= e^{zL_C(-1)}e^{wL_C(-1)}\mathcal{W}(x,e^{\pi i}w)y \\ &= \mathcal{W}(x,e^{2\pi i}z)y.\end{aligned}$$

For the last equality, $w = -z$ as a complex coordinate; both translation operators now act on C. Their product is the identity: for each fixed output weight, its coefficient of $L_C(-1)^n$ is $\sum_{j=0}^{n} z^j w^{n-j}/(j!(n-j)!) = (z+w)^n/n!$. This is a finite sum in every fixed-weight projection, so no rearrangement of an unconditionally convergent infinite series is being assumed.

The position $e^{2\pi i}z$ is numerically z, but the continued logarithm is not reset. For homogeneous $x \in X_\alpha$, $y \in Y_\beta$, equation (2.13) applied to $\mathcal{W}$ gives on the γ-weight component the factor $e^{2\pi i(\gamma-\alpha-\beta)}$ relative to the initial branch; the pullbacks preserve these weights. No logarithmic polynomials occur for the ordinary objects of $\mathcal{C}_G$. Finite homogeneous decompositions extend the assertion to all x, y. The two commutors already include their chosen upper-half-plane transports. The argument follows from (2.11) and the indicated module structures. □

Theorem 4.3. *The conformal twist satisfies the crossed balancing relation*

$$\theta_{X\boxtimes Y} = J^{gh}_{X,Y}(T_{ghg^{-1}}\theta_X \boxtimes T_g\theta_Y)c_{T_gY,X}\, c_{X,Y}, \quad X \in \mathcal{C}_g,\ Y \in \mathcal{C}_h. \tag{4.2}$$

Proof. Put $q = ghg^{-1}$, $U = X \boxtimes Y$, and $k = gh = qg$. The tensor-product objects in the displayed composite lie in $\mathcal{C}_k$. The constituent arrows have the following types:

$$\begin{gathered}c_{X,Y} : U \longrightarrow T_gY \boxtimes X, \qquad c_{T_gY,X} : T_gY \boxtimes X \longrightarrow T_qX \boxtimes T_gY, \\ T_q(\theta_X) : T_qX \longrightarrow T_qT_gX = T_kX, \qquad T_g(\theta_Y) : T_gY \longrightarrow T_gT_hY = T_kY, \\ J^k_{X,Y} : T_kX \boxtimes T_kY \longrightarrow T_kU.\end{gathered}$$

The successive pullbacks use the underlying action functors, while the tensorator J^k remains part of the composite. The composite on the right of (4.2), denoted by $B_{X,Y}$, is thus a morphism $U \to T_kU$. Since $U \in \mathcal{C}_k$, Lemma 4.1 gives a morphism $\theta_U : U \to T_kU$ of the same type.

We compare the two morphisms by precomposing their completed extensions with the universal formal intertwining operator $\mathcal{Y}_{X,Y}$; all operator identities below are interpreted coefficientwise in $\overline{T_kU}$. The positive full-turn identity (4.1) first gives

$$\overline{c_{T_gY,X}c_{X,Y}}\,\mathcal{Y}_{X,Y}(x,z)y = \mathcal{Y}_{T_qX,T_gY}(x,e^{2\pi i}z)y,$$

with logarithm $L + 2\pi i$ if the initial logarithm of z is L. This equation is in $\overline{T_qX \boxtimes T_gY}$. For twisted-module morphisms $f : A \to A'$ and $p : B \to B'$, the bifunctoriality of the chosen $P(1)$-tensor products gives $\overline{f \boxtimes p}\, I^1_{A,B}(a \otimes b) = I^1_{A',B'}(fa \otimes pb)$. The same identity holds for the associated complete formal operators at every consistently continued logarithm: both sides are obtained from the corresponding natural complete intertwining operators and agree coefficientwise by the $L(0)$-grading and the evaluation–coefficient correspondence of Section 2. In particular, with $f = T_q(\theta_X)$ and $p = T_g(\theta_Y)$, it gives

$$\begin{aligned}&\overline{T_q(\theta_X) \boxtimes T_g(\theta_Y)}\, \mathcal{Y}_{T_qX,T_gY}(x, e^{2\pi i}z)y\\ &\qquad = \mathcal{Y}_{T_kX,T_kY}(T_q(\theta_X)x, e^{2\pi i}z)T_g(\theta_Y)y.\end{aligned}$$

The map $f \boxtimes p$ is selected by this universal property.

First take $x \in X_\alpha$, $y \in Y_\beta$. All pullbacks preserve $L(0)$, so their underlying graded vectors satisfy $T_q(\theta_X)x = e^{2\pi i\alpha}x, T_g(\theta_Y)y = e^{2\pi i\beta}y$. The two right-hand inputs in the preceding display are now vectors of T_kX and T_kY, respectively. Applying the completed tensorator and its defining formal normalization (2.12), with $k = gh$, yields the identity in $\overline{T_kU}$

$$\begin{aligned}\overline{B_{X,Y}}\, \mathcal{Y}_{X,Y}(x, z)y &= \overline{J^k_{X,Y}}\, \mathcal{Y}_{T_kX,T_kY}(T_q(\theta_X)x, e^{2\pi i}z)T_g(\theta_Y)y\\ &= e^{2\pi i(\alpha+\beta)}\, \overline{J^k_{X,Y}}\, \mathcal{Y}_{T_kX,T_kY}(x, e^{2\pi i}z)y\\ &= e^{2\pi i(\alpha+\beta)}\, \mathcal{Y}_{X,Y}(x, e^{2\pi i}z)y.\end{aligned}$$

In the last term the coefficients are viewed in T_kU through its fixed underlying weight-space identification. This use of (2.12) is valid on the continued branch because the identity is a normalization of complete *formal* operators, not merely an equality of numerical $P(1)$ evaluations. The extension $\overline{J^k_{X,Y}}$ preserves each conformal weight since $J^k_{X,Y}$ is a twisted-module morphism.

For a weight γ of U, equation (2.13) gives, on the initial branch, $\pi_\gamma \mathcal{Y}_{X,Y}(x, z)y = z^{\gamma-\alpha-\beta}v_\gamma(x, y)$. The positively continued branch in the preceding equality therefore satisfies

$$\begin{aligned}\pi_\gamma \overline{B_{X,Y}}\, \mathcal{Y}_{X,Y}(x, z)y &= e^{2\pi i(\alpha+\beta)}e^{2\pi i(\gamma-\alpha-\beta)}\pi_\gamma \mathcal{Y}_{X,Y}(x, z)y\\ &= e^{2\pi i\gamma}\pi_\gamma \mathcal{Y}_{X,Y}(x, z)y\\ &= \pi_\gamma \overline{\theta_U}\, \mathcal{Y}_{X,Y}(x, z)y.\end{aligned}$$

Here π_γ refers throughout to the γ-weight space of the common output T_kU, which is the same graded vector space as U. Thus the last scalar is exactly the weightwise action of $e^{2\pi i L_U(0)} : U \to T_kU$. The positive full-turn factor is essential: a negative full turn would produce the inverse factor and would not prove (4.2) with this twist. Ordinary grading restriction eliminates logarithmic polynomial terms and makes both completed morphisms and every weight projection well defined. Equality of all weight projections implies equality in the weight-product completion.

Every vector of an ordinary module is a finite sum of its homogeneous-weight components, so the preceding identity holds for arbitrary $x \in X$, $y \in Y$ by linearity. Evaluating the resulting formal equality at $z = 1$ with $\log 1 = 0$ gives

$$\overline{B_{X,Y}}\, I^1_{X,Y} = \overline{\theta_U}\, I^1_{X,Y} : X \otimes Y \longrightarrow \overline{T_kU}.$$

The binary representing property of the chosen twisted $P(1)$-tensor product identifies

$$\mathrm{Hom}_V(U, T_kU) \xrightarrow{\sim} \mathcal{I}_{P(1),0}\binom{T_kU}{X\ Y}, \qquad f \longmapsto \overline{f}\, I^1_{X,Y}.$$

Its injectivity, applied to the two morphisms $B_{X,Y}$ and θ_U, proves (4.2). This argument uses only the canonical operator, its naturality, the tensorator normalization, Lemmas 4.1 and 4.2, and the imported binary representing property. For objects containing several homogeneous G-components, the balancing identity is extended additively from pairs of homogeneous summands; it does not assign a single degree to an inhomogeneous object. □

5. Contragredients and the ribbon axiom

5.1. **Restricted contragredients and categorical duals.** We use the right-dual evaluations and coevaluations of Section 2. The rigidity result in [7] supplies the categorical duals; the contragredient-pairing lemma of [7] supplies, independently of categorical duality, natural morphisms $e_X : X \boxtimes DX \to \mathbf{1}$ which are nonzero for $X \neq 0$. The restricted-dual pullback comparison $\tau_{k,X}$ is the one fixed in Section 2. We use the pairing naturality and its G-covariance as explicit inputs from [7].

Proposition 5.1. *The pairings e_X give natural isomorphisms $\phi_X : DX \xrightarrow{\sim} X^\vee$, characterized by $e_X = \mathrm{ev}_X(\mathrm{id}_X \boxtimes \phi_X)$. Under these isomorphisms, the conformal twist satisfies the following identity, with no implicit object identification: for $X \in \mathcal{C}_g$,*

$$\theta_{DX} = T_{g^{-1}}(D\theta_X) \circ \big(T_{g^{-1}}(\tau_{g,X})\big)^{-1} : DX \longrightarrow T_{g^{-1}}DX. \tag{5.1}$$

Moreover, the identifications ϕ_X intertwine the restricted and categorical duals of the monoidal G-action:

$$\phi_{T_kX} = b_{k,X} \circ T_k(\phi_X) \circ \tau_{k,X} : D(T_kX) \longrightarrow (T_kX)^\vee. \tag{5.2}$$

Consequently the categorical dual-twist identity (2.9) holds.

Proof. We work first with $X \in \mathcal{C}_g$. The two potential duals are separate objects of $\mathcal{C}_{g^{-1}}$: DX is the weightwise restricted contragredient, whereas $X^\vee$ is the chosen categorical right dual, with $\mathrm{ev}_X : X \boxtimes X^\vee \to \mathbf{1}$ and $\mathrm{coev}_X : \mathbf{1} \to X^\vee \boxtimes X$. The contravariant functors D and $(-)^\vee$ are equivalences on the appropriate sectors: $D^2 \cong \mathrm{id}$ by the restricted bidual comparison of Section 2, and categorical right duality by rigidity.

For any U, the right-dual adjunction [17, Definition 2.10.2 and Proposition 2.10.8(b)] identifies $\mathrm{Hom}(U, X^\vee)$ with $\mathrm{Hom}(X \boxtimes U, \mathbf{1})$ by $h \mapsto \mathrm{ev}_X(\mathrm{id}_X \boxtimes h)$. Its inverse sends $p : X \boxtimes U \to \mathbf{1}$ to the following morphism:

$$\begin{aligned} U &\xrightarrow{l_U^{-1}} \mathbf{1} \boxtimes U \xrightarrow{\mathrm{coev}_X \boxtimes \mathrm{id}_U} (X^\vee \boxtimes X) \boxtimes U \\ &\xrightarrow{a_{X^\vee,X,U}} X^\vee \boxtimes (X \boxtimes U) \xrightarrow{\mathrm{id}_{X^\vee} \boxtimes p} X^\vee \boxtimes \mathbf{1} \xrightarrow{r_{X^\vee}} X^\vee. \end{aligned}$$

Indeed, substituting this composite in $\mathrm{ev}_X(\mathrm{id}_X \boxtimes -)$ and applying associator and unitor naturality reduces it to p by the first right-dual triangle identity. Conversely, substituting $p = \mathrm{ev}_X(\mathrm{id}_X \boxtimes h)$ reduces the inverse composite to h by the second triangle identity. Both reductions use the right-dual convention of Section 2, including the displayed orientations of a, l, r; no strict associativity is assumed.

Let $\phi_X : DX \to X^\vee$ be the unique mate of the pairing $e_X : X \boxtimes DX \to \mathbf{1}$ constructed in [7]. Thus $e_X = \mathrm{ev}_X(\mathrm{id}_X \boxtimes \phi_X)$. Since the mate bijection is linear, $e_X \neq 0$ implies $\phi_X \neq 0$. If X is simple, DX is simple under the restricted-contragredient anti-equivalence, and $X^\vee$ is simple under categorical right-dual anti-equivalence. Both belong to $\mathcal{C}_{g^{-1}}$, so ϕ_X is an isomorphism by simplicity. This uses the nonvanishing of e_X from [7, Lemma 10.1], not any prior identification of the two duals.

We establish naturality before extending invertibility beyond simple objects. If $f : X \to Y$ lies in one homogeneous sector, the pairing identity imported from [7] is

$$e_Y(f \boxtimes \mathrm{id}_{DY}) = e_X(\mathrm{id}_X \boxtimes Df) : X \boxtimes DY \longrightarrow \mathbf{1}.$$

The definition of $f^\vee : Y^\vee \to X^\vee$ gives $\mathrm{ev}_Y(f \boxtimes \mathrm{id}_{Y^\vee}) = \mathrm{ev}_X(\mathrm{id}_X \boxtimes f^\vee)$, both with source $X \boxtimes Y^\vee$. Substituting the mate formulas therefore yields

$$\mathrm{ev}_X(\mathrm{id}_X \boxtimes f^\vee\phi_Y) = \mathrm{ev}_X(\mathrm{id}_X \boxtimes \phi_X Df) : X \boxtimes DY \longrightarrow \mathbf{1}.$$

Injectivity of the right-dual mate correspondence gives $\phi_X Df = f^\vee \phi_Y$ as an identity of morphisms $DY \to X^\vee$. No ϕ_X^{-1} has been used here.

For arbitrary homogeneous X, choose a finite decomposition $X = \bigoplus_{i=1}^m S_i$ with S_i simple, writing $\iota_i : S_i \to X$, $p_i : X \to S_i$ for its inclusions and projections. Repetitions among the S_i are

allowed. Contravariance gives $D\iota_i : DX \to DS_i$ and $Dp_i : DS_i \to DX$, with $D\iota_i Dp_i = \mathrm{id}_{DS_i}$, $D\iota_i Dp_j = 0$ for $i \neq j$, and $\sum_i Dp_i D\iota_i = \mathrm{id}_{DX}$. The pairing naturality, applied to ι_i and then precomposed with $\mathrm{id}_{S_i} \boxtimes Dp_j$, gives the precise block relation

$$e_X(\iota_i \boxtimes Dp_j) = \begin{cases} e_{S_i}, & i = j, \\ 0, & i \neq j, \end{cases} : S_i \boxtimes DS_j \longrightarrow \mathbf{1}.$$

In particular, it excludes off-diagonal blocks even for isomorphic summands and their multiplicity spaces. Applying this naturality identity to p_i gives $\phi_X Dp_i = p_i^\vee \phi_{S_i}$. Consequently

$$\phi_X = \sum_i p_i^\vee \phi_{S_i} D\iota_i, \qquad \phi_X^{-1} = \sum_i Dp_i \phi_{S_i}^{-1} \iota_i^\vee.$$

The relations $p_i\iota_i = \mathrm{id}_{S_i}$, $p_i\iota_j = 0$ for $i \neq j$, and $\sum_i \iota_i p_i = \mathrm{id}_X$, together with their contravariant duals, show directly that the two displayed sums are inverse. Hence ϕ_X is an isomorphism for every homogeneous X, including $X = 0$. Since it is the unique mate of the intrinsically defined e_X, the isomorphism does not depend on the decomposition.

Now set $\mathrm{ev}_X^D = e_X$ and transport the right-dual coevaluation along ϕ_X:

$$\mathrm{coev}_X^D = (\phi_X^{-1} \boxtimes \mathrm{id}_X)\,\mathrm{coev}_X : \mathbf{1} \longrightarrow DX \boxtimes X.$$

All arrows are in the neutral degree when appropriate: $X \boxtimes DX$ and $DX \boxtimes X$ have degree 1. With the associator a and unitors l, r from Section 2, the two proposed right-dual triangle composites are

$$\begin{aligned} T_X^D &= l_X(e_X \boxtimes \mathrm{id}_X)a_{X,DX,X}^{-1}(\mathrm{id}_X \boxtimes \mathrm{coev}_X^D)r_X^{-1} : X \longrightarrow X, \\ T_{DX}^D &= r_{DX}(\mathrm{id}_{DX} \boxtimes e_X)a_{DX,X,DX}(\mathrm{coev}_X^D \boxtimes \mathrm{id}_{DX})l_{DX}^{-1} : DX \longrightarrow DX. \end{aligned}$$

For the first composite, the intermediate objects are $X \boxtimes \mathbf{1}$, $X \boxtimes (DX \boxtimes X)$, $(X \boxtimes DX) \boxtimes X$ and $\mathbf{1} \boxtimes X$. Substitution and associator naturality give the explicit cancellation

$$\begin{aligned} T_X^D &= l_X(\mathrm{ev}_X \boxtimes \mathrm{id}_X)((\mathrm{id}_X \boxtimes \phi_X) \boxtimes \mathrm{id}_X)a_{X,DX,X}^{-1}(\mathrm{id}_X \boxtimes (\phi_X^{-1} \boxtimes \mathrm{id}_X))(\mathrm{id}_X \boxtimes \mathrm{coev}_X)r_X^{-1} \\ &= l_X(\mathrm{ev}_X \boxtimes \mathrm{id}_X)a_{X,X^\vee,X}^{-1}(\mathrm{id}_X \boxtimes \mathrm{coev}_X)r_X^{-1} = \mathrm{id}_X\,. \end{aligned}$$

The second composite passes through $\mathbf{1} \boxtimes DX$, $(DX \boxtimes X) \boxtimes DX$, $DX \boxtimes (X \boxtimes DX)$ and $DX \boxtimes \mathbf{1}$. Naturality of a, l, r and the same evaluation/coevaluation substitutions yield, with every intervening occurrence of $\phi_X \phi_X^{-1}$ cancelled,

$$\begin{aligned} \phi_X T_{DX}^D &= r_{X^\vee}(\mathrm{id}_{X^\vee} \boxtimes \mathrm{ev}_X)a_{X^\vee,X,X^\vee}(\mathrm{coev}_X \boxtimes \mathrm{id}_{X^\vee})l_{X^\vee}^{-1}\phi_X \\ &= \phi_X. \end{aligned}$$

The last equality is the second right-dual triangle identity for $X^\vee$. Since ϕ_X is invertible, $T_{DX}^D = \mathrm{id}_{DX}$. Thus $(DX, e_X, \mathrm{coev}_X^D)$ is a right dual of X with both triangle identities, for all homogeneous X. The naturality proved above is the fully typed identity

$$\phi_X \circ Df = f^\vee \circ \phi_Y : DY \longrightarrow X^\vee. \tag{5.3}$$

We next compare the two kinds of dual under pullback. For $X \in \mathcal{C}_g$, the objects $D(T_kX)$, $T_k(DX)$, $T_k(X^\vee)$, and $(T_kX)^\vee$ all lie in $\mathcal{C}_{kg^{-1}k^{-1}}$. In particular, the comparison $\tau_{k,X} : D(T_kX) \to T_k(DX)$ is distinct from the categorical action–dual comparison $b_{k,X} : T_k(X^\vee) \to (T_kX)^\vee$.

The map $\tau_{k,X}$ is the inverse of the weightwise identity $\vartheta_{k,X} : T_k(DX) \to D(T_kX)$ constructed in Section 2. Indeed, writing $u^{\mathrm{op}}(z) = e^{zL(1)}(-z^{-2})^{L(0)}u$, the two restricted-dual vertex operators have the same matrix coefficients against $w \in X$:

$$\begin{aligned} \langle Y_{D(T_kX)}(u,z)\varphi, w\rangle &= \langle \varphi, Y_X(k^{-1}u^{\mathrm{op}}(z), z^{-1})w\rangle \\ &= \langle Y_{T_k(DX)}(u,z)\varphi, w\rangle. \end{aligned}$$

Here $k^{-1}u^{\mathrm{op}}(z) = (k^{-1}u)^{\mathrm{op}}(z)$, since k fixes the conformal vector. Hence $\tau_{k,X}$ is a twisted-module isomorphism, rather than merely an identification of graded vector spaces. For a morphism $f : X \to Y$ in a fixed sector its contravariant naturality is

$$\tau_{k,X}\, D(T_k f) = T_k(Df)\, \tau_{k,Y} : D(T_k Y) \longrightarrow T_k(DX).$$

Since all τ's are identity maps on these restricted-dual weight spaces, the strict underlying action gives, as a composite $D(T_kT_lX) \to T_kD(T_lX) \to T_kT_l(DX)$, $\tau_{kl,X} = T_k(\tau_{l,X})\,\tau_{k,T_lX}, \tau_{1,X} = \mathrm{id}_{DX}$. No monoidal tensorator enters this restricted-dual comparison.

The pairing construction in [7] is independent of the chosen categorical dual: in its contragredient-pairing lemma, its skew and contragredient transforms of the twisted module vertex operator are mapped to V using the normalized invariant form and the vacuum comparison j_g^{-1}. Their universal binary factorization gives $e_X : X \boxtimes DX \to \mathbf{1}$. In particular, the covariance identity is an additional property of this construction, not a consequence of pairing naturality alone. Written with $\tau_{k,X} = \vartheta_{k,X}^{-1}$, it is

$$e_{T_kX} = j_k^{-1}\, T_k(e_X)\, J^k_{X,DX}(\mathrm{id}_{T_kX} \boxtimes \tau_{k,X}) : T_kX \boxtimes D(T_kX) \longrightarrow \mathbf{1}.$$

The four successive arrows on the right have targets $T_kX \boxtimes T_k(DX)$, $T_k(X \boxtimes DX)$, $T_k\mathbf{1}$, and $\mathbf{1}$, respectively. Thus the inverse unit map j_k^{-1} and the tensorator J^k cannot be suppressed. The comparison is understood in the same $P(1)$ and logarithmic normalization as the pairing in [7] and the action tensorator of Section 2.

For completeness, the comparison $b_{k,X}$ referred to in Lemma 5.3 is independently defined by the right-dual evaluation mate of

$$j_k^{-1}T_k(\mathrm{ev}_X)J^k_{X,X^\vee} : T_kX \boxtimes T_k(X^\vee) \longrightarrow \mathbf{1}.$$

The strong monoidal equivalence T_k carries the evaluation and coevaluation of X to a right-dual pair for T_kX; uniqueness of right duals makes its mate $b_{k,X}$ an isomorphism. Its defining comparison is (5.6). The mate $b_{k,X}$ is defined from J^k, j_k, and rigidity; the identification $DX \simeq X^\vee$ is established separately in Proposition 5.1.

We may therefore calculate in $\mathrm{Hom}(T_kX \boxtimes D(T_kX), \mathbf{1})$, using first (5.6), then naturality of J^k, and finally the mate identity $e_X = \mathrm{ev}_X(\mathrm{id}_X \boxtimes \phi_X)$:

$$\begin{aligned}
&\mathrm{ev}_{T_kX}(\mathrm{id}_{T_kX} \boxtimes b_{k,X}T_k(\phi_X)\tau_{k,X}) \\
&\quad = j_k^{-1}T_k(\mathrm{ev}_X)J^k_{X,X^\vee}(\mathrm{id}_{T_kX} \boxtimes T_k(\phi_X)\tau_{k,X}) \\
&\quad = j_k^{-1}T_k(\mathrm{ev}_X)T_k(\mathrm{id}_X \boxtimes \phi_X)J^k_{X,DX}(\mathrm{id}_{T_kX} \boxtimes \tau_{k,X}) \\
&\quad = j_k^{-1}T_k(e_X)J^k_{X,DX}(\mathrm{id}_{T_kX} \boxtimes \tau_{k,X}) = e_{T_kX}.
\end{aligned}$$

The two candidate maps to $(T_kX)^\vee$ have common source $D(T_kX)$; their evaluations are both e_{T_kX}. The right-dual adjunction $\mathrm{Hom}(D(T_kX), (T_kX)^\vee) \cong \mathrm{Hom}(T_kX \boxtimes D(T_kX), \mathbf{1})$ is injective. Hence (5.2) follows, retaining $\tau_{k,X}$ and $b_{k,X}$ as the distinct comparison maps.

The compatibility with morphisms is contravariant. For $f : X \to Y$, the comparison maps satisfy

$$\begin{aligned}
\tau_{k,X}D(T_kf) &= T_k(Df)\tau_{k,Y}, \\
b_{k,X}T_k(f^\vee) &= (T_kf)^\vee b_{k,Y}.
\end{aligned}$$

Together with $\phi_X Df = f^\vee \phi_Y$ these equations identify both routes from $D(T_kY)$ to $(T_kX)^\vee$; for example,

$$\begin{aligned}
&b_{k,X}T_k(\phi_X)\tau_{k,X}D(T_kf) \\
&\quad = b_{k,X}T_k(f^\vee\phi_Y)\tau_{k,Y} = (T_kf)^\vee b_{k,Y}T_k(\phi_Y)\tau_{k,Y}.
\end{aligned}$$

The identities extend to finite direct sums by additivity, including repeated isomorphic simple summands.

It remains to verify compatibility with group multiplication, without treating the monoidal action as strict. The defining evaluation of the composite $T_kT_l(X^\vee) \xrightarrow{T_k(b_{l,X})} T_k((T_lX)^\vee) \xrightarrow{b_{k,T_lX}}$

$(T_kT_lX)^\vee$ is

$$\begin{aligned}
&\mathrm{ev}_{T_kT_lX}(\mathrm{id}\boxtimes b_{k,T_lX}T_k(b_{l,X}))\\
&\quad= j_k^{-1}T_k(\mathrm{ev}_{T_lX})J^k_{T_lX,(T_lX)^\vee}(\mathrm{id}\boxtimes T_k(b_{l,X}))\\
&\quad= j_k^{-1}T_k(j_l^{-1})T_{kl}(\mathrm{ev}_X)T_k(J^l_{X,X^\vee})J^k_{T_lX,T_l(X^\vee)}\\
&\quad= j_{kl}^{-1}T_{kl}(\mathrm{ev}_X)J^{kl}_{X,X^\vee}.
\end{aligned}$$

Here the second equality uses naturality of J^k; the last uses the actual unit and tensorator coherences $j_{kl}=T_k(j_l)j_k$ and $J^{kl}_{X,X^\vee}=T_k(J^l_{X,X^\vee})J^k_{T_lX,T_l(X^\vee)}$. Evaluation-mate uniqueness now proves $b_{kl,X}=b_{k,T_lX}T_k(b_{l,X}), b_{1,X}=\mathrm{id}_{X^\vee}$. Consequently, applying (5.2) first to l and then to k gives the same map as applying it to kl:

$$\begin{aligned}
\phi_{T_kT_lX}&=b_{k,T_lX}T_k(\phi_{T_lX})\tau_{k,T_lX}\\
&=b_{k,T_lX}T_k(b_{l,X})T_{kl}(\phi_X)T_k(\tau_{l,X})\tau_{k,T_lX}\\
&=b_{kl,X}T_{kl}(\phi_X)\tau_{kl,X}.
\end{aligned}$$

In particular this is valid without commuting k and l. At $k=1$ the three comparisons reduce to identities; on the vacuum object the normalized invariant form and j_k give the same evaluation comparison, so the identity is compatible with the unit constraints. Drinfeld isomorphism, or any subsequent ribbon result.

Now fix $X\in\mathcal{C}_g$ and put $h=g^{-1}$. By the strict underlying pullback convention, $T_hT_g=T_1=\mathrm{id}$, whereas the monoidal tensorators of these functors need not be identities. The contravariant restricted dual and the contravariant categorical right dual give, respectively,

$$\begin{gathered}
\theta_X: X\longrightarrow T_gX,\qquad D\theta_X: D(T_gX)\longrightarrow DX,\\
\tau_{g,X}: D(T_gX)\xrightarrow{\sim} T_g(DX),\qquad T_h(\tau_{g,X}): T_hD(T_gX)\xrightarrow{\sim} DX,\\
T_h(D\theta_X): T_hD(T_gX)\longrightarrow T_h(DX),\\
\theta_{DX}: DX\longrightarrow T_h(DX).
\end{gathered}$$

Here $D(T_gX)$, $T_g(DX)$ and DX are all in $\mathcal{C}_h$; so are the sources and targets of the displayed morphisms after applying T_h, since $gh=hg=1$ for this fixed element g. In particular, $(T_h\tau_{g,X})^{-1}$ has source DX and target $T_hD(T_gX)$; the composite on the right of (5.1) is therefore a morphism $DX\to T_h(DX)$, with the same source and target as θ_{DX}.

To compute it, let $\varphi\in(DX)_\lambda=X^*_\lambda$. The pullback T_gX has the same conformal operator as X, since $g\omega=\omega$, and thus $(D(T_gX))_\lambda=X^*_\lambda$ canonically as graded vector spaces. By the definition of the restricted contravariant transpose, for $x\in X_\lambda$,

$$\langle D\theta_X(\varphi),x\rangle=\langle\varphi,\theta_Xx\rangle=e^{2\pi i\lambda}\langle\varphi,x\rangle.$$

This is a complex-linear transpose: it is neither the inverse operator nor the conjugate-linear adjoint, so its scalar is $e^{2\pi i\lambda}$, not $e^{-2\pi i\lambda}$ or its complex conjugate. The opposite-vertex definition (2.2) makes $L_{DX}(0)$ the weightwise transpose of $L_X(0)$; hence φ has weight λ in DX. Moreover, $\tau_{g,X}$ is the weightwise identity twisted-module isomorphism verified above: its inverse after applying T_h sends $\varphi\in(DX)_\lambda$ to the *same functional* in $\big(T_hD(T_gX)\big)_\lambda$. The next arrow $T_h(D\theta_X)$ maps this functional to $e^{2\pi i\lambda}\varphi\in(T_hDX)_\lambda$. This is exactly the action of $\theta_{DX}=e^{2\pi iL_{DX}(0)}$ on that weight space, proving (5.1) as an equality of twisted-module morphisms. Ordinary objects are algebraic direct sums of their weight spaces, so no completion or convergence argument is needed. The same calculation applies to every finite semisimple direct sum, including the zero object, entirely within the restricted-dual construction. The categorical right-dual comparison follows below.

We now compare the categorical twists, keeping the two dual functors separate. The maps involved have types

$$\begin{gathered}\phi_X : DX \xrightarrow{\sim} X^\vee, \qquad \phi_{T_gX} : D(T_gX) \xrightarrow{\sim} (T_gX)^\vee,\\ b_{g,X} : T_g(X^\vee) \xrightarrow{\sim} (T_gX)^\vee, \qquad \theta_X^\vee : (T_gX)^\vee \longrightarrow X^\vee,\\ T_h(b_{g,X}) : X^\vee \longrightarrow T_h((T_gX)^\vee),\\ T_h(\theta_X^\vee) : T_h((T_gX)^\vee) \longrightarrow T_h(X^\vee),\\ \theta_{X^\vee} : X^\vee \longrightarrow T_h(X^\vee), \qquad T_h(\phi_X) : T_h(DX) \longrightarrow T_h(X^\vee).\end{gathered}$$

The first source of $T_h(b_{g,X})$ is literally $X^\vee$ because $T_hT_g = \mathrm{id}$ as underlying functors. In particular, all morphisms in the dual-twist comparison are in $\mathcal{C}_h$. Equation (5.2) at $k = g$ gives, after composition on the right with $\tau_{g,X}^{-1}$,

$$b_{g,X}T_g(\phi_X) = \phi_{T_gX}\tau_{g,X}^{-1} : T_g(DX) \longrightarrow (T_gX)^\vee.$$

Applying T_h and the strict underlying action, without replacing either $b_{g,X}$ or $\tau_{g,X}$ by an identity, yields the equality of maps $DX \to T_h((T_gX)^\vee)$, $T_h(b_{g,X})\phi_X = T_h(\phi_{T_gX})(T_h\tau_{g,X})^{-1}$. Since $\theta_X : X \to T_gX$ is a morphism in the *same* sector $\mathcal{C}_g$, the contravariant naturality (5.3), applied with $f = \theta_X$, is

$$\theta_X^\vee\phi_{T_gX} = \phi_X D\theta_X : D(T_gX) \longrightarrow X^\vee.$$

The preceding two equalities and (5.1) give, now as successive equalities in $\mathrm{Hom}(DX, T_h(X^\vee))$,

$$\begin{aligned}&T_h(\theta_X^\vee)T_h(b_{g,X})\phi_X\\ &\quad= T_h(\theta_X^\vee\phi_{T_gX})(T_h\tau_{g,X})^{-1}\\ &\quad= T_h(\phi_X D\theta_X)(T_h\tau_{g,X})^{-1}\\ &\quad= T_h(\phi_X)T_h(D\theta_X)(T_h\tau_{g,X})^{-1}\\ &\quad= T_h(\phi_X)\theta_{DX}.\end{aligned}$$

Finally, naturality of the conformal crossed twist on the h-sector, applied to the module isomorphism $\phi_X : DX \to X^\vee$, gives

$$T_h(\phi_X)\theta_{DX} = \theta_{X^\vee}\phi_X : DX \longrightarrow T_h(X^\vee).$$

Since ϕ_X is invertible by the first part of this proposition, cancelling it on the right proves (2.9). This is precisely the type-correct expression of the third condition of [15, Lemma 2.3] in the right-dual convention of Definition 2.2: Kirillov writes $\theta_{V^*} = R_{g^{-1}}(\theta_V^*)$ while suppressing the canonical identification between the dual of a pulled-back object and the pullback of a dual. In the present conventions that identification is $T_h(b_{g,X})$. □

5.2. **The crossed Drinfeld isomorphism.**

Definition 5.2. For homogeneous $X \in \mathcal{C}_g$, put

$$f_X := \mathrm{ev}_X \circ c_{X^\vee,T_gX} : X^\vee \boxtimes T_gX \longrightarrow \mathbf{1}.$$

This is well typed: $X^\vee \in \mathcal{C}_{g^{-1}}$, so the crossed braiding has target $T_{g^{-1}}(T_gX) \boxtimes X^\vee = X \boxtimes X^\vee$, where $T_{g^{-1}}T_g = T_1 = \mathrm{id}$ is the underlying action identity for this single-object comparison. The monoidal tensorators enter in the subsequent monoidal calculations. The resulting f_X has target $\mathbf{1}$ and is a morphism in the neutral component.

Since $X^{\vee\vee}$ is the chosen right dual of $X^\vee$, with $\mathrm{ev}_{X^\vee} : X^\vee \boxtimes X^{\vee\vee} \to \mathbf{1}$ and $\mathrm{coev}_{X^\vee} : \mathbf{1} \to X^{\vee\vee} \boxtimes X^\vee$, the right-dual adjunction gives the natural bijection

$$\begin{aligned}\mathrm{Hom}(X^\vee \boxtimes T_gX, \mathbf{1}) &\xrightarrow{\sim} \mathrm{Hom}(T_gX, X^{\vee\vee}),\\ p &\longmapsto r_{X^{\vee\vee}}(\mathrm{id}_{X^{\vee\vee}} \boxtimes p)a_{X^{\vee\vee},X^\vee,T_gX}(\mathrm{coev}_{X^\vee} \boxtimes \mathrm{id}_{T_gX})l_{T_gX}^{-1}.\end{aligned}$$

Its inverse sends $u : T_g X \to X^{\vee\vee}$ to $\mathrm{ev}_{X^\vee}(\mathrm{id}_{X^\vee} \boxtimes u)$. This is the right-dual adjunction of [17, Proposition 2.10.8(b)], applied to $X^\vee$; it is not the left-dual adjunction. Define $\mathfrak{u}_X : T_g X \longrightarrow X^{\vee\vee}$ to be the mate of f_X. Equivalently, the defining mate identity is

$$X^\vee \boxtimes T_g X \xrightarrow{c_{X^\vee, T_g X}} T_{g^{-1}}(T_g X) \boxtimes X^\vee = X \boxtimes X^\vee \xrightarrow{\mathrm{ev}_X} \mathbf{1}. \tag{5.4}$$

After suppressing only the canonical associators and unitors, the adjunction formula reads

$$\mathfrak{u}_X = (\mathrm{id}_{X^{\vee\vee}} \boxtimes \mathrm{ev}_X)(\mathrm{id}_{X^{\vee\vee}} \boxtimes c_{X^\vee, T_g X})(\mathrm{coev}_{X^\vee} \boxtimes \mathrm{id}_{T_g X}). \tag{5.5}$$

Indeed, the successive objects in this composite are $T_g X$, $X^{\vee\vee} \boxtimes X^\vee \boxtimes T_g X$, $X^{\vee\vee} \boxtimes X \boxtimes X^\vee$, and $X^{\vee\vee}$; the second step uses $T_{g^{-1}} T_g X = X$, and the last uses $\mathrm{ev}_X : X \boxtimes X^\vee \to \mathbf{1}$. Thus the mate is a morphism $T_g X \to X^{\vee\vee}$ in $\mathcal{C}_g$.

Lemma 5.3. *Let X be any object and $k \in G$. There is a unique natural isomorphism $b_{k,X} : T_k(X^\vee) \xrightarrow{\sim} (T_k X)^\vee$ whose evaluation comparison is*

$$\mathrm{ev}_{T_k X}(\mathrm{id}_{T_k X} \boxtimes b_{k,X}) = j_k^{-1} T_k(\mathrm{ev}_X) J^k_{X, X^\vee}. \tag{5.6}$$

It induces the double-dual comparison

$$\chi^{(2)}_{k,X} := (b^{-1}_{k,X})^\vee b_{k, X^\vee} : T_k(X^{\vee\vee}) \xrightarrow{\sim} (T_k X)^{\vee\vee}. \tag{5.7}$$

These comparisons satisfy the coherence equations for the strict underlying G-action, with the nontrivial tensorators J^k retained.

Proof. The right side of (5.6) has domain $T_k X \boxtimes T_k(X^\vee)$ and codomain $\mathbf{1}$. By the representing property of the chosen right dual $(T_k X)^\vee$, it is the evaluation pairing of a unique $b_{k,X}$. Strong monoidality of T_k and preservation of duals show this map is invertible. Naturality and coherence follow from uniqueness of these representing morphisms and the identity $J^{kl}_{X,Y} = T_k(J^l_{X,Y}) J^k_{T_l X, T_l Y}$; they are not asserted with the J's replaced by identity maps. In (5.7), the first factor $b_{k,X^\vee}$ has target $(T_k X^\vee)^\vee$, while $(b^{-1}_{k,X})^\vee$ has source $(T_k X^\vee)^\vee$ and target $(T_k X)^{\vee\vee}$. Thus the composite is correctly typed. □

5.3. **The pivotal structure and its coherence.**

Proposition 5.4. *Under the rigidity and crossed-hexagon identities of* [7, Theorem 1.1] *and Theorem 4.3 and Proposition 5.1, the natural maps $\delta_X := \mathfrak{u}_X \theta_X : X \xrightarrow{\sim} X^{\vee\vee}$ form the pivotal structure corresponding to the conformal crossed twist in Kirillov's Lemma 2.3. In particular, writing $\mu_{X,Y} : X^{\vee\vee} \boxtimes Y^{\vee\vee} \xrightarrow{\sim} (X \boxtimes Y)^{\vee\vee}$ for the canonical monoidal structure of the double-right-dual functor, one has*

$$\delta_{X \boxtimes Y} = \mu_{X,Y}(\delta_X \boxtimes \delta_Y), \tag{5.8}$$

$$\delta_{T_k X} = \chi^{(2)}_{k,X}\, T_k(\delta_X), \tag{5.9}$$

$$\delta_{X^\vee} = (\delta_X^{-1})^\vee. \tag{5.10}$$

The equalities include the indicated canonical duality identifications and the original action tensorators.

Proof. We first verify the typing, naturality and invertibility of the explicit Drinfeld construction directly. The right-dual mate construction in Definition 5.2 is natural with respect to X in each fixed sector. To prove this, let $f : X \to Y$ be a morphism with $X, Y \in \mathcal{C}_g$, and put $f_X = \mathrm{ev}_X\, c_{X^\vee, T_g X}$ and $f_Y = \mathrm{ev}_Y\, c_{Y^\vee, T_g Y}$. The two morphisms $\mathfrak{u}_Y T_g(f)$ and $f^{\vee\vee} \mathfrak{u}_X$ have source $T_g X$ and target $Y^{\vee\vee}$. Under the right-dual mate bijection for $Y^\vee$, their respective mates belong

to $\mathrm{Hom}(Y^\vee \boxtimes T_g X, \mathbf{1})$ and are $f_Y(\mathrm{id}_{Y^\vee} \boxtimes T_g(f))$ and $f_X(f^\vee \boxtimes \mathrm{id}_{T_g X})$. These are equal, since

$$\begin{aligned} f_Y(\mathrm{id}_{Y^\vee} \boxtimes T_g(f)) &= \mathrm{ev}_Y\, c_{Y^\vee, T_g Y}(\mathrm{id}_{Y^\vee} \boxtimes T_g(f)) \\ &= \mathrm{ev}_Y(T_{g^{-1}} T_g(f) \boxtimes \mathrm{id}_{Y^\vee}) c_{Y^\vee, T_g X} \\ &= \mathrm{ev}_Y(f \boxtimes \mathrm{id}_{Y^\vee}) c_{Y^\vee, T_g X} \\ &= \mathrm{ev}_X(\mathrm{id}_X \boxtimes f^\vee) c_{Y^\vee, T_g X} \\ &= \mathrm{ev}_X\, c_{X^\vee, T_g X}(f^\vee \boxtimes \mathrm{id}_{T_g X}) \\ &= f_X(f^\vee \boxtimes \mathrm{id}_{T_g X}). \end{aligned}$$

The second line is naturality of the crossed braiding in its second input; the third uses the strict underlying identity $T_{g^{-1}} T_g(f) = f$; the fourth is naturality of evaluation; and the fifth is naturality of the crossed braiding in its first input. The mate correspondence is injective, so this proves $\mathfrak{u}_Y\, T_g(f) = f^{\vee\vee}\, \mathfrak{u}_X$.

Now suppose S is a nonzero simple object of $\mathcal{C}_g$. The right-dual triangle identities imply $\mathrm{ev}_S \neq 0$: otherwise the triangle composite equal to id_S would vanish. Since $c_{S^\vee, T_g S}$ is an isomorphism, $f_S = \mathrm{ev}_S\, c_{S^\vee, T_g S}$ is nonzero; otherwise composing with the inverse crossed braiding would give $\mathrm{ev}_S = 0$. The mate correspondence is a linear isomorphism, hence $\mathfrak{u}_S \neq 0$. The pullback T_g is an equivalence and therefore preserves simple objects, and right duality is a contravariant equivalence in a rigid fusion category, so $S^{\vee\vee}$ is simple as well. A nonzero morphism between simple objects in a semisimple $\mathbb{C}$-linear category is an isomorphism; thus $\mathfrak{u}_S$ is invertible.

Every object of a homogeneous component is a finite direct sum of simple objects because Xu's construction gives a fusion category [7, Theorem 1.1]. For a decomposition $X \cong \bigoplus_i S_i$, naturality with each inclusion $\iota_i : S_i \to \bigoplus_i S_i$ gives $\mathfrak{u}_{\oplus_i S_i} T_g(\iota_i) = \iota_i^{\vee\vee} \mathfrak{u}_{S_i}$. Since T_g and the double-dual functor preserve finite biproducts, these equations identify $\mathfrak{u}_{\oplus_i S_i}$ with the direct sum of the maps $\mathfrak{u}_{S_i}$ under the canonical biproduct identifications. This handles repeated simple summands as well: every inclusion is respected, so no off-diagonal component remains. Conjugating by the chosen decomposition isomorphism and using naturality proves that $\mathfrak{u}_X$ is an isomorphism independently of the decomposition. For the zero object the unique map $0 \to 0$ is its identity and is an isomorphism. Since θ is natural in each homogeneous sector, $\theta_Y f = T_g(f)\theta_X$. Consequently

$$\begin{aligned} \delta_Y f &= \mathfrak{u}_Y \theta_Y f = \mathfrak{u}_Y T_g(f) \theta_X \\ &= f^{\vee\vee} \mathfrak{u}_X \theta_X = f^{\vee\vee} \delta_X. \end{aligned}$$

Both $\mathfrak{u}_X$ and θ_X are isomorphisms, so δ_X is an isomorphism and is natural on every homogeneous sector. On a finite direct sum of sectors, the maps assemble as the direct sum of the homogeneous maps; no single degree is assigned to a nonhomogeneous object.

For the unit, use the canonical right-dual identification $\mathbf{1}^\vee \cong \mathbf{1}$, represented by $\mathrm{ev}_{\mathbf{1}} = r_{\mathbf{1}}$ and $\mathrm{coev}_{\mathbf{1}} = l_{\mathbf{1}}^{-1}$. The crossed unit identity $r_Y c_{\mathbf{1},Y} = l_Y$ at $Y = \mathbf{1}$, together with the unit coherence $l_{\mathbf{1}} = r_{\mathbf{1}}$, identifies $c_{\mathbf{1},\mathbf{1}}$ with the identity under $\mathbf{1} \boxtimes \mathbf{1} \cong \mathbf{1}$. Substituting these maps into (5.5) and using the triangle identities shows that $\mathfrak{u}_{\mathbf{1}}$ is the canonical unit double-dual map, which is the identity under this normalization. Finally, $\theta_{\mathbf{1}} = j_1 = \mathrm{id}_{\mathbf{1}}$ by Definition 2.2 and (2.4). Hence $\delta_{\mathbf{1}} = \mathrm{id}_{\mathbf{1}}$ under the canonical unit identifications. If the initially fixed right-dual representative of $\mathbf{1}$ is not literally $\mathbf{1}$, the same statement is read through its unique duality-preserving canonical isomorphism.

For G-equivariance, fix $X \in \mathcal{C}_g$ and $k \in G$. Put $Z_k := T_k(T_g X) = T_{kg} X = T_{kgk^{-1}}(T_k X)$, where both equalities are strict identities of the underlying action functors. Thus $T_k X \in \mathcal{C}_{kgk^{-1}}$, while $T_k(X^\vee)$ and $(T_k X)^\vee$ lie in $\mathcal{C}_{kg^{-1}k^{-1}}$. The two crossed braidings $c_{T_k(X^\vee), Z_k}, c_{(T_k X)^\vee, Z_k}$ therefore have the same acted-on first factor: indeed, $T_{kg^{-1}k^{-1}} Z_k = T_k X$. This fixes the group-element order needed below; the subsequent calculation retains the displayed order of all group elements.

Transport the defining pairing by setting

$$\widetilde{f}_{k,X} := j_k^{-1}\, T_k(f_X)\, J^k_{X^\vee, T_g X} : T_k(X^\vee) \boxtimes T_k(T_g X) \longrightarrow \mathbf{1}.$$

Here $J^k_{X^\vee, T_g X}$ first maps to $T_k(X^\vee \boxtimes T_g X)$, $T_k(f_X)$ then maps to $T_k \mathbf{1}$, and $j_k^{-1} : T_k \mathbf{1} \to \mathbf{1}$ is applied last. In particular, j_k^{-1} cannot be omitted.

Apply (2.5) to $A = X^\vee \in \mathcal{C}_{g^{-1}}$ and $B = T_g X$. Since $T_{g^{-1}} T_g X = X$, it gives the concrete identity

$$\begin{aligned} T_k(c_{X^\vee, T_g X}) J^k_{X^\vee, T_g X} &= J^k_{T_{g^{-1}} T_g X, X^\vee} c_{T_k(X^\vee), T_k(T_g X)} \\ &= J^k_{X, X^\vee} c_{T_k(X^\vee), T_k(T_g X)}. \end{aligned}$$

The first crossed-braiding input has degree g^{-1}, so the J^k on the right has ordered subscripts $(X, X^\vee)$, not $(X^\vee, X)$. The common target is $T_k(X \boxtimes X^\vee)$. Consequently, using $f_X = \mathrm{ev}_X\, c_{X^\vee, T_g X}$ and then the actual evaluation comparison (5.6), we obtain

$$\begin{aligned} \widetilde{f}_{k,X} &= j_k^{-1} T_k(\mathrm{ev}_X) J^k_{X, X^\vee} c_{T_k(X^\vee), T_k(T_g X)} \\ &= \mathrm{ev}_{T_k X}(\mathrm{id}_{T_k X} \boxtimes b_{k,X}) c_{T_k(X^\vee), T_k(T_g X)}. \end{aligned}$$

The second equality is precisely $\mathrm{ev}_{T_k X}(\mathrm{id}_{T_k X} \boxtimes b_{k,X}) = j_k^{-1} T_k(\mathrm{ev}_X) J^k_{X, X^\vee}$; its domain is $T_k X \boxtimes T_k(X^\vee)$, and $b_{k,X} : T_k(X^\vee) \to (T_k X)^\vee$ is not being identified with an identity map.

We next compare this transported pairing with the pairing defining $\mathfrak{u}_{T_k X}$. By the strict action identities, $T_{kgk^{-1}}(T_k X) = Z_k$. Naturality of the crossed braiding in its first input, applied to $b^{-1}_{k,X} : (T_k X)^\vee \to T_k(X^\vee)$, says

$$c_{T_k(X^\vee), Z_k}(b^{-1}_{k,X} \boxtimes \mathrm{id}_{Z_k}) = (\mathrm{id}_{T_k X} \boxtimes b^{-1}_{k,X}) c_{(T_k X)^\vee, Z_k}.$$

Both sides have source $(T_k X)^\vee \boxtimes Z_k$ and target $T_k X \boxtimes T_k(X^\vee)$. The first tensor factor in the target of the crossed braiding is $T_k X$, because $T_{kg^{-1}k^{-1}} Z_k = T_k X$. Hence the preceding identity and $b_{k,X} b^{-1}_{k,X} = \mathrm{id}_{(T_k X)^\vee}$ give the exact equality $f_{T_k X} = \widetilde{f}_{k,X}(b^{-1}_{k,X} \boxtimes \mathrm{id}_{Z_k})$. The comparison follows from first-variable naturality of the crossed braiding and the evaluation comparison.

It remains to compute the mate of $\widetilde{f}_{k,X}$ rather than merely asserting that mates commute with transport. By the defining right-dual mate identity for $\mathfrak{u}_X$, $f_X = \mathrm{ev}_{X^\vee}(\mathrm{id}_{X^\vee} \boxtimes \mathfrak{u}_X)$, with the fixed associators and unitors inserted. Applying T_k, using naturality of J^k for $\mathrm{id}_{X^\vee}$ and $\mathfrak{u}_X$, and then applying (5.6) to $X^\vee$, yields

$$\begin{aligned} \widetilde{f}_{k,X} &= j_k^{-1} T_k(\mathrm{ev}_{X^\vee}) T_k(\mathrm{id}_{X^\vee} \boxtimes \mathfrak{u}_X) J^k_{X^\vee, T_g X} \\ &= j_k^{-1} T_k(\mathrm{ev}_{X^\vee}) J^k_{X^\vee, X^{\vee\vee}}(\mathrm{id}_{T_k(X^\vee)} \boxtimes T_k(\mathfrak{u}_X)) \\ &= \mathrm{ev}_{T_k(X^\vee)}(\mathrm{id}_{T_k(X^\vee)} \boxtimes b_{k,X^\vee})(\mathrm{id}_{T_k(X^\vee)} \boxtimes T_k(\mathfrak{u}_X)) \\ &= \mathrm{ev}_{T_k(X^\vee)}(\mathrm{id}_{T_k(X^\vee)} \boxtimes b_{k,X^\vee} T_k(\mathfrak{u}_X)). \end{aligned}$$

In the second line the tensorator naturality identity is

$$T_k(\mathrm{id}_{X^\vee} \boxtimes \mathfrak{u}_X) J^k_{X^\vee, T_g X} = J^k_{X^\vee, X^{\vee\vee}}(\mathrm{id}_{T_k(X^\vee)} \boxtimes T_k(\mathfrak{u}_X)).$$

In the third line we used the evaluation comparison for $X^\vee$, namely

$$\mathrm{ev}_{T_k(X^\vee)}(\mathrm{id}_{T_k(X^\vee)} \boxtimes b_{k,X^\vee}) = j_k^{-1} T_k(\mathrm{ev}_{X^\vee}) J^k_{X^\vee, X^{\vee\vee}}.$$

Thus the right-dual mate of $\widetilde{f}_{k,X}$, for the right dual $(T_k(X^\vee))^\vee$ of $T_k(X^\vee)$, is exactly

$$\widetilde{u}_{k,X} := b_{k,X^\vee} T_k(\mathfrak{u}_X) : Z_k \longrightarrow (T_k(X^\vee))^\vee.$$

This computation accounts for both the action tensorator and the unit comparison, and uses no further action compositor.

For completeness, precomposing a pairing in its first variable has the following effect on its right-dual mate. If $a : A' \to A$, $p = \mathrm{ev}_A(\mathrm{id}_A \boxtimes v) : A \boxtimes Z \to \mathbf{1}$, and $p' = p(a \boxtimes \mathrm{id}_Z)$, then the mate of p' is $a^\vee v$. Indeed, the defining evaluation square for the right-dual morphism $a^\vee : A^\vee \to (A')^\vee$ is

$$\mathrm{ev}_{A'}(\mathrm{id}_{A'} \boxtimes a^\vee) = \mathrm{ev}_A(a \boxtimes \mathrm{id}_{A^\vee}) : A' \boxtimes A^\vee \longrightarrow \mathbf{1},$$

so the pairing itself can be rewritten as

$$\begin{aligned} p' &= \mathrm{ev}_A(a \boxtimes v) \\ &= \mathrm{ev}_A(a \boxtimes \mathrm{id}_{A^\vee})(\mathrm{id}_{A'} \boxtimes v) \\ &= \mathrm{ev}_{A'}(\mathrm{id}_{A'} \boxtimes a^\vee)(\mathrm{id}_{A'} \boxtimes v) \\ &= \mathrm{ev}_{A'}(\mathrm{id}_{A'} \boxtimes a^\vee v). \end{aligned}$$

This is exactly the mate identity; uniqueness of mates completes the argument. Apply this with $A = T_k(X^\vee)$, $A' = (T_kX)^\vee$, $a = b_{k,X}^{-1}$, and $v = b_{k,X^\vee} T_k(\mathfrak{u}_X)$. The mate of $f_{T_kX} = \widetilde{f}_{k,X}(b_{k,X}^{-1} \boxtimes \mathrm{id}_{Z_k})$ is therefore the following identity, with the original label retained:

$$\mathfrak{u}_{T_kX} = (b_{k,X}^{-1})^\vee b_{k,X^\vee} T_k(\mathfrak{u}_X) = \chi^{(2)}_{k,X} T_k(\mathfrak{u}_X). \tag{5.11}$$

The directions are

$$T_k(\mathfrak{u}_X) : Z_k \to T_k(X^{\vee\vee}), \qquad b_{k,X^\vee} : T_k(X^{\vee\vee}) \to (T_k(X^\vee))^\vee,$$
$$(b_{k,X}^{-1})^\vee : (T_k(X^\vee))^\vee \to (T_kX)^{\vee\vee}.$$

Thus $\chi^{(2)}_{k,X}$ has source $T_k(X^{\vee\vee})$ and target $(T_kX)^{\vee\vee}$, while the whole composite has source $Z_k = T_{kgk^{-1}}(T_kX) = T_k(T_gX)$ and target $(T_kX)^{\vee\vee}$, exactly as required. The only identifications of action objects used here are strict underlying identities; all monoidal comparison maps J^k, j_k, and $b_{k,X}$ remain displayed.

Finally, the crossed-twist covariance in Definition 2.2(i) states $\theta_{T_kX} = T_k(\theta_X)$ under the same strict action identification. Both maps have source T_kX and target Z_k. Consequently the already-proved (5.11) gives the claimed G-equivariance of δ directly:

$$\begin{aligned} \delta_{T_kX} &= \mathfrak{u}_{T_kX} \theta_{T_kX} \\ &= \chi^{(2)}_{k,X} T_k(\mathfrak{u}_X) T_k(\theta_X) \\ &= \chi^{(2)}_{k,X} T_k(\mathfrak{u}_X \theta_X) \\ &= \chi^{(2)}_{k,X} T_k(\delta_X). \end{aligned}$$

Every composite here has source T_kX and target $(T_kX)^{\vee\vee}$. Drinfeld G-equivariance, together with the covariance of θ established in Lemma 4.1, proves (5.9) using (5.11).

For the tensor comparison, fix homogeneous objects $X \in \mathcal{C}_g$ and $Y \in \mathcal{C}_h$, and put $W := X \boxtimes Y$, $D := W^\vee$, $q := ghg^{-1}$, and $\widehat{W} := T_{gh}W$. The two crossed commutors in the balancing formula have the types

$$c_{X,Y} : W \longrightarrow T_gY \boxtimes X, \qquad c_{T_gY,X} : T_gY \boxtimes X \longrightarrow T_qX \boxtimes T_gY,$$

so $B_{X,Y} := c_{T_gY,X} c_{X,Y} : W \longrightarrow T_qX \boxtimes T_gY$. Moreover,

$$T_q\theta_X : T_qX \longrightarrow T_qT_gX = T_{gh}X, \qquad T_g\theta_Y : T_gY \longrightarrow T_gT_hY = T_{gh}Y,$$

and therefore

$$A_{X,Y} := J^{gh}_{X,Y}(T_q\theta_X \boxtimes T_g\theta_Y) : T_qX \boxtimes T_gY \longrightarrow T_{gh}W.$$

Here the equalities $T_qT_g = T_{gh}$ and $T_gT_h = T_{gh}$ concern only the underlying action functors; the displayed $J^{gh}_{X,Y}$ is still required. Thus $A_{X,Y}B_{X,Y} : W \to \widehat{W}$ is exactly θ_W by (2.8). The argument uses the crossed-braiding, rigidity and monoidal-action identities stated in Section 2 and established in Xu's construction [7, Theorem 1.1, Proposition 8.5, Proposition 9.6, Proposition 10.3]; their concrete VOA realization is not reproved here.

We make the right-dual comparison used in the mate calculation explicit. Let $d_{X,Y} : D \xrightarrow{\sim} Y^\vee \boxtimes X^\vee$ be the unique right-dual isomorphism characterized by $\mathrm{ev}_W(\mathrm{id}_W \boxtimes d_{X,Y}^{-1}) = \widehat{\mathrm{ev}}_{X,Y}$, where $\widehat{\mathrm{ev}}_{X,Y} : W \boxtimes (Y^\vee \boxtimes X^\vee) \to \mathbf{1}$ is the composite which, with the indicated associators and

unitors, first applies $\mathrm{ev}_Y : Y \boxtimes Y^\vee \to \mathbf{1}$ and then $\mathrm{ev}_X : X \boxtimes X^\vee \to \mathbf{1}$. Explicitly, it is

$$\begin{aligned}\widehat{\mathrm{ev}}_{X,Y} := {} & \mathrm{ev}_X(\mathrm{id}_X \boxtimes l_{X^\vee})(\mathrm{id}_X \boxtimes(\mathrm{ev}_Y \boxtimes \mathrm{id}_{X^\vee})) \\ & \cdot (\mathrm{id}_X \boxtimes a^{-1}_{Y,Y^\vee,X^\vee}) a_{X,Y,Y^\vee \boxtimes X^\vee}.\end{aligned}$$

Similarly, $\widehat{\mathrm{ev}}_{Y^\vee,X^\vee}$ is the evaluation of $Y^\vee \boxtimes X^\vee$ against its right dual $X^{\vee\vee} \boxtimes Y^{\vee\vee}$, first using $\mathrm{ev}_{X^\vee}$ and then $\mathrm{ev}_{Y^\vee}$. With these choices the canonical monoidal comparison of the double-right-dual functor is

$$\mu_{X,Y} = d^\vee_{X,Y}\, d^{-1}_{Y^\vee,X^\vee} : X^{\vee\vee} \boxtimes Y^{\vee\vee} \longrightarrow W^{\vee\vee}.$$

In particular, no literal equality $D = Y^\vee \boxtimes X^\vee$ is used.

The precise identity to be established is the following, with the original label retained:

$$\mathfrak{u}_W A_{X,Y} B_{X,Y} = \mu_{X,Y}(\delta_X \boxtimes \delta_Y), \qquad \delta_Z = \mathfrak{u}_Z \theta_Z. \tag{5.12}$$

For a map $m : W \to W^{\vee\vee}$, write $m^\flat := \mathrm{ev}_{W^\vee}(\mathrm{id}_{W^\vee} \boxtimes m)$ for its right-dual mate. The two mates initially lie in $\mathrm{Hom}(D \boxtimes W, \mathbf{1})$:

$$\mathcal{L}_{X,Y} := \mathrm{ev}_W\, c_{D,\widehat{W}}(\mathrm{id}_D \boxtimes A_{X,Y} B_{X,Y}), \qquad \mathcal{R}_{X,Y} := \mathrm{ev}_{W^\vee}(\mathrm{id}_D \boxtimes \mu_{X,Y}(\delta_X \boxtimes \delta_Y)).$$

To compare them using the chosen tensor-dual model, precompose both with $d^{-1}_{X,Y} \boxtimes \mathrm{id}_W$. This gives two maps in the same Hom-space $\mathrm{Hom}((Y^\vee \boxtimes X^\vee) \boxtimes W, \mathbf{1})$:

$$\begin{aligned}\widetilde{\mathcal{L}}_{X,Y} &:= \mathcal{L}_{X,Y}(d^{-1}_{X,Y} \boxtimes \mathrm{id}_W) \\ &= \widehat{\mathrm{ev}}_{X,Y} c_{Y^\vee \boxtimes X^\vee, \widehat{W}}(\mathrm{id}_{Y^\vee \boxtimes X^\vee} \boxtimes A_{X,Y} B_{X,Y}), \\ \widetilde{\mathcal{R}}_{X,Y} &:= \mathcal{R}_{X,Y}(d^{-1}_{X,Y} \boxtimes \mathrm{id}_W) \\ &= \widehat{\mathrm{ev}}_{Y^\vee,X^\vee}(\mathrm{id}_{Y^\vee \boxtimes X^\vee} \boxtimes \delta_X \boxtimes \delta_Y).\end{aligned}$$

The first equality uses naturality of the crossed braiding in the first variable and the defining evaluation property of $d_{X,Y}$. The second uses $\mu_{X,Y} = d^\vee_{X,Y} d^{-1}_{Y^\vee,X^\vee}$ and naturality of right-dual transposition. Here $\delta_Z = \mathfrak{u}_Z \theta_Z$. By the defining mate identity (5.4), $\mathrm{ev}_{X^\vee}(\mathrm{id}_{X^\vee} \boxtimes \delta_X) = \mathrm{ev}_X\, c_{X^\vee, T_g X}(\mathrm{id}_{X^\vee} \boxtimes \theta_X)$, and the analogous formula holds for Y. Consequently $\widetilde{\mathcal{R}}_{X,Y}$ is the iterated contraction of the X-pair and then the Y-pair, with no permutation of the right-dual factors.

For precision, the crossed-hexagon substitutions needed to compare $\widetilde{\mathcal{L}}_{X,Y}$ with this contraction are as follows. The second crossed hexagon, (2.7), with $(A, B, Z) = (Y^\vee, X^\vee, \widehat{W})$ and degrees (h^{-1}, g^{-1}, gh), gives

$$\begin{aligned}c_{Y^\vee \boxtimes X^\vee, \widehat{W}} = {} & a_{W,Y^\vee,X^\vee}(c_{Y^\vee, T_h W} \boxtimes \mathrm{id}_{X^\vee}) a^{-1}_{Y^\vee, T_h W, X^\vee} \\ & \cdot (\mathrm{id}_{Y^\vee} \boxtimes c_{X^\vee, \widehat{W}}) a_{Y^\vee, X^\vee, \widehat{W}}.\end{aligned}$$

Indeed, $T_{g^{-1}} \widehat{W} = T_h W$ and $T_{h^{-1} g^{-1}} \widehat{W} = W$; this is the ordered product $(gh)^{-1} = h^{-1} g^{-1}$, not its reversed product. To split the second inputs of the two resulting crossings, use naturality with $J^{gh}_{X,Y}$ and $J^h_{X,Y}$, and apply (2.6) with $(A, U, V) = (X^\vee, T_{gh} X, T_{gh} Y), (A, U, V) = (Y^\vee, T_h X, T_h Y)$. For either substitution, the explicit form of the first crossed hexagon is

$$\begin{aligned}c_{A,U \boxtimes V}\, a_{A,U,V} = {} & (J^{a_0}_{U,V} \boxtimes \mathrm{id}_A) a^{-1}_{T_{a_0} U, T_{a_0} V, A}(\mathrm{id}_{T_{a_0} U} \boxtimes c_{A,V}) \\ & \cdot a_{T_{a_0} U, A, V}(c_{A,U} \boxtimes \mathrm{id}_V),\end{aligned}$$

where $a_0 = \deg(A)$ (so $a_0 = g^{-1}$ in the first substitution and $a_0 = h^{-1}$ in the second). The required object identities are

$$T_{g^{-1}} T_{gh} X = T_h X, \quad T_{g^{-1}} T_{gh} Y = T_h Y, \quad T_{h^{-1}} T_h X = X, \quad T_{h^{-1}} T_h Y = Y.$$

The corresponding tensorator strings are fixed, not suppressed, by the action-coherence equations

$$\begin{aligned}T_{g^{-1}}(J^{gh}_{X,Y}) J^{g^{-1}}_{T_{gh} X, T_{gh} Y} &= J^h_{X,Y}, \\ T_{h^{-1}}(J^h_{X,Y}) J^{h^{-1}}_{T_h X, T_h Y} &= J^1_{X,Y} = \mathrm{id}_{X \boxtimes Y}.\end{aligned}$$

Whenever an action is moved across an evaluation cap, the actual comparison is (5.6), $\mathrm{ev}_{T_kZ}(\mathrm{id}_{T_kZ}\boxtimes b_{k,Z}) = j_k^{-1}T_k(\mathrm{ev}_Z)J^k_{Z,Z^\vee}$, with $(k,Z) = (g,X),(h,Y),(gh,W)$ as appropriate. The unit maps j_k^{-1} are retained; their composites are governed by $j_{kl} = T_k(j_l)j_k$ and $j_1 = \mathrm{id}$. The duality comparison also has to respect the double-dual tensorator. In the present notation the required equation, with all endpoints displayed, is

$$\begin{aligned}(J^k_{X,Y})^{\vee\vee}\mu_{T_kX,T_kY}(\chi^{(2)}_{k,X}\boxtimes\chi^{(2)}_{k,Y})\\ = \chi^{(2)}_{k,W}T_k(\mu_{X,Y})J^k_{X^{\vee\vee},Y^{\vee\vee}}.\end{aligned}$$

Both sides have source $T_k(X^{\vee\vee})\boxtimes T_k(Y^{\vee\vee})$ and target $(T_kW)^{\vee\vee}$. To verify the equation, take right-dual mates: on the left use the evaluation of $T_kX\boxtimes T_kY$ transported by $(J^k_{X,Y})^{\vee\vee}$; on the right use the evaluation of T_kW. Applying (5.6) to X, Y and W, and then the defining evaluation property of $\mu_{X,Y}$, identifies both mate pairings with the same $j_k^{-1}T_k(\widehat{\mathrm{ev}}_{X,Y})$ after the monoidal comparison J^k. The right-dual mate correspondence is injective, so the displayed equation follows. In particular, this is a comparison of evaluation morphisms, not an identification of $b_{k,Z}$ or J^k with an identity.

We finish the mate comparison by passing to the equivariantization. This argument avoids identifying two six-crossing expressions merely from their endpoints. We use the braided structure on $\mathcal{E} := \mathcal{C}^G$ of [17, Theorem 8.24.3 and (8.108)]; throughout this paragraph G is finite. The construction of $\mathcal{E}$ requires a monoidal G-action and a crossed braiding, but neither a pivotal structure nor a twist. The forgetful functor $U : \mathcal{E}\to\mathcal{C}$ is faithful and strict monoidal on the chosen underlying tensor products. Right duals in $\mathcal{E}$ may be chosen with underlying objects, evaluations and coevaluations equal to those in $\mathcal{C}$. More precisely, the equivariant action on $M^\vee$ is obtained from that on M by the explicit composite $T_k(M^\vee)\xrightarrow{b_{k,M}}(T_kM)^\vee\xrightarrow{(r_k^{-1})^\vee}M^\vee$; its compatibility with evaluation is exactly (5.6). Hence the double-dual monoidal comparison is carried by U to $\mu_{M,N}$, with the canonical duality identifications rather than literal identifications of arbitrarily chosen right duals.

For an equivariant object (M,r) write $r_k : T_kM\xrightarrow{\sim}M$, so that $r_{kl} = r_kT_k(r_l)$ under the strict underlying action. The monoidal equivariant structure on $M\boxtimes N$ is $r_k^{M\boxtimes N} = (r_k^M\boxtimes r_k^N)(J^k_{M,N})^{-1}$. For $X = M_g$, let $r_{g,X} : T_gX\to X$ denote the restriction of r_g; its target is M_g, since $T_g\mathcal{C}_g\subset\mathcal{C}_g$. Define an automorphism of M by its homogeneous components, $\widetilde{\theta}_M|_{M_g} := r_{g,X}\theta_X : X\longrightarrow X$. It is a natural automorphism of the identity functor on $\mathcal{E}$. Indeed, if $X = M_g$ and $q = kgk^{-1}$, the crossed-twist covariance, its naturality with respect to $r_k : T_kX\to M_q$, and the equivariant cocycle relation give

$$\begin{aligned}\widetilde{\theta}_M\, r_k|_{T_kX} &= r_{q,M}\,\theta_{M_q}\,r_k|_{T_kX}\\ &= r_{q,M}\,T_q(r_k)\,\theta_{T_kX}\\ &= r_{qk,M}\,T_k(\theta_X) = r_k\,T_k(\widetilde{\theta}_M|_X).\end{aligned}$$

Here $qk = kg$ and $\theta_{T_kX} = T_k(\theta_X)$; no monoidal tensorator was suppressed in this one-object calculation.

We check the ordinary balancing identity rather than assume it. For $(M,r),(N,s)\in\mathcal{E}$, take $X = M_g$, $Y = N_h$, and put $q = ghg^{-1}$. By [17, (8.108)], the ordinary braiding $\widetilde{c}$ on $\mathcal{E}$ has restriction $\widetilde{c}_{M,N}|_{X\boxtimes Y} = (s_g|_{T_gY}\boxtimes\mathrm{id}_X)c_{X,Y}$. Its second application, using naturality of the crossed braiding in the first variable, is the correctly ordered composite

$$(\widetilde{c}_{N,M}\widetilde{c}_{M,N})|_{X\boxtimes Y} = (r_q|_{T_qX}\boxtimes s_g|_{T_gY})c_{T_gY,X}c_{X,Y}.$$

On the other hand, the tensor equivariant structure and (2.8) yield

$$\begin{aligned}\widetilde{\theta}_{M\boxtimes N}|_{X\boxtimes Y} &= r_{gh}^{M\boxtimes N}\,\theta_{X\boxtimes Y}\\ &= (r_{gh,M}\boxtimes s_{gh,N})(T_q\theta_X\boxtimes T_g\theta_Y)c_{T_gY,X}c_{X,Y}.\end{aligned}$$

The displayed $J^{gh}_{X,Y}$ in $\theta_{X\boxtimes Y}$ cancels only against $(J^{gh}_{M,N})^{-1}$ in the actual tensor equivariant structure, by naturality of J^{gh}; the individual tensorators are not identities. The preceding

equivariance computation, applied respectively to $r_q : T_qX \to M_{qgq^{-1}}$ and $s_g : T_gY \to N_q$, gives

$$\widetilde{\theta}_M\, r_q|_{T_qX} = r_{qg,M} T_q\theta_X = r_{gh,M} T_q\theta_X, \qquad \widetilde{\theta}_N\, s_g|_{T_gY} = s_{gh,N} T_g\theta_Y.$$

Consequently, on every $X \boxtimes Y$ and hence on $M \boxtimes N$,

$$\widetilde{\theta}_{M\boxtimes N} = (\widetilde{\theta}_M \boxtimes \widetilde{\theta}_N)\widetilde{c}_{N,M}\widetilde{c}_{M,N}.$$

The crossed unit condition gives $\widetilde{\theta}_{\mathbf{1}} = \mathrm{id}_{\mathbf{1}}$. Thus $\widetilde{\theta}$ is a twist in the precise sense of [17, Definition 8.10.1, (8.32)]; the alternative order $\widetilde{c}_{N,M}\widetilde{c}_{M,N}(\widetilde{\theta}_M \boxtimes \widetilde{\theta}_N)$ agrees by naturality.

Let $\widetilde{\mathfrak{u}}_M : M \to M^{\vee\vee}$ denote the ordinary Drinfeld isomorphism associated with $\widetilde{c}$ and our chosen *right* duals. On $X = M_g$ its right-dual mate is $\mathrm{ev}_X(r_{g^{-1},M}|_{T_{g^{-1}}X} \boxtimes \mathrm{id}_{X^\vee})c_{X^\vee,X}$. The equivariant cocycle relation implies $r_{g^{-1},M}|_{T_{g^{-1}}X} = T_{g^{-1}}(r_{g,X}^{-1})$; hence crossed-braiding naturality identifies this pairing with $\mathrm{ev}_X\, c_{X^\vee,T_gX}(\mathrm{id}_{X^\vee} \boxtimes r_{g,X}^{-1})$. By the defining mate of $\mathfrak{u}_X$ and its uniqueness,

$$(*) \qquad U(\widetilde{\mathfrak{u}}_M)|_X = \mathfrak{u}_X r_{g,X}^{-1}, \qquad U(\widetilde{\mathfrak{u}}_M\widetilde{\theta}_M)|_X = \mathfrak{u}_X\theta_X = \delta_X.$$

The componentwise identity holds also for direct sums, and is compatible with the chosen double-dual identifications.

For clarity about the ordinary categorical input, apply [17, Proposition 8.9.3, (8.31)] to the *reversed tensor convention*: a right dual in $\mathcal{E}$ is a left dual in its reversed monoidal category, whose braiding in the reversed order is $\widetilde{c}_{N,M}$. With the tensor factors returned to our order, the proposition is the typed identity

$$\widetilde{\mathfrak{u}}_{M\boxtimes N}\widetilde{c}_{N,M}\widetilde{c}_{M,N} = \mu_{M,N}(\widetilde{\mathfrak{u}}_M \boxtimes \widetilde{\mathfrak{u}}_N).$$

This equation involves the canonical double-right-dual comparison $\mu_{M,N}$. Combining it with the just-proved balancing formula and the naturality of the twist yields, by actual composition and cancellation of the invertible double braiding,

$$(**) \qquad \widetilde{\mathfrak{u}}_{M\boxtimes N}\widetilde{\theta}_{M\boxtimes N} = \mu_{M,N}(\widetilde{\mathfrak{u}}_M\widetilde{\theta}_M \boxtimes \widetilde{\mathfrak{u}}_N\widetilde{\theta}_N).$$

This is the ordinary Drinfeld–twist tensor identity, obtained from (8.31) and (8.32).

It remains to transfer (**) to arbitrary homogeneous objects of $\mathcal{C}$, not just those carrying an equivariant structure. For $X \in \mathcal{C}$, the induction object

$$\mathrm{Ind}\, X := \bigoplus_{k\in G} T_kX$$

has the equivariant structure which reindexes $T_lT_kX = T_{lk}X$ by the strict underlying action. Its underlying object contains X as the $k = 1$ direct summand. Let $i_X : X \to U(\mathrm{Ind}\, X)$ and $p_X : U(\mathrm{Ind}\, X) \to X$ be the canonical inclusion and projection, with $p_Xi_X = \mathrm{id}_X$. The maps $\delta_Z = \mathfrak{u}_Z\theta_Z$ are natural on each homogeneous component by the previously established naturality of $\mathfrak{u}$ and θ; on general finite G-graded objects they extend by direct sum. Therefore the identities (*) and (**) for $(\mathrm{Ind}\, X, \mathrm{Ind}\, Y)$, composed with $i_X \boxtimes i_Y$ and with the corresponding double-dual comparison, give by naturality $\delta_{X\boxtimes Y} = \mu_{X,Y}(\delta_X \boxtimes \delta_Y)$. Explicitly, the equality after precomposition with $i_X \boxtimes i_Y$ is detected by the split projection $p_X \boxtimes p_Y$, and the naturality of μ and δ identifies its restriction with the displayed equality on $X \boxtimes Y$. The split inclusion and projection make the argument valid for arbitrary X, Y in $\mathcal{C}$. Finally, by the already established (2.8), $\theta_W = A_{X,Y}B_{X,Y}$, and hence $\mathfrak{u}_WA_{X,Y}B_{X,Y} = \delta_W = \mu_{X,Y}(\delta_X \boxtimes \delta_Y)$. This proves (5.12) independently of (5.8); the latter is its immediate consequence. Applying the injective right-dual mate correspondence and precomposing with $d_{X,Y}^{-1} \boxtimes \mathrm{id}_W$ also proves the previously displayed transported-mate equality $\widetilde{\mathcal{L}}_{X,Y} = \widetilde{\mathcal{R}}_{X,Y}$. Thus the original six-crossing comparison follows from the equivariantization construction and the ordinary braided Drinfeld identity, using the crossed-braiding and monoidal-action data of [7, Theorem 1.1, Proposition 8.5, Proposition 9.6].

Finally, fix $X \in \mathcal{C}_g$. The two maps in (5.10) have the same source and target. Indeed, $X^\vee \in \mathcal{C}_{g^{-1}}$, and

$$\delta_{X^\vee} = \mathfrak{u}_{X^\vee}\theta_{X^\vee} : X^\vee \longrightarrow X^{\vee\vee\vee}, \qquad (\delta_X^{-1})^\vee : X^\vee \longrightarrow X^{\vee\vee\vee}.$$

More explicitly, $\theta_X^{-1} : T_g X \to X$ and $\mathfrak{u}_X^{-1} : X^{\vee\vee} \to T_g X$, so the contravariance of the chosen right-dual functor gives

$$(\delta_X^{-1})^\vee = (\mathfrak{u}_X^{-1})^\vee (\theta_X^{-1})^\vee, \quad X^\vee \xrightarrow{(\theta_X^{-1})^\vee} (T_g X)^\vee \xrightarrow{(\mathfrak{u}_X^{-1})^\vee} X^{\vee\vee\vee}.$$

The categorical dual-twist identity (2.9) is an equality in $\mathrm{Hom}(X^\vee, T_{g^{-1}} X^\vee)$, with the action–dual comparison $b_{g,X}$ retained. It concerns the twist itself and is not an identification of $\theta_X^\vee$ with $(\theta_X^{-1})^\vee$. The required comparison of the complete maps δ follows from the already established monoidality, by the uniqueness of right-dual mates.

Write $D^2 Z = Z^{\vee\vee}$ and let $\nu : \mathbf{1} \xrightarrow{\sim} D^2\mathbf{1}$ be the canonical unit comparison of the double-right-dual functor. Its monoidal structure is the canonical $\mu_{U,V}$ occurring in (5.8). With the fixed right-dual representatives and their canonical comparison isomorphisms, the defining evaluation-preservation identity for this strong monoidal functor is

$$\mathrm{ev}_{X^{\vee\vee}} = \nu^{-1} (\mathrm{ev}_X)^{\vee\vee} \mu_{X,X^\vee} : X^{\vee\vee} \boxtimes X^{\vee\vee\vee} \longrightarrow \mathbf{1}.$$

Here $D^2(X^\vee) = X^{\vee\vee\vee}$ is the chosen right dual of $D^2 X = X^{\vee\vee}$. This evaluation identity is part of the canonical monoidal structure on D^2, not a consequence of the desired pivotal-duality formula; it follows directly by transporting ev_X through the two right-dual adjunctions. The associators, unitors and canonical identifications of the iterated right duals are understood in this identity.

Naturality of δ for the evaluation morphism $\mathrm{ev}_X : X \boxtimes X^\vee \to \mathbf{1}$, together with $\delta_{\mathbf{1}} = \nu$ and the already proved (5.8), gives two morphisms $X \boxtimes X^\vee \to \mathbf{1}$ which are equal:

$$\begin{aligned} \mathrm{ev}_X &= \nu^{-1} (\mathrm{ev}_X)^{\vee\vee} \delta_{X \boxtimes X^\vee} \\ &= \nu^{-1} (\mathrm{ev}_X)^{\vee\vee} \mu_{X,X^\vee} (\delta_X \boxtimes \delta_{X^\vee}) \\ &= \mathrm{ev}_{X^{\vee\vee}} (\delta_X \boxtimes \delta_{X^\vee}). \end{aligned}$$

On the other hand, right-dual naturality for the inverse isomorphism $\delta_X^{-1} : X^{\vee\vee} \to X$ states precisely that

$$\mathrm{ev}_{X^{\vee\vee}} (\mathrm{id}_{X^{\vee\vee}} \boxtimes (\delta_X^{-1})^\vee) = \mathrm{ev}_X (\delta_X^{-1} \boxtimes \mathrm{id}_{X^\vee}) : X^{\vee\vee} \boxtimes X^\vee \longrightarrow \mathbf{1}.$$

Precomposing with $\delta_X \boxtimes \mathrm{id}_{X^\vee}$ therefore yields $\mathrm{ev}_X = \mathrm{ev}_{X^{\vee\vee}} (\delta_X \boxtimes (\delta_X^{-1})^\vee)$. Consequently the two maps $\delta_{X^\vee}, (\delta_X^{-1})^\vee : X^\vee \to X^{\vee\vee\vee}$ have identical right-dual pairings after precomposition with the isomorphism δ_X. Since $X^{\vee\vee\vee}$ is the right dual of $X^{\vee\vee}$, the right-dual mate correspondence is injective. Cancelling δ_X in this correspondence gives $\delta_{X^\vee} = (\delta_X^{-1})^\vee$, as required. This argument applies to every homogeneous object, including zero and finite direct sums, and uses neither a prior pivotal-duality assertion nor any ribbon or modularity theorem. It is the evaluation proof of the general duality property of a monoidal pivotal transformation [17, Definition 4.7.7 and Exercise 4.7.9]. The relation between the ordinary ribbon twist and equality of left and right traces is given in [17, Proposition 8.10.12]. □

5.4. **Verification of the crossed ribbon axiom.**

Theorem 5.5. *Let V be strongly rational and $G \leq \mathrm{Aut}(V)$ finite, and let $\mathcal{C}_G$ be Xu's faithfully G-graded G-crossed braided fusion category* [7, Theorem 1.1]*, with action and crossed-commutor normalizations* [7, Proposition 8.5, Proposition 9.6] *and the pairing and rigidity data* [7, Lemma 10.1, Proposition 10.3]*. Then the conformal twist equips $\mathcal{C}_G$ with a G-crossed ribbon fusion structure in Kirillov's sense.*

Proof. We compare the structure maps with the four conditions in [15, Lemma 2.3], using its definition of a G-equivariant fusion category in Section 2. The underlying faithfully graded rigid fusion category, strong monoidal G-action, and crossed braiding are provided by Xu's construction [7, Theorem 1.1, Proposition 8.5, Proposition 9.6, Proposition 10.3]. The action has strict underlying functorial multiplication $T_k T_l = T_{kl}$ and $T_1 = \mathrm{id}$, but its tensor and unit comparisons $J^k_{A,B} : T_k A \boxtimes T_k B \to T_k(A \boxtimes B)$ and $j_k : \mathbf{1} \to T_k \mathbf{1}$ are retained throughout.

For the unit condition, the vacuum module $V = \mathbf{1}$ lies in $\mathcal{C}_1$, and Lemma 4.1 gives $\theta_{\mathbf{1}} = e^{2\pi i L_V(0)} = \mathrm{id}_{\mathbf{1}} = j_1$. In particular, this uses $j_1 = \mathrm{id}$, not an identification of j_k with the identity for arbitrary k. For $X \in \mathcal{C}_g$, $\theta_X : X \to T_g X$ is an invertible natural morphism by that lemma. If $k \in G$, put $g' = kgk^{-1}$. Both

$$\theta_{T_k X} : T_k X \longrightarrow T_{g'} T_k X, \qquad T_k(\theta_X) : T_k X \longrightarrow T_k T_g X$$

have target $T_{kg}X$, since $g'k = kg$, and they are equal by Lemma 4.1. This establishes the action covariance condition without discarding any monoidal structure of T_k.

For balancing, let $Y \in \mathcal{C}_h$ and $q = ghg^{-1}$. The ordered crossed braidings have types

$$c_{X,Y} : X \boxtimes Y \longrightarrow T_g Y \boxtimes X,$$
$$c_{T_g Y, X} : T_g Y \boxtimes X \longrightarrow T_q X \boxtimes T_g Y.$$

Here $T_g Y \in \mathcal{C}_q$, and the second target has degree $qg = gh$. The two twisted input morphisms are $T_q(\theta_X) : T_q X \to T_q T_g X = T_{gh} X$ and $T_g(\theta_Y) : T_g Y \to T_g T_h Y = T_{gh} Y$. Thus the following is an equality in $\mathrm{Hom}(X \boxtimes Y, T_{gh}(X \boxtimes Y))$ by Theorem 4.3:

$$\theta_{X \boxtimes Y} = J^{gh}_{X,Y}\big(T_q(\theta_X) \boxtimes T_g(\theta_Y)\big) c_{T_g Y, X} c_{X,Y}.$$

This is Kirillov's second condition: his $R_{gY,X} R_{X,Y}$ is the indicated *ordered* pair of crossed braidings, and the omitted canonical identification from the separate output twists to the twist of $X \boxtimes Y$ is precisely the strong-monoidal comparison $J^{gh}_{X,Y}$ together with $T_q T_g = T_{gh} = T_g T_h$. The group factors remain in their prescribed order, and both arrows are the crossed braidings displayed above.

The right-dual condition has the same content as Kirillov's third condition after restoring its suppressed action–dual comparison. Indeed, for $X \in \mathcal{C}_g$, its chosen right dual lies in $\mathcal{C}_{g^{-1}}$, and Lemma 5.3 supplies the natural isomorphism $b_{g,X} : T_g(X^\vee) \xrightarrow{\sim} (T_g X)^\vee$. Consequently the two factors in $\theta_{X^\vee} = T_{g^{-1}}(\theta_X^\vee)\, T_{g^{-1}}(b_{g,X})$ have respective types

$$T_{g^{-1}}(b_{g,X}) : X^\vee = T_{g^{-1}} T_g(X^\vee) \longrightarrow T_{g^{-1}}((T_g X)^\vee),$$
$$T_{g^{-1}}(\theta_X^\vee) : T_{g^{-1}}((T_g X)^\vee) \longrightarrow T_{g^{-1}}(X^\vee).$$

Thus both sides lie in $\mathrm{Hom}(X^\vee, T_{g^{-1}} X^\vee)$; the equality is (2.9), proved in Proposition 5.1. Under the canonical identification $(T_g X)^\vee \simeq T_g(X^\vee)$, this is exactly $\theta_{X^\vee} = T_{g^{-1}}(\theta_X^\vee)$ in [15, Lemma 2.3], with the contravariant dual $\theta_X^\vee : (T_g X)^\vee \to X^\vee$ in its correct direction.

There is also the pivotal datum required by Kirillov's definition of a G-equivariant fusion category. With the same chosen right duals, Proposition 5.4 constructs $\delta_X = \mathfrak{u}_X \theta_X : X \xrightarrow{\sim} X^{\vee\vee}$. It proves naturality and the unit normalization, together with

$$\delta_{X \boxtimes Y} = \mu_{X,Y}(\delta_X \boxtimes \delta_Y),$$
$$\delta_{T_k X} = \chi^{(2)}_{k,X} T_k(\delta_X),$$
$$\delta_{X^\vee} = (\delta_X^{-1})^\vee.$$

Here $\mu_{X,Y}$ is the canonical monoidal structure of the double right dual, while $\chi^{(2)}_{k,X} = (b^{-1}_{k,X})^\vee b_{k,X^\vee}$ is the action–double-dual comparison; neither is being treated as an identity map. These identities give the monoidal, equivariant and dual-compatible double-dual structure required in [15, Section 2 and Lemma 2.3]. The crossed twist identities above are established independently of any assertion about the ribbon structure of the equivariantization.

All these identities are natural and additive in homogeneous objects. On an arbitrary object $M = \bigoplus_g M_g$ of $\mathcal{C}_G$, the crossed twist is the componentwise family $\bigoplus_g \theta_{M_g} : \bigoplus_g M_g \to \bigoplus_g T_g M_g$; it is not identified with the action of a single group element on M. Since tensor product, duality, the action and crossed braidings preserve finite direct sums, the conditions just checked extend componentwise to all objects. This verifies Kirillov's four twist conditions and the required pivotal compatibility, proving the assertion. □

6. Modularity and equivariantization

6.1. Equivariantization and nondegeneracy.

Definition 6.1. The *equivariantization* $\mathcal{B} = \mathcal{C}_G^G$ has objects $A = (\underline{A}, (u_k^A)_{k\in G})$, where $u_k^A : T_k\underline{A} \xrightarrow{\sim} \underline{A}$, $u_1^A = \mathrm{id}$, and $u_{kl}^A = u_k^A T_k(u_l^A)$. A morphism $f : A \to B$ is a morphism $f : \underline{A} \to \underline{B}$ satisfying $fu_k^A = u_k^B T_k(f)$ for all $k \in G$. Writing $\underline{A} = \bigoplus_{g\in G} A_g$, the grading and the action imply $u_k^A : T_k A_g \xrightarrow{\sim} A_{kgk^{-1}}$ on each homogeneous summand. In particular, the restriction $u_g^A : T_g A_g \to A_g$ is defined, and no equivariant object is assumed homogeneous.

The tensor product has underlying object $\underline{A} \boxtimes \underline{B}$ and equivariant structure

$$u_k^{A\boxtimes B} = (u_k^A \boxtimes u_k^B)(J^k_{\underline{A},\underline{B}})^{-1} : T_k(\underline{A} \boxtimes \underline{B}) \longrightarrow \underline{A} \boxtimes \underline{B}.$$

Indeed, the action coherence (2.4), naturality of J^k, and the two cocycle identities give

$$\begin{aligned} u_k^{A\boxtimes B} T_k(u_l^{A\boxtimes B}) &= (u_k^A \boxtimes u_k^B)(T_k(u_l^A) \boxtimes T_k(u_l^B))(J^{kl}_{\underline{A},\underline{B}})^{-1} \\ &= u_{kl}^{A\boxtimes B}. \end{aligned}$$

The identity at $k = 1$ follows from $J^1 = \mathrm{id}$. The tensor unit is $(\mathbf{1}, (j_k^{-1})_k)$; its cocycle is $j_{kl}^{-1} = j_k^{-1} T_k(j_l^{-1})$. The monoidal action axioms for J^k, j_k make the original associator and unitors equivariant morphisms.

For homogeneous $X = A_g$ and $Y = B_h$, put $q = ghg^{-1}$. The braiding is defined componentwise by

$$\widetilde{c}_{A,B}|_{X\boxtimes Y} = (u_g^B|_{T_g Y} \boxtimes \mathrm{id}_X) c_{X,Y} : X \boxtimes Y \longrightarrow B_q \boxtimes X \subseteq \underline{B} \boxtimes \underline{A}.$$

For a nonhomogeneous second argument it is the direct sum of these arrows; the target is not a fixed $Y \boxtimes X$ unless the grading happens to be preserved by conjugation. The formula is extended by finite direct sums. Naturality, invertibility, equivariance and both ordinary hexagons follow from the crossed-braiding and action axioms (2.5)–(2.7) with their unit constraints and J^k retained; this is the braided equivariantization construction [16, Definition 4.41][17, Theorem 8.24.3, (8.108)].

Choose the right dual $M^\vee$ of $M = \underline{A}$, with evaluation $\mathrm{ev}_M : M \boxtimes M^\vee \to \mathbf{1}$ and coevaluation $\mathrm{coev}_M : \mathbf{1} \to M^\vee \boxtimes M$. Its equivariant action is

$$\widehat{u}_k = ((u_k^A)^{-1})^\vee b_{k,M} : T_k(M^\vee) \xrightarrow{\sim} M^\vee.$$

Here $b_{k,M} : T_k(M^\vee) \to (T_k M)^\vee$ is the canonical evaluation comparison of Lemma 5.3; the displayed composite uses $((u_k^A)^{-1})^\vee : (T_k M)^\vee \to M^\vee$. Its cocycle follows from the contravariant naturality of b and $b_{kl,M} = b_{k,T_l M} T_k(b_{l,M})$:

$$\begin{aligned} \widehat{u}_k T_k(\widehat{u}_l) &= ((u_k^A)^{-1})^\vee b_{k,M} T_k(((u_l^A)^{-1})^\vee) T_k(b_{l,M}) \\ &= ((u_k^A)^{-1})^\vee (T_k((u_l^A)^{-1}))^\vee b_{kl,M} = \widehat{u}_{kl}. \end{aligned}$$

Also $\widehat{u}_1 = \mathrm{id}$. More precisely, the defining evaluation comparison and right-dual naturality yield

$$\begin{aligned} &\mathrm{ev}_M(u_k^A \boxtimes \widehat{u}_k) \\ &\quad = \mathrm{ev}_{T_k M}(\mathrm{id}_{T_k M} \boxtimes b_{k,M}) = j_k^{-1} T_k(\mathrm{ev}_M) J^k_{M,M^\vee}. \end{aligned}$$

Thus ev_M is equivariant, since the unit action is j_k^{-1}. The corresponding transported coevaluation satisfies $(\widehat{u}_k \boxtimes u_k^A)(J^k_{M^\vee,M})^{-1} T_k(\mathrm{coev}_M) j_k = \mathrm{coev}_M$. To see this identity, apply the two right-dual triangle identities to the transported evaluation above: both sides are coevaluations for the same chosen evaluation and hence coincide by right-dual uniqueness. Consequently evaluation, coevaluation and both triangle identities are morphisms in $\mathcal{B}$. These constructions identify $\mathcal{B}$ with Kirillov's orbifold category $\mathcal{C}/G$ in [15, Definition 3.1 and Theorem 3.9], with the monoidal and dual comparisons suppressed in that source made explicit here.

For $A \in \mathcal{B}$ with homogeneous decomposition $\underline{A} = \bigoplus_{g\in G} A_g$, define

$$\Theta_A = \bigoplus_{g\in G} (u_g^A|_{T_g A_g})\, \theta_{A_g}. \tag{6.1}$$

Because g fixes its own conjugacy class element, each summand in (6.1) is an automorphism of A_g. These automorphisms are natural in equivariant morphisms. For $X = A_g$, $k \in G$, and $q = kgk^{-1}$, the twist covariance, twist naturality for u_k^A, and the cocycle give, as maps $T_k X \to A_q$,

$$\begin{aligned}\Theta_A u_k^A|_{T_k X} &= u_q^A \theta_{A_q} u_k^A = u_q^A T_q(u_k^A)\theta_{T_k X}\\ &= u_{qk}^A T_k(\theta_X) = u_k^A T_k(u_g^A \theta_X) = u_k^A T_k(\Theta_A|_X).\end{aligned}$$

Hence Θ_A is an equivariant natural isomorphism. Its unit value is the identity. For $X = A_g$, $Y = B_h$, put $q = ghg^{-1}$. The crossed balancing identity (2.8) and the inverse tensorator in the equivariant tensor product give

$$\begin{aligned}\Theta_{A\boxtimes B}|_{X\boxtimes Y} &= (u_{gh}^A \boxtimes u_{gh}^B)(J_{X,Y}^{gh})^{-1}\theta_{X\boxtimes Y}\\ &= (u_{gh}^A T_q\theta_X \boxtimes u_{gh}^B T_g\theta_Y) c_{T_g Y,X} c_{X,Y}.\end{aligned}$$

Thus $(J_{X,Y}^{gh})^{-1}$ and $J_{X,Y}^{gh}$ cancel only after their domains are correctly matched. On the other hand, using the degrees g, q of the successive first inputs of the two ordinary braidings and crossed-braiding naturality for $u_g^B : T_g Y \to B_q$, we obtain

$$\begin{aligned}&\widetilde{c}_{B,A}\widetilde{c}_{A,B}|_{X\boxtimes Y}\\ &\quad= (u_q^A \boxtimes \mathrm{id}_{B_q}) c_{B_q,X}(u_g^B \boxtimes \mathrm{id}_X) c_{X,Y}\\ &\quad= (u_q^A \boxtimes u_g^B) c_{T_g Y,X} c_{X,Y}.\end{aligned}$$

The already established equivariance of Θ, together with $u_q^A T_q(u_g^A) = u_{qg}^A$ and $u_g^B T_g(u_h^B) = u_{gh}^B$, gives the following identities with the indicated sources and targets:

$$\Theta_A u_q^A|_{T_q X} = u_{gh}^A T_q\theta_X, \qquad \Theta_B u_g^B|_{T_g Y} = u_{gh}^B T_g\theta_Y,$$

where $qg = gh$. Substituting these two identities in the previous crossing computation yields $\Theta_{A\boxtimes B} = (\Theta_A \boxtimes \Theta_B)\widetilde{c}_{B,A}\widetilde{c}_{A,B}$. The proof uses neither a prior ribbon structure on $\mathcal{B}$ nor any nondegeneracy statement.

It remains to check the right-dual twist identity independently. Let $X = A_g$ and write $u_g^X : T_g X \to X$ and $u_{g^{-1}}^X : T_{g^{-1}} X \to X$ for the corresponding restrictions of the equivariant structure. The right-dual summand $X^\vee$ has degree g^{-1}. By the crossed dual-twist identity (2.9), contravariant naturality of b, and its group coherence, the following are equal morphisms $X^\vee \to X^\vee$:

$$\begin{aligned}\Theta_{A^\vee}|_{X^\vee} &= \widehat{u}_{g^{-1}}\theta_{X^\vee}\\ &= ((u_{g^{-1}}^X)^{-1})^\vee b_{g^{-1},X} T_{g^{-1}}(\theta_X^\vee) T_{g^{-1}}(b_{g,X})\\ &= ((u_{g^{-1}}^X)^{-1})^\vee (T_{g^{-1}}\theta_X)^\vee b_{g^{-1},T_g X} T_{g^{-1}}(b_{g,X})\\ &= ((u_{g^{-1}}^X)^{-1})^\vee (T_{g^{-1}}\theta_X)^\vee\\ &= \big((T_{g^{-1}}\theta_X)(u_{g^{-1}}^X)^{-1}\big)^\vee = (u_g^X\theta_X)^\vee.\end{aligned}$$

The third equality is the naturality relation $b_{g^{-1},X} T_{g^{-1}}(\theta_X^\vee) = (T_{g^{-1}}\theta_X)^\vee b_{g^{-1},T_g X}$; the fourth is the actual coherence $b_{g^{-1},T_g X} T_{g^{-1}}(b_{g,X}) = b_{1,X} = \mathrm{id}_{X^\vee}$. For the final equality, twist naturality for $u_{g^{-1}}^X : T_{g^{-1}} X \to X$, followed by u_g^X, gives

$$u_g^X \theta_X u_{g^{-1}}^X = u_g^X T_g(u_{g^{-1}}^X)\theta_{T_{g^{-1}}X} = T_{g^{-1}}(\theta_X),$$

where $u_g^X T_g(u_{g^{-1}}^X) = \mathrm{id}_X$ and $\theta_{T_{g^{-1}}X} = T_{g^{-1}}(\theta_X)$. The identity extends to arbitrary equivariant objects by finite direct sums and canonical chosen-dual comparisons. We have thereby proved $\Theta_{A^\vee} = (\Theta_A)^\vee$.

Thus $\mathcal{B}$ is a braided ribbon fusion category, with the braiding, unit twist, balancing and right-dual twist compatibility verified above. Kirillov's [15, Theorem 3.9] produces this twist and braiding, and [17, Proposition 8.10.12] supplies the corresponding spherical pivotal structure.

Remark 6.2. In [16, Definition 4.41] the data are phrased as a braided G-crossed fusion category and the group action is written $g(X)$; in our notation it is T_gX. Their crossed-braiding map has the same source and target as $c_{X,Y}$, and their G-grading is our external direct sum. The identities referred to as action compatibility and crossed hexagons in that source are established by Xu's construction [7, Theorem 1.1, Proposition 8.5, Proposition 9.6]; it fixes the positive-half-turn normalization used here. Proposition 4.56(ii) of [16] gives the nondegeneracy criterion in terms of faithful grading and nondegeneracy of the identity component. Kirillov's Theorem 3.9 supplies the braided ribbon structure on $\mathcal{C}/G$, and his Theorem 10.5 identifies modularity with invertibility of the extended S-operator.

Proposition 6.3. *Let V be strongly rational and $G \leq \mathrm{Aut}(V)$ finite, and let $\mathcal{C}_G$ be the faithfully G-graded G-crossed braided fusion category constructed in* [7, Theorem 1.1]. *Then its equivariantization $\mathcal{B}$ is a nondegenerate braided fusion category.*

Proof. The strongly rational hypotheses imply the CFT-type and self-contragredience conditions, complete reducibility, and C_2-cofiniteness required by Huang's rigidity and modularity theorem [14]. Hence $\mathrm{Rep}(V)$ is a nondegenerate braided fusion category. By Theorem 3.3 and Xu's binary $P(z)$-representing, product–iterate, unit, positive-half-turn commutor, and parallel-transport identities [7, Theorem 4.3, Theorem 8.3, Proposition 8.5, Proposition 9.6], Proposition 3.4 gives a braided monoidal equivalence $F : \mathcal{C}_1 \xrightarrow{\sim} \mathrm{Rep}(V)$. Such an equivalence preserves double braidings and identifies the Müger centers, so $\mathcal{Z}_2(\mathcal{C}_1) \simeq_{\mathrm{br}} \mathcal{Z}_2(\mathrm{Rep}(V)) \simeq_{\mathrm{br}} \mathrm{Vec}$.

Xu's Theorem 1.1 supplies a braided G-crossed fusion category $\mathcal{C}_G$ in the sense of [16, Definition 4.41], for the finite group G, with a faithful grading $\mathcal{C}_G = \bigoplus_{g\in G} \mathcal{C}_g$. In particular, $\mathcal{C}_g \neq 0$ for every $g \in G$ is part of the theorem's faithful-grading conclusion. The equivariantization constructed above is the braided fusion category $\mathcal{B} = \mathcal{C}_G^G$. By [16, Proposition 4.56(ii)], this category is nondegenerate if and only if its neutral component is nondegenerate and its G-grading is faithful. Both conditions hold, and therefore $\mathcal{Z}_2(\mathcal{B}) \simeq_{\mathrm{br}} \mathrm{Vec}$. □

Proof of Theorem 1.1. The first assertion is Theorem 5.5. For the second, the equivariantization $\mathcal{B} = \mathcal{C}_G^G$ is a fusion category with the braiding and right duals constructed in Definition 6.1. Its ordinary twist (6.1) satisfies balancing and right-dual compatibility, so $\mathcal{B}$ is a ribbon fusion category; this is the structure of [15, Theorem 3.9], with the action tensorators and duality comparisons made explicit above. Proposition 6.3 gives $\mathcal{Z}_2(\mathcal{B}) \simeq \mathrm{Vec}$. Thus $\mathcal{B}$ is a nondegenerate ribbon fusion category, hence a modular tensor category in the convention of Section 2. For a ribbon fusion category, this is equivalent to nonsingularity of its ordinary S-matrix, the premodular criterion of [16]; thus it is also modular in the ordinary sense used by Kirillov.

We apply [15, Theorem 10.5] to the G-equivariant fusion category $\mathcal{C}_G$ underlying Theorem 5.5. Its finite semisimple rigid structure, crossed braiding, monoidal G-action, and equivariant double-dual identification $\delta : \mathrm{id} \xrightarrow{\sim} (-)^{\vee\vee}$ supply the data required in [15, Definition 2.1]; the equivariantization $\mathcal{B}$ agrees with Kirillov's orbifold category $\mathcal{C}_G/G$. In the notation of [15, Section 7], the corresponding extended Verlinde space has components

$$\widetilde{\mathcal{V}}_{k,g} = \bigoplus_{[X]\in \mathrm{Irr}(\mathcal{C}_g)} \mathrm{Hom}_{\mathcal{C}_G}(X, T_kX).$$

Since $T_kX \in \mathcal{C}_{kgk^{-1}}$, a summand vanishes unless $kg = gk$; even for commuting k, g, it vanishes unless $T_kX \simeq X$. Thus the nonzero components have precisely Kirillov's commuting-pair grading. With his unnormalized, categorical-trace definition [15, (9.3)], the operator acts as $\widetilde{S} : \widetilde{\mathcal{V}}_{k,g} \to \widetilde{\mathcal{V}}_{g^{-1},k}$: his R_k is our T_k, and the canonical action and right-dual comparisons are those fixed in Sections 2 and 5. This is the extended operator specified in Definition 2.3, not the ordinary S-matrix of $\mathcal{B}$ or a genus-one VOA trace transformation. Kirillov's Theorem 10.5 states that his extended operator is invertible if and only if this orbifold category is modular. Applying its implication from the modularity of $\mathcal{B}$ proves invertibility of $\widetilde{S}$ on the entire extended Verlinde

space, and hence the asserted G-crossed modularity. Finally, we compute the Frobenius–Perron dimensions. By Proposition 3.4, there is a braided strong monoidal equivalence

$$F : (\mathcal{C}_1, \boxtimes_D, c^D, a^D) \xrightarrow{\sim} (\mathrm{Rep}(V), \boxtimes_H, c^H, a^H).$$

A tensor equivalence induces an isomorphism of Grothendieck rings, identifies their simple-object classes, and carries the unique positive Frobenius–Perron dimension character to the corresponding character. Consequently it preserves Frobenius–Perron dimensions. Using also the convention from Section 2 that the dimension character of $\mathcal{C}_G$ restricts to the intrinsic one on its neutral fusion subcategory, we obtain

$$\mathrm{FPdim}(\mathcal{C}_1) = \sum_{[X]\in \mathrm{Irr}(\mathcal{C}_1)} \mathrm{FPdim}_{\mathcal{C}_G}(X)^2 = \sum_{[M]\in \mathrm{Irr}(\mathrm{Rep}(V))} \mathrm{FPdim}_{\mathrm{Rep}(V)}(M)^2 = \mathrm{FPdim}(\mathrm{Rep}(V)).$$

The grading of $\mathcal{C}_G$ is faithful by Xu's Theorem 1.1, and G is finite. By [18, Proposition 8.20], in a fusion category faithfully graded by a finite group, the Frobenius–Perron dimensions of all homogeneous components are equal; these component dimensions are computed using the FP-dimension character of the whole fusion category. Hence, for every $g \in G$, $\mathrm{FPdim}(\mathcal{C}_g) = \mathrm{FPdim}(\mathcal{C}_1) = \mathrm{FPdim}(\mathrm{Rep}(V))$. Here $\mathrm{FPdim}(\mathcal{C}_g)$ is the sum of squared ambient Frobenius–Perron dimensions of the simple objects in that sector, as defined in Section 2.

Because the grading is an external direct-sum decomposition, $\mathrm{Hom}_{\mathcal{C}_G}(X_g, Y_h) = 0$ for $g \neq h$, and every object is a finite direct sum of homogeneous objects, each simple-object class belongs to exactly one sector. Thus $\mathrm{Irr}(\mathcal{C}_G) = \bigsqcup_{g\in G} \mathrm{Irr}(\mathcal{C}_g)$, where the union retains the external sector labels. By the definition of the component dimensions in Section 2,

$$\begin{aligned}\mathrm{FPdim}(\mathcal{C}_G) &= \sum_{[X]\in \mathrm{Irr}(\mathcal{C}_G)} \mathrm{FPdim}_{\mathcal{C}_G}(X)^2 \\ &= \sum_{g\in G} \mathrm{FPdim}(\mathcal{C}_g) = |G|\,\mathrm{FPdim}(\mathcal{C}_1) = |G|\,\mathrm{FPdim}(\mathrm{Rep}(V)).\end{aligned}$$

Finally, T is an action of the finite group G on the fusion category $\mathcal{C}_G$ by strong monoidal autoequivalences, with tensorators J^k and unit comparisons j_k as fixed in Section 2. The underlying functors are strict, but these monoidal comparisons are part of the action data. The equivariantization formula in [16, Proposition 4.26] gives, for $\mathcal{B} = \mathcal{C}_G^G$,

$$\begin{aligned}\mathrm{FPdim}(\mathcal{B}) &= |G|\,\mathrm{FPdim}(\mathcal{C}_G) \\ &= |G|^2\,\mathrm{FPdim}(\mathrm{Rep}(V)).\end{aligned}$$

This completes the proof. □

6.2. **Dimensions and simple-object counts.**

Corollary 6.4. *There is a canonical Tannakian subcategory $\mathcal{E} \simeq \mathrm{Rep}(G)$ of $\mathcal{B}$. Writing $\mathcal{E}'$ for its Müger centralizer (the full subcategory of objects having trivial double braiding with every object of $\mathcal{E}$), one has $\mathcal{E}' \simeq_{\mathrm{br}} \mathcal{C}_1^G, \mathrm{FPdim}(\mathcal{E}') = |G|\,\mathrm{FPdim}(\mathrm{Rep}(V))$. The twist on $\mathcal{E}$ is trivial.*

Proof. For a finite-dimensional representation $\rho : G \to \mathrm{GL}(W)$, let $I(W) = W \otimes \mathbf{1}$ denote the corresponding finite direct sum of copies of the tensor unit. Since T_k is $\mathbb{C}$-linear and additive, there is a canonical direct-sum identification $\iota_{k,W} : T_k I(W) \xrightarrow{\sim} W \otimes T_k\mathbf{1}$. Give $I(W)$ the equivariant structure

$$u_k^{I(W)} = (\rho(k) \otimes j_k^{-1})\,\iota_{k,W} : T_k I(W) \longrightarrow I(W).$$

This has $u_1^{I(W)} = \mathrm{id}$ and satisfies the cocycle: under the displayed identifications, the equality $u_k^{I(W)} T_k(u_l^{I(W)}) = u_{kl}^{I(W)}$ is exactly $\rho(k)\rho(l) = \rho(kl)$ together with $j_{kl}^{-1} = j_k^{-1} T_k(j_l^{-1})$. Thus $W \mapsto (I(W), (u_k^{I(W)}))$ defines a functor $\mathcal{I} : \mathrm{Rep}(G) \to \mathcal{B}$.

For representations W, U, the canonical tensor constraint $\phi_{W,U} : I(W) \boxtimes I(U) \xrightarrow{\sim} I(W \otimes U)$ is induced by the associators and unitors. It is equivariant because

$$\phi_{W,U}(u_k^{I(W)} \boxtimes u_k^{I(U)})(J^k_{I(W),I(U)})^{-1} = u_k^{I(W\otimes U)} T_k(\phi_{W,U});$$

this is precisely the monoidal unit coherence for J^k and j_k, with $\rho_{W\otimes U}(k) = \rho_W(k)\otimes\rho_U(k)$. In particular, the inverse tensorator in the equivariant tensor product is retained. The trivial representation maps to the equivariant tensor unit $(\mathbf{1}, (j_k^{-1})_k)$, so $\mathcal{I}$ is strong monoidal. Moreover,

$$\mathrm{Hom}_{\mathcal{B}}(I(W), I(U)) \cong \{f \in \mathrm{Hom}_{\mathbb{C}}(W,U) : f\rho_W(k) = \rho_U(k)f \text{ for all } k \in G\} = \mathrm{Hom}_G(W,U),$$

where we used $\mathrm{End}_{\mathcal{C}}(\mathbf{1}) = \mathbb{C}$ and the equivariant-morphism condition. Hence $\mathcal{I}$ is fully faithful.

Both $I(W)$ and $I(U)$ are supported in degree 1. The equivariantized braiding formula and the crossed unit identities therefore identify $\widetilde{c}_{I(W),I(U)}$ with the ordinary flip $W\otimes U \to U\otimes W$. Thus $\mathcal{I}$ is braided and its image $\mathcal{E}$ is symmetric, with the canonical symmetric tensor equivalence $\mathrm{Rep}(G) \simeq \mathcal{E}$; in particular, $\mathcal{E}$ is Tannakian. This canonical embedding is relative to the fixed action (T_k, J^k, j_k) and the equivariantization in Definition 6.1.

The twist on this subcategory is trivial. Indeed, for $E = I(W)$ the underlying object is supported in degree 1, so $\Theta_E = u_1^E\theta_{I(W)} = \theta_{I(W)} = \mathrm{id}_E$. Here $u_1^E = \mathrm{id}$, and $\theta_{I(W)} = \mathrm{id}$ follows by additivity and naturality from $\theta_{\mathbf{1}} = j_1 = \mathrm{id}$ in Lemma 4.1. Thus the crossed twist $\theta_{I(W)} : I(W) \to T_1 I(W)$ is identified with the ordinary equivariantized twist $\Theta_E : E \to E$ only through $T_1 = \mathrm{id}$ and the displayed equivariant structure.

We next compute the monodromy. Let $A = (\underline{A}, (u_k^A)) \in \mathcal{B}$, with $\underline{A} = \bigoplus_{g\in G} A_g$, and let $X = A_g$ be homogeneous. By the equivariantized braiding formula,

$$\widetilde{c}_{A,E_\rho}|_{X\boxtimes I(W)} = (u_g^{E_\rho}\boxtimes \mathrm{id}_X)c_{X,I(W)}, \qquad \widetilde{c}_{E_\rho,A}|_{I(W)\boxtimes X} = (u_1^A\boxtimes\mathrm{id})c_{I(W),X} = c_{I(W),X}.$$

The factor j_g^{-1} in $u_g^{E_\rho} = (\rho(g)\otimes j_g^{-1})\iota_{g,W}$ cancels the unit comparison in $c_{X,\mathbf{1}}$ by the crossed unit identity $l_X(j_g^{-1}\boxtimes\mathrm{id}_X)c_{X,\mathbf{1}} = r_X$. Consequently, after the canonical unit identifications, the composite is

$$\widetilde{c}_{E_\rho,A}\widetilde{c}_{A,E_\rho}\big|_{X\boxtimes I(W)} = \mathrm{id}_X\boxtimes\rho(g).$$

In particular, the action is $\rho(g)$, not $\rho(g^{-1})$; the order of the two braidings and the convention $c_{X,Y} : X\boxtimes Y \to T_gY\boxtimes X$ determine this direction.

For a general A, this monodromy is the block-diagonal direct sum $\bigoplus_{g\in G}(\mathrm{id}_{A_g}\boxtimes\rho(g))$ under the canonical decomposition of $\underline{A}\boxtimes I(W)$. Hence distinct degrees cannot cancel each other. If $A \in \mathcal{E}'$, choose the left regular representation ρ_{reg} of G. For every $g \neq 1$, $\rho_{\mathrm{reg}}(g)e_1 = e_g \neq e_1$, so $\rho_{\mathrm{reg}}(g) \neq \mathrm{id}$; the displayed monodromy can be the identity only if $A_g = 0$. Conversely, if A is supported in degree 1, then $\rho(1) = \mathrm{id}$ for every representation and the monodromy with every object of $\mathcal{E}$ is the identity. Since $T_k\mathcal{C}_1 = \mathcal{C}_1$, the equivariant objects supported in degree 1 are exactly $\mathcal{C}_1^G$. Thus, as full braided subcategories of the fixed equivariantization, $\mathcal{E}' = \mathcal{C}_1^G$, which gives the canonical braided equivalence in the statement. This is the direct monodromy description of [16, Proposition 4.56(i)]. The centralizer identity follows in the finite-group equivariantization with its braided G-crossed structure.

Finally, $\mathcal{C}_1$ is a fusion category, and each T_k restricts to a strong monoidal autoequivalence of $\mathcal{C}_1$: it preserves degree 1, with tensorator $J^k_{X,Y} : T_kX\boxtimes T_kY \to T_k(X\boxtimes Y)$ and unit comparison $j_k : \mathbf{1} \to T_k\mathbf{1}$. These data, including their coherence, define the action used in $\mathcal{C}_1^G$. Since G is finite, [16, Proposition 4.26] applies and gives $\mathrm{FPdim}(\mathcal{C}_1^G) = |G|\,\mathrm{FPdim}(\mathcal{C}_1)$. The braided strong monoidal equivalence of Proposition 3.4 preserves Frobenius–Perron dimensions, so $\mathrm{FPdim}(\mathcal{C}_1) = \mathrm{FPdim}(\mathrm{Rep}(V))$. Combining these identities with $\mathcal{E}' = \mathcal{C}_1^G$ yields $\mathrm{FPdim}(\mathcal{E}') = |G|\,\mathrm{FPdim}(\mathrm{Rep}(V))$, as required. □

Corollary 6.5. *For every $g \in G$, the number of irreducible g-twisted modules is*

$$|\,\mathrm{Irr}(\mathcal{C}_g)| = \big|\{[M] \in \mathrm{Irr}(\mathrm{Rep}(V)) : [T_gM] = [M]\}\big|. \tag{6.2}$$

In particular, every twisted sector contains an irreducible object.

Proof. By Theorem 1.1(ii), the extended $\widetilde{S}$-operator of Definition 2.3 is invertible. This is Kirillov's modularity condition for the underlying G-equivariant fusion category, which has finitely many simple isomorphism classes because Xu's Theorem 1.1 gives a fusion category.

Kirillov's fixed-point lemma [15, Lemma 10.7] therefore applies. We spell out the graded-space calculation, in particular the inverse in the second index transformation.

With the same indexing as in Kirillov's definition and extended-S formula [15, Equation (9.3)], our extended Verlinde space is the finite direct sum of the pieces

$$\widetilde{\mathcal{V}}_{k,h} = \bigoplus_{[X]\in \mathrm{Irr}(\mathcal{C}_h)} \mathrm{Hom}_{\mathcal{C}_G}(X, T_k X), \qquad \widetilde{S}(\widetilde{\mathcal{V}}_{k,h}) \subseteq \widetilde{\mathcal{V}}_{h^{-1},k}.$$

The index transformation $(k,h) \mapsto (h^{-1},k)$ is a bijection (and preserves the support on commuting pairs). Since $\widetilde{S}$ is invertible on the whole extended Verlinde space and each source piece maps into its uniquely assigned target piece, its restriction is an isomorphism onto that target piece: the inverse operator maps each target piece back to its unique source piece. In particular,

$$\widetilde{S}|_{\widetilde{\mathcal{V}}_{1,g}} : \widetilde{\mathcal{V}}_{1,g} \xrightarrow{\sim} \widetilde{\mathcal{V}}_{g^{-1},1}.$$

Thus $\widetilde{S}$ sends $\widetilde{\mathcal{V}}_{1,g}$ to $\widetilde{\mathcal{V}}_{g^{-1},1}$, with the sector label inverted.

Since $T_1 = \mathrm{id}$, the definition gives

$$\begin{aligned} \dim \widetilde{\mathcal{V}}_{1,g} &= \sum_{[X]\in \mathrm{Irr}(\mathcal{C}_g)} \dim \mathrm{Hom}_{\mathcal{C}_G}(X, X) \\ &= |\,\mathrm{Irr}(\mathcal{C}_g)|. \end{aligned}$$

Here each X is simple in the finite semisimple category $\mathcal{C}_G$, so $\mathrm{End}_{\mathcal{C}_G}(X) = \mathbb{C}$; the direct sum is indexed by simple isomorphism classes, and no Hom-spaces between distinct simples enter this definition.

For the target piece, $T_{g^{-1}}$ preserves the neutral component and sends simple objects to simple objects. Hence

$$\begin{aligned} \dim \widetilde{\mathcal{V}}_{g^{-1},1} &= \sum_{[M]\in \mathrm{Irr}(\mathcal{C}_1)} \dim \mathrm{Hom}_{\mathcal{C}_G}(M, T_{g^{-1}} M) \\ &= \big|\{[M] \in \mathrm{Irr}(\mathcal{C}_1) : [T_{g^{-1}} M] = [M]\}\big|. \end{aligned}$$

Indeed, the summand is zero when M and $T_{g^{-1}}M$ are not isomorphic, and is one-dimensional when they are isomorphic, by simplicity and Schur's lemma.

The fixed isomorphism classes for T_g and $T_{g^{-1}}$ coincide. If $T_g M \simeq M$, applying $T_{g^{-1}}$ and using the underlying action identity $T_{g^{-1}} T_g = T_1 = \mathrm{id}$ gives $T_{g^{-1}} M \simeq M$. Conversely, applying T_g to an isomorphism $T_{g^{-1}} M \simeq M$ gives $T_g M \simeq M$. If the action identities are expressed using their canonical natural isomorphisms, the same argument gives the identical statement on isomorphism classes. Consequently,

$$\big|\{[M] \in \mathrm{Irr}(\mathcal{C}_1) : [T_{g^{-1}} M] = [M]\}\big| = \big|\{[M] \in \mathrm{Irr}(\mathcal{C}_1) : [T_g M] = [M]\}\big|.$$

This inverse-action identification reconciles the stated g-fixed count with the actual target of Equation (9.3).

Finally, Proposition 3.4 gives a tensor equivalence $F : \mathcal{C}_1 \xrightarrow{\sim} \mathrm{Rep}(V)$ whose underlying functor is the identity on modules and module morphisms. It therefore induces a bijection on simple isomorphism classes. The action T_g on the neutral sector is the pullback action $Y_{T_g M}(v,z) = Y_M(g^{-1}v, z)$, which is the same pullback action on ordinary V-modules under this identity-on-modules comparison. Thus the bijection induced by F intertwines the actions on isomorphism classes, and it identifies the two fixed-point counts. Combining the block isomorphism and the dimension calculations proves (6.2).

The final nonemptiness assertion also follows directly from the faithful grading in Xu's Theorem 1.1: every $\mathcal{C}_g$ is nonzero, and a nonzero object in a finite semisimple graded component has a simple constituent in that component. Independently, the right-hand fixed-point set is nonempty because the vacuum module is preserved up to isomorphism by every pullback T_g. □

Corollary 6.6. *If V is additionally holomorphic (so V is its only irreducible untwisted module), $\mathcal{C}_G$ is equivalent, as a faithfully G-graded fusion category, to Vec_G^{ω} for some $[\omega] \in H^3(G, \mathbb{C}^\times)$. Here*

Vec_G^ω denotes finite-dimensional G-graded vector spaces with associator twisted by a normalized 3-cocycle ω. Its equivariantization is a modular category of Frobenius–Perron dimension $|G|^2$.

Proof. Since V is holomorphic, V is the only simple object of $\mathrm{Rep}(V)$. Under the standing strong-rationality hypotheses, $\mathrm{Rep}(V)$ is a fusion category with tensor unit V, so $\mathrm{Rep}(V) \simeq_\otimes \mathrm{Vec}$ and $\mathrm{FPdim}(\mathrm{Rep}(V)) = 1$. Proposition 3.4 is a strong tensor equivalence; hence $\mathcal{C}_1$ has the single simple isomorphism class $[\mathbf{1}]$ and $\mathrm{FPdim}(\mathcal{C}_1) = 1$.

By Xu's Theorem 1.1, $\mathcal{C}_G$ is a fusion category faithfully graded by the finite group G. The ENO grading theorem [18, Proposition 8.20] therefore gives $\mathrm{FPdim}(\mathcal{C}_g) = \mathrm{FPdim}(\mathcal{C}_1) = 1$ for every $g \in G$. As in Section 2, $\mathrm{FPdim}(\mathcal{C}_g)$ is the sum of squares of the ambient dimensions $\mathrm{FPdim}_{\mathcal{C}_G}(X)$ over the simple objects labeled by $\mathcal{C}_g$.

Faithfulness implies that each $\mathcal{C}_g$ is nonzero, and hence each finite semisimple homogeneous summand contains a simple object. If X is simple, then $\mathbf{1}$ occurs in $X \otimes X^\vee$ with multiplicity $\dim \mathrm{Hom}(\mathbf{1}, X \otimes X^\vee) = \dim \mathrm{End}(X) = 1$. Positivity and multiplicativity of Frobenius–Perron dimension imply $\mathrm{FPdim}_{\mathcal{C}_G}(X)^2 = \mathrm{FPdim}_{\mathcal{C}_G}(X \otimes X^\vee) \geq 1$. Since $\mathrm{FPdim}(\mathcal{C}_g) = \sum_{[X] \in \mathrm{Irr}(\mathcal{C}_g)} \mathrm{FPdim}_{\mathcal{C}_G}(X)^2 = 1$, there is exactly one simple isomorphism class in each $\mathcal{C}_g$. Write its representative as X_g; then $\mathrm{FPdim}(X_g) = 1$. The object $X_g \otimes X_g^\vee$ has FP dimension one and contains $\mathbf{1}$, so it is isomorphic to $\mathbf{1}$; similarly $X_g^\vee \otimes X_g \cong \mathbf{1}$. Thus each X_g is invertible.

Choose $X_1 = \mathbf{1}$. The grading places $X_g \otimes X_h$ in $\mathcal{C}_{gh}$, and multiplicativity gives it FP dimension one. Since the only simple class in that component is represented by X_{gh}, whose FP dimension is one, the semisimple decomposition of $X_g \otimes X_h$ contains exactly one copy of X_{gh}. Thus $X_g \otimes X_h \cong X_{gh}$. In particular, $X_{g^{-1}}$ is a two-sided tensor inverse of X_g. Consequently the simple invertible isomorphism classes are indexed by G, and their tensor product corresponds to group multiplication.

For every $g, h \in G$, choose an isomorphism $\phi_{g,h} : X_g \otimes X_h \xrightarrow{\sim} X_{gh}$, taking $\phi_{1,g}$ and $\phi_{g,1}$ to be the canonical left and right unitors. Let α denote the associator of $\mathcal{C}_G$. Define isomorphisms from $(X_g \otimes X_h) \otimes X_k$ to X_{ghk} by

$$L_{g,h,k} = \phi_{gh,k} \circ (\phi_{g,h} \otimes \mathrm{id}_{X_k}),$$
$$R_{g,h,k} = \phi_{g,hk} \circ (\mathrm{id}_{X_g} \otimes \phi_{h,k}) \circ \alpha_{X_g, X_h, X_k}.$$

Because X_{ghk} is simple over $\mathbb{C}$, there is a unique scalar $\omega(g,h,k) \in \mathbb{C}^\times$ such that $R_{g,h,k} = \omega(g,h,k) L_{g,h,k}$. The unit coherence identities for the associator and the chosen unitors imply that $\omega(g,h,k) = 1$ whenever any one of g, h, k is the identity. Thus ω is normalized.

The pentagon identity for α equates the two composites from $(((X_g \otimes X_h) \otimes X_k) \otimes X_\ell)$ to $X_g \otimes (X_h \otimes (X_k \otimes X_\ell))$. Substituting $R_{a,b,c} = \omega(a,b,c) L_{a,b,c}$ into those composites yields the 3-cocycle identity

$$\omega(h,k,\ell)\, \omega(g,hk,\ell)\, \omega(g,h,k) = \omega(gh,k,\ell)\, \omega(g,h,k\ell).$$

If the choices are changed to $\phi'_{g,h} = b(g,h) \phi_{g,h}$, with b normalized to preserve the chosen unitors, then

$$\omega'(g,h,k) = \omega(g,h,k)\, \frac{b(h,k) b(g,hk)}{b(g,h) b(gh,k)} = \omega(g,h,k)\, (\delta b)(g,h,k).$$

Hence the cohomology class $[\omega] \in H^3(G, \mathbb{C}^\times)$ is independent of these choices.

With the convention that the associator of Vec_G^ω on homogeneous degrees (g,h,k) is multiplication by $\omega(g,h,k)$ from $((g \otimes h) \otimes k)$ to $g \otimes (h \otimes k)$, the chosen maps $\phi_{g,h}$ define a strong monoidal functor $\mathrm{Vec}_G^\omega \to \mathcal{C}_G$ sending the one-dimensional simple object of degree g to X_g. The identity $R_{g,h,k} = \omega(g,h,k) L_{g,h,k}$ is exactly its associativity coherence, and the chosen unitors give its unit coherence. The functor preserves the grading and is fully faithful and essentially surjective, since it gives a bijection between simple isomorphism classes and preserves their endomorphism spaces. It is therefore an equivalence of faithfully G-graded fusion categories.

Finally, Theorem 1.1(ii) gives the modularity of the equivariantization $\mathcal{C}_G^G$. By Theorem 1.1(iii), $\mathrm{FPdim}(\mathcal{C}_G^G) = |G|^2 \,\mathrm{FPdim}(\mathrm{Rep}(V)) = |G|^2$. □

Use of AI tools. ChatGPT (OpenAI) was used in mathematical exploration, literature searching, preparation of preliminary drafts, proof checking, language editing, and LaTeX typesetting. The author assumes full responsibility for the mathematical arguments, references, and final text.

References

[1] T. Abe, G. Buhl and C. Dong, *Rationality, regularity, and C_2-cofiniteness*, Trans. Amer. Math. Soc. **356** (2004), 3391–3402, doi:10.1090/S0002-9947-03-03413-5.

[2] C. Dong, H. Li and G. Mason, *Twisted representations of vertex operator algebras*, Math. Ann. **310** (1998), 571–600, arXiv:q-alg/9509005.

[3] C. Dong, H. Li and G. Mason, *Modular-invariance of trace functions in orbifold theory and generalized moonshine*, Comm. Math. Phys. **214** (2000), 1–56, doi:10.1007/s002200000242.

[4] Y.-Z. Huang, *Some open problems in mathematical two-dimensional conformal field theory*, Contemp. Math. **695** (2017), 123–138, doi:10.1090/conm/695/13999.

[5] S. Carnahan and M. Miyamoto, *Regularity of fixed-point vertex operator subalgebras*, arXiv:1603.05645.

[6] R. McRae, *Twisted modules and G-equivariantization in logarithmic conformal field theory*, Comm. Math. Phys. **383** (2021), 1939–2019, arXiv:1910.13226.

[7] S. Xu, *G-crossed braided tensor categories of twisted modules*, arXiv:2610.29130v1 [math.QA] (2026).

[8] J. Du and Y.-Z. Huang, *Twisted intertwining operators and tensor products of (generalized) twisted modules*, SIGMA **22** (2026), 086, 64 pp., doi:10.3842/SIGMA.2026.086.

[9] I. B. Frenkel, Y.-Z. Huang and J. Lepowsky, *On axiomatic approaches to vertex operator algebras and modules*, Mem. Amer. Math. Soc. **104** (1993), no. 494, doi:10.1090/memo/0494.

[10] Y.-Z. Huang and J. Yang, *Associative algebras for (logarithmic) twisted modules for a vertex operator algebra*, Trans. Amer. Math. Soc. **371** (2019), 3747–3786, doi:10.1090/tran/7490.

[11] J. Yang, *On associative algebras, modules and twisted modules for vertex operator algebras*, J. Algebra **440** (2015), 354–378, doi:10.1016/j.jalgebra.2015.05.026.

[12] Y.-Z. Huang and J. Lepowsky, *Tensor products of modules for a vertex operator algebra and vertex tensor categories*, in *Lie Theory and Geometry*, Progr. Math. **123**, Birkhäuser, 1994, 349–383.

[13] Y.-Z. Huang, *Differential equations and intertwining operators*, Commun. Contemp. Math. **7** (2005), 375–400, doi:10.1142/S0219199705001799.

[14] Y.-Z. Huang, *Rigidity and modularity of vertex tensor categories*, Commun. Contemp. Math. **10** (2008), suppl. 1, 871–911, doi:10.1142/S0219199708003083.

[15] A. Kirillov, Jr., *On G-equivariant modular categories*, arXiv:math/0401119.

[16] V. Drinfeld, S. Gelaki, D. Nikshych and V. Ostrik, *On braided fusion categories I*, Selecta Math. (N.S.) **16** (2010), 1–119, doi:10.1007/s00029-010-0017-z.

[17] P. Etingof, S. Gelaki, D. Nikshych and V. Ostrik, *Tensor Categories*, Math. Surveys Monogr. **205**, Amer. Math. Soc., 2015, Section 8.24.

[18] P. Etingof, D. Nikshych and V. Ostrik, *On fusion categories*, Ann. of Math. (2) **162** (2005), 581–642, arXiv:math/0203060.

School of Mathematical Sciences, Anhui University, Hefei 230601, Anhui, China
Email address: shunxu@ahu.edu.cn